

Air Pollution and Behavior: Evidence from Wearable Devices

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Abstract

Avoiding air pollution may protect health while discouraging activities that improve health. We examine this tradeoff using millions of daily wearable observations linked to pollution data. Instrumenting pollution with wind direction, we find that pollution reduces physical activity, shortens sleep, and lowers running performance. Declines in activity are followed by next-day increases, suggesting that people partly adjust when they are active rather than simply forgoing activity. These findings reveal both the scope for adaptation through rescheduling and its limits: pollution affects participation in exercise, performance during exercise, and sleep, extending its economic consequences beyond conventional measures of illness and productivity.

JEL Codes: Q53, I12, Q51

Keywords: Air pollution; physical activity; sleep; wearable devices; avoidance behavior

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1 Introduction

Few relationships in environmental economics are as thoroughly documented as the harm from breathing polluted air. Exposure to fine particulate matter increases mortality (Chay and Greenstone, 2003; Currie and Neidell, 2005; Deryugina et al., 2019; Molitor, Mullins and White, 2023; Miller, Molitor and Zou, 2024), raises morbidity (Neidell, 2004; Schlenker and Walker, 2016; Heft-Neal et al., 2023; Barwick et al., 2026), impairs cognition and learning (Zhang, Chen and Zhang, 2018; Krebs and Luechinger, 2024; La Nauze and Severnini, 2025), and reduces worker productivity (Graff Zivin and Neidell, 2012; Chang et al., 2016; Archsmith, Heyes and Saberian, 2018; Chang et al., 2019).¹ Among these consequences, the effects on health have received particular attention. Pollution can affect health both through physiological harm from exposure and through changes in daily behavior. Individuals may reduce outdoor exercise, postpone activities, or remain indoors when air quality deteriorates. These “avoidance” responses can limit exposure, but they may also displace activities that contribute to health. Pollution can also disrupt sleep or make physical exertion more difficult, even without deliberate changes in behavior. Understanding these responses matters for interpreting the costs of pollution: avoidance can attenuate observed health effects, while forgone exercise, disrupted routines, and lost recreational opportunities impose costs of their own.

An important question is how pollution changes daily behavior, including how much people move, how intensely they exercise, and how they sleep. Existing measures of avoidance, such as attendance at outdoor recreation sites, capture particular activities or settings. A decline in outdoor activity need not imply less physical activity overall if people substitute toward indoor exercise. Conversely, unchanged attendance or time spent outdoors may conceal reductions in the duration or intensity of activity. Pollution may also impair performance during exertion or disrupt sleep, responses that measures of outdoor attendance cannot capture. Moreover, much physical activity also occurs outside formal exercise, through walking and other routine activities that may be difficult to observe in conventional data. The timing of these responses also matters: a decline in activity on a polluted day may represent exercise forgone or activity postponed until another day. Distinguishing these possibilities requires following behavior beyond the day of exposure. Measuring these dimensions alongside sleep, and tracing their responses over time, provides a more complete account of how pollution changes daily life and the activities that contribute to health.

We address these questions using longitudinal Fitbit records from the National Institutes of Health’s All of Us Research Program, the world’s largest integrated health database

¹Aguilar-Gomez et al. (2022) survey a literature now numbering in the hundreds of papers.

(Nunn, 2026), covering approximately 51 million participant-days of physical activity and 37 million nights of sleep between 2009 and 2024. These records measure daily steps, walking and running minutes, activity intensity, and sleep duration and stages. Minute-level step counts reveal how activity is distributed throughout the day and, for running in particular, allow us to measure duration and cadence (steps per minute), which we use as a measure of performance. Because the devices record movement in both indoor and outdoor settings, we measure total activity after any substitution between them, although we cannot distinguish the settings themselves. We link these records to participants’ demographic and socioeconomic characteristics, electronic health records, and daily pollution and weather measures at their residential three-digit ZIP code (ZIP3) locations.

Estimating the effects of pollution presents two empirical challenges. First, exposure is not randomly assigned. Individuals sort into residential locations that differ in air quality (Banzhaf and Walsh, 2008; Bayer, Keohane and Timmins, 2009), and their health, income, and preferences may also influence activity and sleep. Within locations, changes in economic activity and other conditions may affect both pollution and these outcomes. Second, pollution concentrations measured at monitoring stations may differ from those where participants live, work, and spend time, introducing measurement error in assigned exposure. Participant fixed effects allow us to compare activity and sleep within individuals across days, removing persistent differences across individuals associated with residential sorting. To address remaining confounding and measurement error, we use an instrumental variable strategy based on changes in wind direction. Building on Deryugina et al. (2019) and La Nauze and Severnini (2025), we construct an indicator for wind arriving from the direction associated with the highest pollution concentration in each region and calendar month. We construct each observation’s instrument using wind directions selected from other years of data, separating direction selection from estimation through cross-fitting. Along with participant fixed effects, our regressions include ZIP3-by-month-by-year and calendar-date fixed effects and controls for weather and daylight. Under the assumption that wind direction affects activity and sleep only through pollution and is uncorrelated with pollution measurement error conditional on these controls, this approach identifies responses to wind-induced changes in $\text{PM}_{2.5}$. We examine both changes in concentration and differences between days above and at or below $12 \mu\text{g}/\text{m}^3$, a benchmark chosen to correspond to the EPA’s annual $\text{PM}_{2.5}$ standard in place during most of our study period.² We then extend the analysis using distributed lag models to assess whether initial reductions in activity and sleep persist or are offset over

²The EPA adopted the $12 \mu\text{g}/\text{m}^3$ annual standard in 2012 and lowered it to $9 \mu\text{g}/\text{m}^3$ in 2024. We use this benchmark to distinguish higher- and lower-pollution days, rather than to measure violations of the annual standard.

the following seven to fourteen days.

Our IV estimates show reductions in both everyday movement and exercise. A one-standard-deviation increase in $\text{PM}_{2.5}$, approximately $6.1 \mu\text{g}/\text{m}^3$, reduces daily steps by 99. Participants take 327 fewer steps on days with concentrations above $12 \mu\text{g}/\text{m}^3$ than on days at or below that threshold, a reduction equivalent to 4.2% of mean daily steps. Walking, running, and total active time also decline, while sedentary time increases. Nearly all of the reduction in active time occurs in light activity, indicating that participants reduce routine movement as well as vigorous exercise on more polluted days. When participants run, they do so for less time and at a lower cadence. Shorter runs and lower cadence may reflect both greater difficulty exercising (leading to reduced performance) and participants choosing to exert less effort, as well as compositional changes in who runs on polluted days.

Participants also sleep less on more polluted days. A one-standard-deviation increase in $\text{PM}_{2.5}$ reduces sleep duration by approximately 1.9 minutes. On days with concentrations above $12 \mu\text{g}/\text{m}^3$, participants sleep 6.4 fewer minutes that night than on days with concentrations at or below the threshold. In the sample of participants for whom we observe sleep-stage measurements, the decline in sleep is concentrated in light and rapid eye movement (REM) sleep stages, with no statistically significant change in deep sleep. These reductions may reflect physiological effects of pollution, changes in sleep schedules due to changing opportunity costs of sleep, or both.

We then examine whether participants offset reductions in activity and sleep by shifting them to other times. Within the day, activity reductions are concentrated in the afternoon and evening, with no evidence of offsetting increases in other periods. Across days, however, an initial decline in steps is followed by a next-day increase, and sleep also shows some recovery. Cumulative responses over seven and fourteen days are not statistically distinguishable from zero. These patterns are consistent with participants making up some of the lost activity and sleep on subsequent days, although the estimates are not precise enough to determine whether they make up the full amount. These findings on behavioral dynamics highlight why responses on the day of exposure alone may not capture the full change in activity and sleep.

Wildfire smoke provides a complementary setting for studying responses to salient air pollution. We link participants' wearable records to daily wildfire-attributable $\text{PM}_{2.5}$ concentrations and estimate how activity changes on the exposure day and over subsequent days. Participants reduce their steps, walking, and running as smoke concentrations rise, without fully making up these reductions over the following seven or fourteen days. These more persistent reductions motivate an illustrative, back-of-the-envelope calculation of the recreational costs of smoke exposure. Combining our estimates with observed smoke concen-

trations during June 6–8, 2023, when Canadian wildfire smoke blanketed the northeastern United States, yields projected cumulative losses of approximately 23 walking minutes and 0.69 running minutes per person across New York, New Jersey, and Pennsylvania. Applying recreational valuations and extrapolating to the region’s population gives approximately \$309 million in forgone recreational value.

Our paper makes several contributions to the literature. First, we add to a large literature examining behavioral responses to air pollution, which are often used to estimate the benefits of improved air quality (Bartik, 1988). Studies of pharmaceuticals, protective masks, and air purifiers establish defensive expenditures as an important component of the demand for air quality (Deschênes, Greenstone and Shapiro, 2017; Zhang and Mu, 2018; Ito and Zhang, 2020), while research on pollution disclosure shows that information induces protective purchases and avoidance of outdoor exposure (Barwick et al., 2024; Wang and Zhang, 2023). Studies of outdoor recreation and wildfire-related mobility examine how pollution and air-quality information affect visits to outdoor destinations and time spent at home (Neidell, 2009; Keiser, Lade and Rudik, 2018; Burke et al., 2022; Heft-Neal et al., 2023; Berestycki and Chen, 2026), while Fan (2024) examines participation in outdoor exercise, exercise duration, and substitution indoors among users of a popular exercise app in China. We build on this evidence by measuring total movement throughout observed days, including activity that would not appear in logged exercise sessions and any movement maintained indoors. Additionally, whereas earlier studies relied on reported time outdoors and survey measures of physical activity and sleep (Bresnahan, Dickie and Gerking, 1997; Deschênes et al., 2020), we exploit repeated device measurements that do not require participants to recall or characterize their activity, avoiding reporting errors associated with recall bias and social desirability (Adams et al., 2005; Lauderdale et al., 2008). This broader measurement is consequential: most of the reduction in active time occurs in light activity, indicating that pollution responses extend beyond formal exercise and recreation to routine movement. Thus, our findings reveal changes in a health-related behavior that protective expenditures, destination visits, and exercise participation alone cannot fully capture.

Second, we demonstrate the importance of measuring cumulative changes in activity and sleep rather than focusing only on the day of exposure. Measuring dynamic effects has long been central to the pollution and temperature literatures: research on mortality displacement, or “harvesting,” asks whether an initial increase in deaths partly reflects deaths brought forward by days or weeks (Schwartz, 2000; Deschênes and Moretti, 2009). A related question arises for behavior: do participants forgo activity on polluted days or postpone it? Fan (2024) examines this possibility and finds that subsequent increases in exercise approximately offset additional reductions after the exposure day, leaving the contemporaneous

estimate as a reasonable approximation to the net exercise effect. We find a different pattern for total daily movement. An initial reduction in steps is followed by a next-day increase, and cumulative estimates over seven and fourteen days are not statistically distinguishable from zero. Sleep also shows subsequent recovery. Although these estimates are not precise enough to determine whether participants make up the full amount, they suggest that contemporaneous reductions can overstate cumulative losses in our setting. Such recovery need not imply costless adjustment, since participants may exercise or sleep at less preferred times, but it cautions against treating every contemporaneous reduction as activity or sleep permanently forgone.

Third, we extend evidence on pollution and human performance to everyday exercise. Existing studies document productivity losses among agricultural workers, indoor manufacturing workers, and call-center employees (Graff Zivin and Neidell, 2012; Chang et al., 2016, 2019), as well as impaired cognitive and athletic performance (La Nauze and Severnini, 2025; Krebs and Luechinger, 2024; Künn, Palacios and Pestel, 2023; Mullins, 2018). We examine running, where individuals can adjust whether to participate, how long to continue, and how intensely to exert themselves. A 10 $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ reduces running cadence by approximately 0.8% of its sample mean, alongside a reduction in running duration. For comparison, Archsmith, Heyes and Saberian (2018) find that the same increase in $\text{PM}_{2.5}$, measured over twelve hours, raises baseball umpires’ probability of an incorrect call by 2.6%. Our findings extend this literature beyond workplaces and competitive settings, where incentives to maintain performance may be stronger, to voluntary exercise that contributes to health and well-being.

Fourth, we contribute to research on sleep as an economically important use of time and a potential determinant of productivity and health (Biddle and Hamermesh, 1990; Gibson and Shrader, 2018; Giuntella and Mazzonna, 2019; Avery, Giuntella and Jiao, 2025; Giuntella, Saccardo and Sadoff, 2026; Harris, 2026*b*). Pollution may affect sleep through physiological responses or by changing its opportunity cost: fewer appealing opportunities for outdoor leisure, for example, may encourage people to spend more time sleeping (Harris, 2026*c*). Existing research uses nighttime social-media activity to measure pollution-induced sleeplessness (Heyes and Zhu, 2019) and wearable records to examine sleep in China and Germany (Zhou et al., 2023; Janzen and Bigler, 2026). We build on this evidence by using wind-induced pollution variation to examine sleep duration, time in bed, and time spent in light, REM, and deep sleep. Observing these dimensions together shows how changes in total sleep are distributed across stages and accompanied by changes in time in bed. Following participants over subsequent nights also reveals recovery that is not apparent from the exposure-night response alone. To the best of our knowledge, we are the first to causally

estimate these sleep stage effects as well as cumulative responses.

Taken together, our findings offer two candidate pathways through which pollution may affect outcomes beyond those directly examined here. Reduced physical activity may contribute to pollution-related obesity and other health consequences, a possibility explored by Deschênes et al. (2020). Sleep supports learning by consolidating newly acquired information into memory and helps sustain attention and cognitive performance during waking hours. Experimental evidence shows that restricting sleep can produce lapses in attention and impair working memory, with deficits accumulating across successive nights (Rasch and Born, 2013; Van Dongen et al., 2003). These functions matter for tasks that require workers to remain alert, retain information, and respond accurately. Pollution-induced sleep loss may therefore contribute to the cognitive and workplace productivity losses documented elsewhere. We do not establish that sleep mediates these outcomes or that the reductions we estimate produce the impairments observed under experimental sleep restriction. Nevertheless, directly observing movement and rest helps connect evidence on pollution exposure to the everyday activities through which some of its wider consequences may arise.

2 Data

2.1 All of Us Data

Fitbit data. Our primary data are longitudinal Fitbit records from the National Institutes of Health’s All of Us Research Program, a nationwide initiative that combines participant surveys, physical measurements, electronic health records, and other health information (Denny et al., 2019). As of July 2026, nearly 750,000 individuals have shared data with the program, and over 60,000 of them have shared Fitbit device data.³ Participants contribute Fitbit data, automatically and continuously upon consent, either from devices they already owned (or those they purchase after enrollment) or from devices provided through the Wearables Enhancing All of Us Research (WEAR) study.⁴ Shared records can include activity recorded before enrollment, allowing our sample period to span January 2009 through November 2024. Observation histories vary with when participants began using their Fitbit device and how consistently they wear them.

We examine daily (and hourly) steps, walking and running minutes, time spent at different activity intensities, calories expended, sleep duration, and time spent in each sleep

³See <https://www.joinallofus.org/article/the-all-of-us-research-program-enters-the-multiphomics-era-with-landmark-data-release> for information on the most recent data release.

⁴See <https://support.researchallofus.org/hc/en-us/articles/20281023493908-Resources-for-Using-Fitbit-Data> for the program’s Fitbit data collection documentation.

stage. We construct walking and running measures from minute-level step counts following Harris (2026a). A walking minute contains more than 30 and at most 130 steps, while a running minute contains more than 130 steps. Fitbit classifies activity intensity using estimated metabolic equivalents (METs), where one MET represents the metabolic rate at rest, which Fitbit’s proprietary algorithm estimates based on a user’s age, sex, height, weight, and other metrics. Its categories distinguish sedentary time (less than 1.5 METs), lightly active time (1.5–3 METs), fairly active time (3–6 METs), and very active time (more than 6 METs) (Bailey et al., 2025). We examine minutes in each category and total active minutes, defined as the sum of lightly, fairly, and very active minutes. Fitbit estimates daily calories expended by combining an estimate of calories burned at rest with recorded activity and, when available, heart-rate data. The measure therefore includes calories burned both at rest and during activity.⁵ Fitbit estimates time spent in light, deep, and rapid eye movement (REM) sleep using movement and heart-rate patterns, including beat-to-beat variation in heart rate.⁶ We examine minutes spent in each stage during the main sleep episode, restricting these measures to nights with valid sleep-stage records. For these nights, we also examine total time in bed during the main sleep episode, which includes both time asleep and time awake. The physical activity measures capture activity in both indoor and outdoor settings, which we cannot distinguish in the data. Thus, they allow us to assess changes in total physical activity, including any offsetting increases in indoor activity when outdoor conditions deteriorate.

Electronic health records. The All of Us data link participants’ physical activity records to electronic health records, which we use to examine whether activity responses differ for participants with evidence of respiratory conditions. We identify participants with a recorded asthma or chronic obstructive pulmonary disease diagnosis or a record of respiratory medication, including rescue bronchodilators, inhaled corticosteroids, and other maintenance treatments. Participants with electronic health records but no such evidence form the comparison group; those without electronic health records are classified separately. Medication records do not necessarily establish that prescriptions were filled or medications taken.

Participant characteristics. Participants report their demographic and socioeconomic characteristics in the All of Us baseline survey, including gender, race, ethnicity, household income, educational attainment, and employment status. The survey also collects residential

⁵See <https://support.google.com/googlehealth/answer/14237111> for Fitbit’s documentation of its calorie estimates.

⁶See <https://support.google.com/googlehealth/answer/14236712> for Fitbit’s documentation of its sleep-stage estimates.

Table 1: Summary statistics

Variable	Participants	Mean	SD	P25	Median	P75
A. Fitbit outcomes						
Daily steps	61,338	7,838.2	5,032.9	4,150.0	6,906.0	10,562.0
Walking minutes	61,338	85.6	58.4	43.0	76.0	117.0
Running minutes	61,338	1.6	8.3	0.0	0.0	0.0
Active minutes	61,338	259.0	119.0	177.0	249.0	329.0
Sedentary minutes	61,338	817.7	268.7	630.0	758.0	1,025.0
Lightly active minutes	61,338	218.7	100.4	151.0	209.0	276.0
Fairly active minutes	61,338	19.6	31.3	0.0	9.0	26.0
Very active minutes	61,338	20.7	34.1	0.0	5.0	30.0
Calories expended (kcal/day)	61,338	2,320.1	736.9	1,857.0	2,207.0	2,657.0
Sleep duration (minutes)	59,055	389.5	110.5	336.0	403.0	458.0
Time in bed (minutes)	59,055	442.8	124.3	384.0	457.0	517.0
Light sleep (minutes)	56,138	259.9	69.1	214.0	258.0	303.0
Deep sleep (minutes)	56,138	59.7	26.9	41.0	59.0	77.0
REM sleep (minutes)	56,138	80.7	36.1	55.0	80.0	105.0
B. Participant characteristics						
Age at first observation (years)	61,338	50.8	16.4	37.0	51.6	64.6
Observed days	61,338	834.1	889.9	207.0	440.0	1,181.0
Height (cm)	56,007	168.4	9.5	161.8	167.6	175.3
Weight (kg)	55,849	83.9	23.0	67.3	80.3	96.2
Body mass index (kg/m ²)	55,729	29.6	7.6	24.2	28.0	33.3
C. Environment						
PM _{2.5} (μg/m ³)	61,338	8.1	6.1	4.7	6.9	9.9
Wildfire PM _{2.5} (μg/m ³)	57,519	1.0	4.7	0.0	0.0	0.0
Overall AQI	53,522	49.2	20.0	36.0	47.0	58.0
Maximum temperature (°F)	61,338	66.6	19.8	52.3	69.9	82.3
Precipitation (mm)	61,338	2.7	7.0	0.0	0.0	1.7
Daylight (hours)	61,338	12.3	2.0	10.5	12.3	14.0
Wind speed (m/s)	61,338	3.1	1.4	2.0	2.8	3.9
D. Participant demographics (share of participants)						
Gender		Race/ethnicity		Household income		
Female	0.661	Black	0.078	<25k	0.150	
		White	0.704	25-75k	0.319	
		Asian	0.049	75-150k	0.298	
		Hispanic	0.111	>150k	0.168	
		Other	0.030	Missing/unknown	0.064	
		Missing/unknown	0.028			
Educational attainment		Employment status				
HS or less	0.096	Employed	0.482			
Some college	0.265	Retired	0.233			
BA+	0.631	Other employment	0.155			
Missing/unknown	0.008	Missing/unknown	0.129			

Notes: The table presents descriptive statistics for Fitbit wearers in the *All of Us* Research Program. Panel A summarizes Fitbit outcomes. Panel B describes participant characteristics. Height and weight use the earliest valid adult measurements from health records; BMI is calculated using height and weight measured on the same date. Panel C summarizes pollution and weather variables. Panel D reports participant shares by demographic characteristic, including missing responses in the denominator. Panels A and C weight person-days equally; Panels B and D weight participants equally. Sleep measures use the independently eligible sleep sample; stage durations require valid stage records. Participant counts reflect available observations for each variable.

ZIP codes, which are available to researchers at the three-digit ZIP code (ZIP3) level. We use residential ZIP3 codes to link Fitbit records to pollution and weather conditions, holding each participant’s recorded location fixed throughout the observation period.⁷ We calculate age from the recorded date of birth and obtain height and weight from measurement records, using these to calculate body mass index (BMI). Panels B and D of Table 1 summarize participant characteristics and the number of observed days per participant, weighting each participant equally.

Participation in All of Us, consent to share wearable records, and continued device use make the Fitbit sample selective. WEAR broadens participation by providing devices at no cost to groups historically underrepresented in biomedical research. Appendix Table A.1 compares WEAR participants with those classified as bringing their own devices (BYOD). Providing devices reduces an ownership barrier but does not eliminate selection, so our estimates need not represent average responses among U.S. adults.

Sample restrictions. For physical activity analyses, we require positive step counts in at least six distinct hours of the day and exclude daily step totals above the pooled 99.99th percentile among days meeting this coverage requirement. These restrictions reduce the influence of incomplete recording and extreme values, which are likely data errors. For sleep analyses, we use the period Fitbit identifies as the participant’s primary sleep rather than shorter periods such as naps. We require sleep durations greater than zero and no more than 16 hours and do not impose the daytime activity coverage requirement. Analyses of sleep stages and time in bed additionally require valid records for those measures. Each regression uses observations with nonmissing outcomes, exposure measures, instruments where applicable, and controls. Sample sizes therefore vary across outcomes and specifications; the tables report the number of observations used in each regression.

2.2 Air Pollution and Wildfire Smoke

We construct daily ambient fine-particulate-matter (PM_{2.5}) concentrations using monitoring data from the U.S. Environmental Protection Agency’s Air Quality System. For each ZIP3 and date, we assign concentrations using up to three available monitoring sites and inverse-distance weights relative to the ZIP3’s population-weighted centroid. We prioritize sites located within the ZIP3. When no such site has an available measurement, we use the nearest available sites within 250 miles of the centroid. If fewer than three sites satisfy the selection rule, we use the available sites; if none do, exposure is missing. This measure

⁷Participants only complete the survey once (at enrollment), so we cannot observe whether people move.

captures ambient conditions near a participant’s recorded residence rather than personal pollution exposure. Geographic aggregation and the distance to available monitors are potential sources of measurement error.

We also explore how individuals respond to wildfire smoke pollution distinct from ambient $\text{PM}_{2.5}$. To distinguish wildfire pollution from total particulate pollution, we use daily estimates of wildfire-attributable $\text{PM}_{2.5}$ from the Stanford Environmental Change and Human Outcomes Lab (Childs et al., 2022). We use the updated data product covering 2006–2023 and aggregate its ZIP Code Tabulation Area estimates to ZIP3 using population weights.⁸ These are modeled estimates of the wildfire contribution to ambient $\text{PM}_{2.5}$ rather than separate monitor measurements.⁹

Additionally, we obtain daily Air Quality Index (AQI) values from the EPA AirNow Data Management Center and link reporting areas to ZIP3s. EPA converts pollutant concentrations into pollutant-specific AQI values using concentration thresholds associated with different levels of health concern. Overall AQI is the highest of the reported pollutant-specific values, so the pollutant determining the index can vary across locations and days. Our primary AQI analyses use this overall measure; we also examine $\text{PM}_{2.5}$ -specific AQI for comparison. We estimate models using AQI as a continuous variable and models comparing its standard categories: good (0–50), moderate (51–100), unhealthy for sensitive groups (101–150), unhealthy (151–200), very unhealthy (201–300), and hazardous (301–500). The categorical models use days with good air quality as the reference group.

2.3 Weather and Daylight

We obtain daily maximum, minimum, and average temperature and precipitation from NOAA’s nClimGrid-Daily county averages for the contiguous United States. We construct ZIP3-level weather measures as population-weighted averages of county values, using the December 2021 HUD ZIP–county crosswalk and 2010 Census ZIP Code Tabulation Area populations to estimate each county’s share of the ZIP3 population. We then match these measures to participants by residential ZIP3 and date.¹⁰

We separately calculate astronomical daylight duration from sunrise and sunset at each ZIP3’s representative coordinates. Daylight duration captures geographic and seasonal vari-

⁸The updated smoke product is documented in Childs et al. (2025).

⁹As discussed by Breshears, Brewer and Earle (2027), satellite observations of smoke plumes do not necessarily indicate smoke at ground level. Modeled wildfire-attributable $\text{PM}_{2.5}$ estimates better characterize surface-level exposure, although they remain subject to measurement error from estimating concentrations away from monitors and distinguishing wildfire smoke from other sources of particulate pollution.

¹⁰See <https://www.ncei.noaa.gov/products/land-based-station/nclimgrid-daily> for NOAA’s documentation.

ation in the opportunity for activity during daylight hours; it does not measure cloud cover, actual solar irradiance, or the attenuation of sunlight by smoke. For within-day analyses, we additionally measure the number of daylight minutes falling within the relevant clock hour or time window.

Finally, we obtain hourly wind components at 10 meters above the surface from ERA5 reanalysis on a $0.25^\circ \times 0.25^\circ$ grid. We average these components across grid cells using their area of overlap with each ZIP3 and construct daily wind measures from midnight to midnight in local time. Daily wind speed is the mean of hourly speeds, while daily wind direction is calculated from the mean eastward and northward wind components. We treat daily wind measures as missing when hourly coverage is incomplete. We also exclude daily wind directions when the average wind components are too small to define a meaningful direction.

3 Empirical Strategy

3.1 Pollution and Physical Activity

We estimate the effects of air pollution on physical activity and sleep using the following equation:

$$y_{it} = \beta P_{z(i)t} + X'_{z(i)t} \gamma + \alpha_i + \lambda_{z(i),m(t)} + \delta_t + \varepsilon_{it}, \quad (1)$$

where y_{it} denotes an activity or sleep outcome for participant i on date t , $P_{z(i)t}$ is daily $\text{PM}_{2.5}$ concentration in the participant’s residential ZIP3, and $m(t)$ identifies the calendar month and year. We examine steps, walking and running minutes, time spent at different activity intensities, calories expended, and sleep duration. For sleep, we estimate the effect of pollution during the day on sleep that night. The coefficient β measures the change in the outcome—in steps, minutes, or calories, as applicable—for a one $\mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$.

Participant fixed effects, α_i , absorb persistent differences in activity, health, and device use across participants. ZIP3-by-month-by-year fixed effects, $\lambda_{z(i),m(t)}$, absorb local conditions that vary across months, including seasonal activity patterns and changes in pollution sources. Calendar-date fixed effects, δ_t , absorb shocks common across locations on a given date, including national holidays and day-of-week patterns. The controls X_{zt} include indicators for daily maximum temperature in five-degree Fahrenheit intervals, precipitation categories, astronomical daylight duration, and daily wind speed.¹¹ Standard errors are

¹¹The lowest temperature category includes values below 0°F , and the highest includes values at or above 110°F . Precipitation categories distinguish zero precipitation, positive amounts up to 0.25 inches, subsequent quarter-inch intervals through one inch, and amounts above one inch. Daylight duration measures the interval

clustered by ZIP3 and calendar date, allowing dependence across participants and dates within a ZIP3 and across locations on the same date.

Equation (1) can be estimated by ordinary least squares (OLS), but measurement error in pollution exposure is a central concern. We assign pollution using monitoring stations near each participant’s residential ZIP3, although concentrations can vary within a ZIP3 and between residences and monitoring sites. Our measure therefore approximates local ambient pollution rather than observing it at each participant’s residence. Classical measurement error would attenuate the OLS coefficient toward zero. The discrepancy need not be classical, however, so the direction of bias is not guaranteed. In addition, day-to-day changes in local economic activity or other unobserved conditions may affect both pollution and physical activity despite the controls and fixed effects.

These concerns motivate an instrumental variable approach based on changes in wind direction (Deryugina et al., 2019; Brewer, Dench and Taylor, 2023; La Nauze and Severnini, 2025). Winds from different directions transport pollution from different sources, generating variation in local concentrations. Following the approach of La Nauze and Severnini (2025), we construct an indicator for wind arriving from the direction associated with the highest pollution concentration. To do this, we first adapt the geographic clustering approach of Deryugina et al. (2019) by grouping ZIP3 areas into 20 regions using k -means clustering of their population-weighted centroid coordinates.¹² Then, within each region, we regress $\text{PM}_{2.5}$ on interactions between four 90-degree wind sectors and calendar-month indicators, controlling for weather, daylight, ZIP3-by-month-by-year fixed effects, and date fixed effects. For each region and calendar month, we select the sector with the largest estimated coefficient and define the instrument as an indicator for wind arriving from that sector.

To separate direction selection from estimation, we divide calendar years into five folds and conduct cross-fitting. For observations in a given fold, we select directions using years in the other four folds. We exclude all observations from years in the given fold, across all locations, along with the seven days immediately before and after each excluded year.¹³ Let W_{zt} denote the observed wind sector, $r(z)$ the geographic region, $c(t)$ the calendar month, and $f(t)$ the fold containing date t . If $\hat{w}_{r,c}^{(-f)}$ is the selected sector learned without fold f , the instrument is

$$Z_{zt} = \mathbf{1}\left\{W_{zt} = \hat{w}_{r(z),c(t)}^{(-f(t))}\right\}. \quad (2)$$

This procedure assigns each observation a wind-direction indicator selected using data from

between sunrise and sunset, rather than realized sunshine or solar irradiance.

¹²Appendix Figure B.2 shows a map of these regions.

¹³Years are assigned cyclically to five folds: for example, 2009, 2014, 2019, and 2024 belong to the same fold.

other years. We then estimate the first stage using all observations:

$$P_{z(i)t} = \pi Z_{z(i)t} + X'_{z(i)t} \eta + a_i + \ell_{z(i),m(t)} + d_t + u_{it}. \quad (3)$$

Our primary estimates use two-stage least squares, combining equations (1) and (3) with the same controls, fixed effects, and observations in both stages. We construct the wind-direction instrument using continuous $\text{PM}_{2.5}$ concentrations and use the same instrument across contemporaneous outcomes and exposure specifications. Selecting the most pollution-producing wind direction and estimating its first-stage relationship using the same observations could overstate instrument relevance, because the selected direction may partly reflect chance variation in pollution. Cross-fitting separates these steps: each observation receives an instrument based on directions selected using other years. This reduces overfitting in instrument construction while allowing all eligible observations to contribute to estimation.

Our instrument follows the selected-direction approach of La Nauze and Severnini (2025), with several modifications. Their main specification selects the wind sector most strongly associated with $\text{PM}_{2.5}$ exceeding $25 \mu\text{g}/\text{m}^3$, using the full estimation sample. They assess sensitivity to this use of the same observations for direction selection and estimation through separate sample-splitting and leave-one-ZIP3-out exercises. We incorporate this separation into our primary specification: the wind direction used to construct each observation’s instrument is selected using years in the other four folds. We then estimate the model using all five folds together. Excluding entire years across all locations ensures that pollution observed elsewhere on the same dates cannot influence the selection of wind directions for those observations. We also select directions using continuous $\text{PM}_{2.5}$ rather than a threshold indicator, providing a common instrument for our continuous and binary exposure specifications.

Identification requires that the selected-direction indicator predict pollution and, conditional on the controls and fixed effects, be uncorrelated with pollution measurement error and affect outcomes only through pollution. Cross-fitting limits overfitting in direction selection but does not establish this exclusion restriction. Wind could directly affect outdoor comfort or transport other pollutants that independently affect activity. We therefore interpret the estimates as responses to wind-induced changes in ambient $\text{PM}_{2.5}$ under these assumptions. Additionally, the design does not convert residential ambient concentrations into personal inhaled exposure. Because Fitbit records activity indoors and outdoors, the estimated response incorporates substitution between these settings; a decline in total activity can reflect avoidance, physical impairment, or both.

For comparison, we also estimate the daily-steps models using the regional wind instruments of Deryugina et al. (2019), adapted to our 20 geographic regions. These instruments

interact indicators for the four wind sectors with region indicators, omitting one wind sector within each region. We compare the two IV approaches for continuous $\text{PM}_{2.5}$ and indicators for $\text{PM}_{2.5} > 12$ and $\text{PM}_{2.5} > 25$, holding all else fixed across approaches. We report first-stage F -statistics for each specification.

3.2 Nonlinear Responses

Alongside continuous $\text{PM}_{2.5}$, we estimate specifications replacing P_{zt} with $\mathbf{1}\{P_{zt} > c\}$, for $c \in \{9, 12, 25\} \mu\text{g}/\text{m}^3$. We focus on $12 \mu\text{g}/\text{m}^3$, the level of the EPA’s annual $\text{PM}_{2.5}$ standard adopted in 2012, and examine $9 \mu\text{g}/\text{m}^3$, the revised annual standard adopted in 2024, and $25 \mu\text{g}/\text{m}^3$ as alternatives. These specifications compare higher- and lower-pollution days without imposing a linear relationship between concentration and the outcome. Each indicator is instrumented directly with the same Z_{zt} in equation (2). Because the EPA standards apply to annual average concentrations, exceeding these benchmarks on a given day does not constitute a regulatory violation.

We also estimate separate IV regressions comparing each of five pollution intervals with a common reference interval, following La Nauze and Severnini (2025). Each regression retains days in the reference interval $[0, 5)$ and one comparison interval: $[5, 10)$, $[10, 15)$, $[15, 20)$, $[20, 25)$, or $[25, \infty) \mu\text{g}/\text{m}^3$. We instrument an indicator for the comparison interval with the common selected-direction instrument. The five coefficients come from separate regressions, each with one endogenous indicator and one excluded instrument. Because the samples differ and are restricted using realized pollution, these estimates need not trace a single causal concentration-response function; their interpretation requires the IV assumptions to hold within each restricted sample.

We also examine nonlinear responses using Air Quality Index (AQI) categories. We consider both overall AQI, defined as the maximum index across reported pollutants, and $\text{PM}_{2.5}$ -specific AQI. For each measure, we include indicators for moderate, unhealthy for sensitive groups, unhealthy, very unhealthy, and hazardous air quality, using good air quality as the reference category. These models retain the same fixed effects and controls. We estimate them by OLS and cluster standard errors by reporting area and calendar date. Differences across categories can capture responses to both pollution levels and public air-quality information; they do not separately identify the effect of an air-quality warning.

3.3 Wildfire Smoke

We next distinguish wildfire-attributable $\text{PM}_{2.5}$ from total particulate pollution. This analysis examines responses to a specific pollution source and whether those responses differ with

previous smoke experience. We estimate

$$y_{it} = \beta_S S_{z(i)t} + X_{z(i)t}^{S'} \gamma_S + \alpha_i + \lambda_{z(i),m(t)} + \delta_t + \varepsilon_{it}, \quad (4)$$

where S_{zt} is wildfire-attributable $\text{PM}_{2.5}$ concentration and X_{zt}^S includes temperature, precipitation, and daylight controls. We include participant, ZIP3-by-month-by-year, and calendar-date fixed effects and cluster standard errors by ZIP3 and calendar date as before. We also replace the linear exposure term with indicators for smoke concentrations in $[0, 5]$, $(5, 25]$, $(25, 50]$, $(50, 100]$, $(100, 400]$, and $(400, \infty)$ $\mu\text{g}/\text{m}^3$, using the lowest interval as the reference category.

We estimate these models by OLS. Their causal interpretation requires wildfire smoke to be uncorrelated with other determinants of activity conditional on the controls and fixed effects. We do not use wind direction to instrument wildfire smoke because wind can also change pollution from non-wildfire sources, which may independently affect activity.

Finally, we estimate the binned model separately across quartiles of a ZIP3's previous smoke-episode count, measured through the preceding calendar year. Prior experience may increase awareness of smoke-related risks and familiarity with protective actions, leading to larger activity reductions during subsequent episodes. Alternatively, repeated exposure may reduce the salience of smoke warnings or encourage investments in indoor exercise options, allowing participants to maintain total activity during smoke events. These competing mechanisms motivate testing whether responses differ between places with more and less prior smoke experience.

3.4 Dynamic Responses

Pollution may change activity on the day of exposure and on subsequent days. Activity forgone on a polluted day might be postponed, producing offsetting increases later, or it might remain unrecovered. To distinguish these possibilities, we extend the contemporaneous model to include an exposure history:

$$y_{it} = \beta_0 D_{z(i)t} + \sum_{\ell=1}^L \beta_\ell D_{z(i),t-\ell} + \tilde{X}_{z(i)t}' \gamma + \alpha_i + \lambda_{z(i),m(t)} + \delta_t + \varepsilon_{it}, \quad (5)$$

where D_{zt} is continuous $\text{PM}_{2.5}$ or the indicator for $\text{PM}_{2.5} > 12$. We report the lag profile and cumulative responses, $C_h = \sum_{\ell=0}^h \beta_\ell$. In the continuous specification, C_h measures the cumulative outcome change through day h following a one-day, one- $\mu\text{g}/\text{m}^3$ increase in pollution. In the indicator specification, it sums the model's responses to a one-day change

from below to above the threshold. Positive responses following an initial decline indicate offsetting activity, while a negative cumulative response indicates that the initial loss is not fully recovered within the horizon examined.

For activity and sleep, we first estimate an unrestricted distributed lag model containing current exposure and seven daily lags. We treat all eight exposure terms as endogenous and instrument them with the corresponding current and lagged wind-direction indicators. For each outcome date, we construct all eight instruments using wind directions selected without any years in that date’s fold. Each lag uses the selected direction for its own calendar month, but all eight instruments use directions selected from the same training sample. When selecting wind directions, we also exclude the 90 days before and after each year omitted from the training sample. This buffer separates direction selection from the exposure histories used in estimation and reduces overlap from pollution episodes spanning year boundaries. We use the resulting wind-direction instruments for both continuous and binary exposure specifications and for all dynamic horizons.

Second, we estimate a smooth distributed lag model, adapting the approach of Barwick et al. (2026). We leave the contemporaneous coefficient unrestricted and represent the lag coefficients using a cubic B-spline, $\beta_\ell = \sum_{k=1}^K \theta_k B_k(\ell)$. For the seven-day horizon, we use four spline terms ($K = 4$) for days 1–7. This reduces the number of lag parameters and imposes smoothness on the response profile, helping stabilize estimation when pollution is highly correlated across consecutive days.

The endogenous regressors are current exposure and weighted sums of exposure over the preceding L days, $V_{k,zt} = \sum_{\ell=1}^L B_k(\ell) D_{z,t-\ell}$, where $B_k(\ell)$ are the spline basis weights. We instrument current exposure with the current selected-direction indicator and each weighted exposure sum with the corresponding weighted sum of past selected-direction indicators. The unrestricted and spline seven-day models use the same observations, controls, fixed effects, and clustering. The controls $\tilde{X}_{zt}$ include current temperature, precipitation, daylight, and wind speed, together with controls for histories of temperature, squared temperature, precipitation, squared precipitation, and wind speed. For activity and sleep, we extend the horizon to 14 days as a sensitivity check, using the spline specification.

For wildfire smoke and AQI, we estimate both unrestricted and spline distributed lag models containing current exposure and seven daily lags. We consider linear specifications and specifications with exposure-category indicators, using the same reference categories as in the contemporaneous models. In the binned spline models, each nonreference category has its own smooth lag-response curve, represented by four spline terms for days 1–7 and a separate contemporaneous coefficient. Smoothness is imposed across lag days, not across exposure categories. These models are estimated by OLS using the same controls, fixed

effects, and clustering described above.

3.5 Within-Day Activity

To examine whether participants substitute activity across times of day when pollution is high, we estimate separate models for steps and walking and running minutes during five windows: overnight (00:00–05:59), morning (06:00–09:59), midday (10:00–13:59), afternoon (14:00–17:59), and evening (18:00–23:59).¹⁴ We use the same daily pollution measure in every window, estimating specifications with continuous $\text{PM}_{2.5}$ and an indicator for $\text{PM}_{2.5} > 12$, each instrumented with the cross-fitted selected-direction indicator. Comparing the window-specific responses allows us to assess whether reductions during some parts of a high-pollution day are offset by increases during others, or instead contribute to lower activity over the day as a whole. Because daily pollution includes concentrations occurring after the earlier windows, we interpret the estimates as changes in the within-day pattern of activity on high-versus low-pollution days, rather than as effects of exposure during a particular window.

3.6 Running Performance

Pollution may affect both whether people undertake exercise and how they perform during it. We therefore examine participation in sustained running, running duration, and cadence, identifying sustained running bouts from minute-level step counts following Harris (2026*a*). We distinguish the probability of recording a qualifying run, total qualifying running minutes, and average cadence conditional on running. Our primary definition requires at least 15 consecutive minutes with more than 130 steps per minute. Alternative definitions vary the duration and cadence thresholds or allow brief interruptions of one or two recorded walking or rest minutes while retaining the minimum number of running minutes. We estimate the continuous-PM and $\text{PM}_{2.5} > 12$ specifications using the same cross-fitted instrument, controls, and fixed effects as in the daily activity models.

Cadence measures steps per minute and provides an observable dimension of running performance, but it does not measure speed directly. Running speed is the product of cadence and step length; we do not observe step length or distance and therefore cannot calculate pace. A runner can change speed without a corresponding change in cadence, or change cadence while maintaining speed. Moreover, these are mostly routine runs rather than competitions in which participants can be assumed to exert maximum effort. As discussed by

¹⁴The hourly records provide step counts, from which we construct walking and running minutes. The other activity outcomes are available only as daily totals and therefore cannot be assigned to these time windows.

Harris (2026a), observed performance therefore reflects both physical capacity, which pollution may diminish, and chosen effort, which pollution may also alter. This combination is relevant to understanding pollution’s economic consequences: labor productivity and many other everyday activities likewise depend on both capacity and effort.¹⁵ Our estimates capture changes in realized running performance along both margins, complementing evidence from high-stakes settings in which incentives to maintain effort are stronger (e.g., Mullins, 2018; Lichter, Pestel and Sommer, 2017).

Selection into running poses a separate concern. Cadence is observed only for participants and days with a qualifying run, and pollution may itself affect whether a run occurs or lasts long enough to qualify. If participants who are more affected by pollution disproportionately forgo runs on polluted days, the observed runs may understate performance losses. Changes in the types of runs undertaken can also affect average cadence, so the direction of selection bias is not predetermined. Individual fixed effects account for persistent differences across runners but do not remove selection into running on a particular day, and instrumenting pollution does not by itself resolve this problem. We therefore interpret the cadence estimates alongside the participation and duration estimates as responses among observed qualifying runs, rather than as an isolated effect on physical capacity.

4 Results

4.1 Daily Physical Activity

Air pollution reduces daily physical activity. Table 2 reports the impacts of $\text{PM}_{2.5}$ on daily steps, walking and running minutes, active and sedentary time, calories expended, and sleep duration. We first examine the OLS estimates in Panels A and C. A $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ is associated with 11 fewer daily steps, while the indicator for concentrations above $12 \mu\text{g}/\text{m}^3$ is associated with 3 fewer steps. The estimated changes in walking, active time, and sedentary time are small and statistically insignificant. As expected, these estimates are substantially attenuated relative to the IV estimates, consistent with findings elsewhere in the pollution literature (e.g., Schlenker and Walker, 2016; Deryugina et al., 2019; La Nauze and Severnini, 2025).

Our preferred IV estimates in Panels B and D imply considerably larger responses. A $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ reduces daily steps by 162, or 2.1% of the sample mean. A one-standard-deviation increase in daily $\text{PM}_{2.5}$ ($6.1 \mu\text{g}/\text{m}^3$) corresponds to approximately 99

¹⁵For instance, Gneezy et al. (2019) highlight how differences in effort may confound comparisons of performance across settings in the educational context.

fewer steps. Walking falls by 1.59 minutes and running by 0.11 minutes per day. Active time falls by 2.27 minutes, while sedentary time rises by 2.61 minutes. On days with $\text{PM}_{2.5}$ above $12 \mu\text{g}/\text{m}^3$, participants take 327 fewer steps, spend 3.20 fewer minutes walking and 0.22 fewer minutes running, and record 4.59 fewer active minutes and 5.28 additional sedentary minutes than on days at or below this threshold. The step reduction on such days represents a 4.2% reduction in daily steps.

Most of the decline in active time occurs in light activity. For context, light-intensity activities include slow walking, cooking, and light household chores; moderate-intensity activities, corresponding to Fitbit’s fairly active category, include brisk walking, slow cycling, or easy hiking; and vigorous activities, corresponding to its very active category, include jogging, running, and swimming laps. Appendix Table C.3 shows a reduction of 2.21 lightly active minutes per $10 \mu\text{g}/\text{m}^3$, accounting for nearly all of the reduction in total active time. Fairly and very active time do not change significantly. The IV estimates also show no significant change in daily calories expended, a measure that includes calories burned at rest. These findings suggest that pollution affects everyday movement, including activity too light to qualify as moderate or vigorous exercise. Measures focused on exercise participation or vigorous activity would therefore miss much of the reduction in total active time.

Our estimates are consistent with evidence that pollution discourages outdoor exercise. Fan (2024) estimates that a $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ reduces the likelihood of outdoor exercise among Chinese fitness-app users by 1.43%, compared with our estimated 2.1% reduction in daily steps. Although these outcomes differ, our estimates capture changes in total daily movement, including activity outside exercise sessions and any substitution between indoor and outdoor settings. Extreme temperatures provide another benchmark. Using an earlier All of Us Fitbit sample, Harris (2026a) estimates approximately 500 fewer daily steps on days with maximum temperatures above 100°F and 1,000 fewer on days below freezing, relative to days with maximum temperatures of $75\text{--}80^\circ\text{F}$. Our estimate of 327 fewer steps on days with $\text{PM}_{2.5}$ above $12 \mu\text{g}/\text{m}^3$, compared with days at or below that threshold, is roughly two-thirds of the extreme-heat response and one-third of the cold-day response.

4.2 Sleep

The effect of air pollution on sleep is theoretically ambiguous. Pollution may disrupt sleep by irritating and inflaming the airways, impairing breathing, and causing awakenings (Zanobetti et al., 2010; Billings et al., 2019). It may also interfere with the neural regulation of sleep through inflammatory responses and changes in neurotransmitter activity (Zanobetti et al., 2010). At the same time, avoiding outdoor activities on polluted days may lower the oppor-

Table 2: Pollution, daily activity, and sleep

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$), OLS</i>							
Estimate	-11.183** (5.209)	-0.073 (0.062)	-0.016*** (0.002)	-0.086 (0.102)	0.080 (0.100)	0.492 (0.523)	-0.130*** (0.045)
Adjusted R^2	0.461	0.449	0.349	0.418	0.436	0.587	0.284
<i>Panel B: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$), IV</i>							
Estimate	-161.755*** (48.547)	-1.585*** (0.583)	-0.109*** (0.025)	-2.274** (0.914)	2.614*** (0.921)	0.469 (4.360)	-3.125*** (0.513)
First-stage F	317.1	317.1	317.1	317.1	317.1	317.1	265.5
Adjusted R^2	0.460	0.449	0.349	0.418	0.436	0.587	0.284
<i>Panel C: $PM_{2.5} > 12$, OLS</i>							
Estimate	-3.125 (4.361)	0.013 (0.052)	-0.012*** (0.004)	0.068 (0.084)	0.012 (0.106)	1.152*** (0.390)	-0.149** (0.062)
Adjusted R^2	0.461	0.449	0.349	0.418	0.436	0.587	0.284
<i>Panel D: $PM_{2.5} > 12$, IV</i>							
Estimate	-326.638*** (97.661)	-3.201*** (1.170)	-0.220*** (0.052)	-4.591** (1.839)	5.279*** (1.841)	0.947 (8.809)	-6.431*** (1.110)
First-stage F	247.7	247.7	247.7	247.7	247.7	247.7	219.1
Adjusted R^2	0.460	0.449	0.349	0.418	0.436	0.587	0.284
Observations	51,033,286	51,033,286	51,033,286	51,033,285	51,033,285	51,032,966	36,952,488
Mean outcome	7,838.291	85.645	1.639	259.022	817.699	2,320.103	389.539

Notes: The dependent variable is given in each column heading. Activity-duration and sleep outcomes are in minutes. Panels A and B report OLS and IV estimates for continuous $PM_{2.5}$; Panels C and D report OLS and IV estimates for $PM_{2.5} > 12$. Continuous-PM estimates are scaled to a 10 $\mu g/m^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu g/m^3$. IV estimates use the cross-fitted selected-direction instrument. OLS and IV regressions use the same estimation sample within each outcome. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

tunity cost of sleep, leading individuals to spend more time in bed. Whether sleep increases or decreases therefore depends on both physiological effects and changes in how individuals allocate their time. Our estimates indicate that, on balance, air pollution shortens sleep.

The IV estimates in Table 2 imply that a $10 \mu\text{g}/\text{m}^3$ increase in daytime $\text{PM}_{2.5}$ reduces sleep that night by 3.13 minutes. A one-standard-deviation increase in $\text{PM}_{2.5}$ corresponds to 1.91 fewer minutes. On days with $\text{PM}_{2.5}$ above $12 \mu\text{g}/\text{m}^3$, participants sleep 6.43 fewer minutes than on days at or below this threshold.

Table 3 examines how this reduction is distributed across sleep stages. Light sleep encompasses the transition into sleep and periods of slower breathing and heart rate; deep sleep supports physical restoration and memory consolidation; and REM sleep contributes to learning, memory, and emotional regulation. These functions overlap, so reductions in light or REM sleep can matter even when deep-sleep duration is unchanged.¹⁶ In the sample with usable stage measurements, a $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ reduces total sleep by 2.30 minutes, comprising 1.29 fewer minutes of light sleep and 1.03 fewer minutes of REM sleep. Deep-sleep duration does not change significantly. The difference between the total-duration estimates here and in Table 2 reflects the different samples.

Pollution also reduces the time participants spend in bed. Fitbit’s time-in-bed measure includes both estimated sleep and awake time within the recorded sleep episode. In Table 3, a $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ reduces time in bed by 2.11 minutes, compared with a 2.30-minute reduction in sleep. The corresponding estimates from the $\text{PM}_{2.5} > 12$ specification are 4.40 and 4.79 minutes. Time asleep therefore falls slightly more than time in bed, a pattern consistent with both a shorter sleep period and more time awake within that period. Although these differences are not statistically significant, the point estimates suggest that changes in sleep schedules may not fully account for the reduction in sleep. This pattern is consistent with physiological effects of pollution on sleep, though it is not evidence of it.

4.3 Timing of Activity Responses

Participants may shift activity within a polluted day or postpone it to subsequent days. We first estimate separate regressions for steps and walking and running minutes during five periods—overnight, morning, midday, afternoon, and evening—using daily pollution. We then examine whether participants make up forgone activity over subsequent days.

Figure 1 shows that participants reduce their steps primarily in the afternoon and evening. On days with $\text{PM}_{2.5}$ above $12 \mu\text{g}/\text{m}^3$, participants take 163 fewer steps between 2:00 and 5:59 p.m. and 135 fewer between 6:00 and 11:59 p.m. than on days at or below this threshold.

¹⁶See <https://www.nichd.nih.gov/health/topics/sleep/conditioninfo/what-happens> for an overview of sleep stages and their functions.

Table 3: Pollution and sleep stages

	Total minutes	Light minutes	Deep minutes	REM minutes	Time in bed
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$)</i>					
Estimate	-2.298*** (0.493)	-1.289*** (0.361)	0.026 (0.116)	-1.035*** (0.175)	-2.111*** (0.552)
First-stage F	211.1	211.1	211.1	211.1	211.1
Adjusted R^2	0.269	0.300	0.302	0.286	0.266
<i>Panel B: $PM_{2.5} > 12$</i>					
Estimate	-4.786*** (1.064)	-2.685*** (0.774)	0.054 (0.242)	-2.155*** (0.380)	-4.396*** (1.181)
First-stage F	178.8	178.8	178.8	178.8	178.8
Adjusted R^2	0.269	0.300	0.302	0.286	0.266
Observations	28,833,638	28,833,638	28,833,638	28,833,638	28,833,630
Mean outcome	400.290	259.910	59.710	80.670	457.676

Notes: The dependent variable is given in each column heading. Total, light, deep, and REM sleep durations and time in bed are measured in minutes. All models require usable measurements for all three stages; the time-in-bed model additionally requires a valid duration between sleep time and 1,440 minutes. Panels distinguish continuous PM and $PM_{2.5} > 12$. Estimates measure changes in sleep that night. Continuous-PM estimates are scaled to a 10 $\mu g/m^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu g/m^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

These reductions represent 7.8% and 8.2% of average steps in the respective windows. Participants also spend 1.61 and 1.45 fewer minutes walking during these periods, equivalent to 6.8% and 8.3% of the corresponding sample means. Participants reduce running at midday as well as in the afternoon and evening. The continuous-PM estimates show the same broad pattern (Appendix Tables D.4–D.8). No period exhibits a statistically significant increase in steps, walking, or running. Thus, we find no evidence that participants offset the afternoon and evening reductions by increasing activity elsewhere within the same day.

The response across days is different. The left panel of Figure 2 shows an initial decline in steps followed by a next-day increase after a one-day rise in $\text{PM}_{2.5}$. For a $10 \mu\text{g}/\text{m}^3$ increase, the unrestricted distributed lag model implies 234 fewer steps on the exposure day and 437 additional steps the following day. The seven-day spline model yields 182 fewer steps initially and 283 additional steps the next day. The right panel shows the cumulative responses, which reach 41 and 13 steps, respectively, through day seven; neither is statistically distinguishable from zero. Walking follows a similar pattern: the spline estimates imply 1.75 fewer walking minutes initially and 3.06 additional minutes the next day (Appendix Figure E.4). These patterns are consistent with intertemporal substitution and resemble the rebound following cold days documented by Harris (2026a).

The left panel of Figure 3 shows that this pattern also holds for the $\text{PM}_{2.5} > 12$ indicator. The unrestricted estimates imply an initial reduction of 548 steps and a next-day increase of 943; the seven-day spline estimates imply a reduction of 405 followed by an increase of 557. The right panel shows cumulative responses of 118 and 18 steps, respectively, through day seven, neither statistically distinguishable from zero. Extending the spline horizon to fourteen days produces cumulative step estimates of 6 for continuous PM and -88 for the $\text{PM}_{2.5} > 12$ indicator, also statistically insignificant (Appendix Tables E.9 and E.10). These estimates are consistent with participants making up for activity lost on polluted days, although the cumulative estimates are not precise enough to determine whether this compensation is complete.

Sleep also exhibits some recovery after the initial decline. Following a one-day increase of $10 \mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$, the seven-day spline model implies 3.43 fewer minutes of sleep that night but 1.70 additional minutes the following night. For the $\text{PM}_{2.5} > 12$ indicator, the corresponding estimates are an initial loss of 7.20 minutes and a next-night increase of 3.55 minutes. In both cases, the cumulative response through day seven is statistically indistinguishable from zero (Appendix Figure E.10 and Table E.10). Extending the $\text{PM}_{2.5} > 12$ model to fourteen days yields a similar pattern: 6.92 fewer minutes initially, 3.24 additional minutes the next night, and no statistically significant cumulative response through day fourteen (Appendix Table E.10). Together with the activity results, these estimates suggest

that responses on the day of exposure can overstate the lasting reductions in activity and sleep. Such adjustments may nevertheless be costly if they shift activity or sleep to less preferred times.

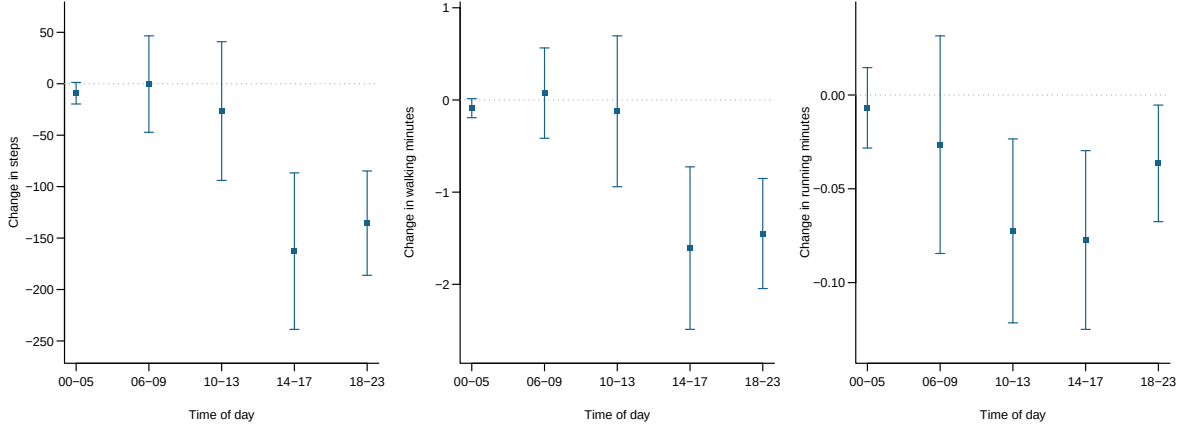


Figure 1: Within-day activity responses to daily pollution. Panels show steps, walking minutes, and running minutes, respectively. The exposure is the daily $\text{PM}_{2.5} > 12$ indicator, instrumented with the cross-fitted selected-direction indicator. Labels denote clock-hour windows, inclusive through minute 59 of the last hour. All models retain the daily controls and fixed effects and add daylight minutes within the window. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

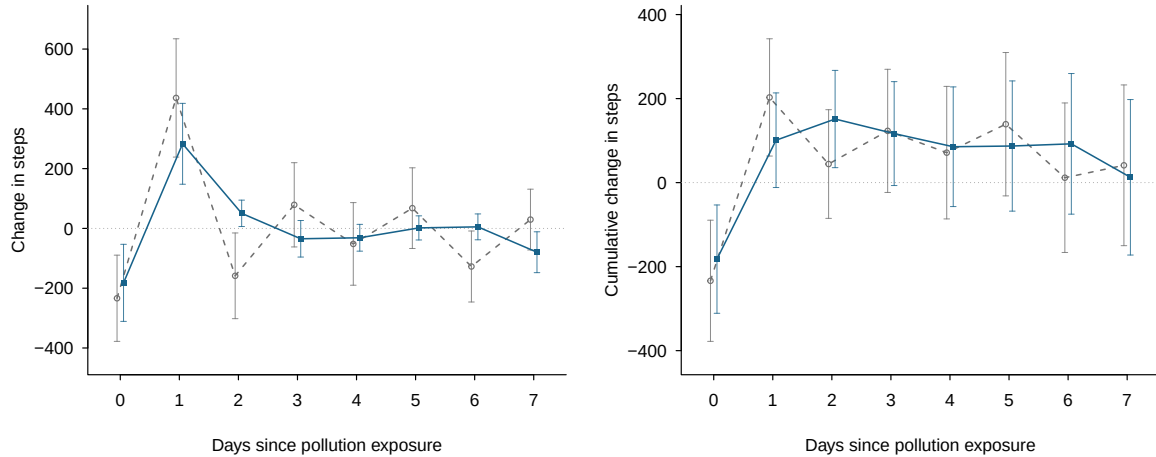


Figure 2: Dynamic step responses: continuous $\text{PM}_{2.5}$. Exposure is a one-day increase of $10 \mu\text{g}/\text{m}^3$. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

4.4 Sustained Running Performance

Pollution may affect both the decision to exercise and performance during exercise. We identify sustained running bouts as periods of at least 15 consecutive minutes with more than

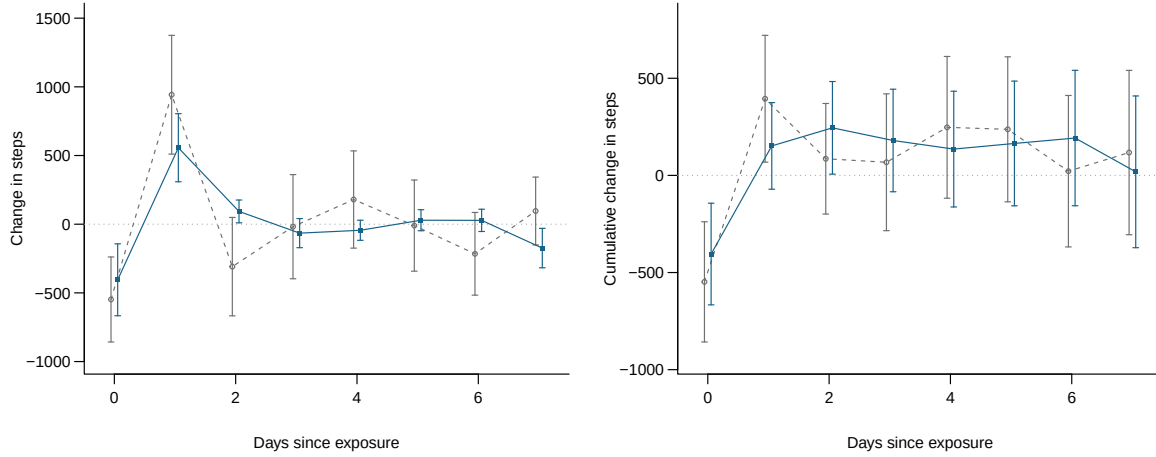


Figure 3: Dynamic step responses: $\text{PM}_{2.5} > 12$. Exposure is a one-day change in the $\text{PM}_{2.5} > 12$ indicator. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

130 steps per minute. Table 4 reports pollution’s effects on the probability of completing such a bout, total time spent in these bouts, and running cadence. On days with $\text{PM}_{2.5}$ above $12 \mu\text{g}/\text{m}^3$, participants are 0.17 percentage points less likely to complete a sustained running bout and spend 0.10 fewer minutes in sustained running than on days at or below this threshold. Among days with at least one such bout, running time falls by 2.33 minutes, or 6.4% of its sample mean. The continuous-PM estimate likewise indicates shorter running duration, with a reduction of 1.17 minutes per $10 \mu\text{g}/\text{m}^3$.

Participants also run at a slower cadence, taking 1.34 fewer steps per minute for each $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$, or 0.82 fewer for a one-standard-deviation increase. On days with concentrations above $12 \mu\text{g}/\text{m}^3$, cadence is 2.67 steps per minute lower than on days at or below this threshold. Together with the shorter running duration, these findings suggest that pollution affects how participants exercise as well as whether they exercise. The estimates may reflect changes in physical capacity, chosen effort, or the composition of observed runs, since duration and cadence are measured only on days when participants complete a sustained running bout.

The estimated reduction in running cadence is broadly robust to alternative definitions of sustained running, although its magnitude and statistical significance vary (Appendix Figure F.13). Estimates remain negative and significant when the minimum uninterrupted running duration is 10, 20, or 30 minutes, but are less precise when we allow one-minute interruptions or lower the cadence threshold to 120 steps per minute.

Hot weather provides a benchmark for the magnitude of the cadence response. Harris

(2026a) finds that participants take 0.42 fewer steps per minute on days with maximum temperatures of 85–90°F than on days reaching 75–80°F. A one-standard-deviation increase in PM_{2.5} reduces cadence by 0.82 steps per minute in our sample, roughly twice that amount. Although the studies differ in their samples and identification strategies, this comparison suggests that pollution can affect running performance by an economically meaningful amount.¹⁷

Table 4: Pollution and sustained running

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>				
Estimate	-0.086** (0.042)	-0.052*** (0.017)	-1.167*** (0.407)	-1.339*** (0.477)
First-stage F	317.1	317.1	409.9	409.9
Adjusted R^2	0.249	0.300	0.495	0.332
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.173** (0.085)	-0.104*** (0.035)	-2.331*** (0.822)	-2.673*** (0.962)
First-stage F	247.7	247.7	273.8	273.8
Adjusted R^2	0.249	0.300	0.494	0.332
Observations	51,033,286	51,033,286	1,033,181	1,033,181
Mean outcome	2.043	0.744	36.499	159.428

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. The primary definition requires at least 15 consecutive minutes above 130 steps per minute. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.5 Alternative Specifications and Pollutant Controls

We next examine whether the activity findings are robust to alternative pollution measures, instrument designs, and controls for other pollutants. First, we assess sensitivity to the pollution threshold by replacing the indicator for PM_{2.5} > 12 with indicators for concentrations exceeding 9 or 25 $\mu\text{g}/\text{m}^3$, each instrumented with the same wind-direction instrument (Appendix Table G.22). The first specification implies that participants take 190 fewer steps

¹⁷The temperature regressions also control for running duration, whereas our cadence regressions allow duration to respond to pollution.

on days above $9 \mu\text{g}/\text{m}^3$ than on days at or below that threshold. Using $25 \mu\text{g}/\text{m}^3$ produces a much larger estimate of 6,004 fewer steps, but the first-stage results raise concerns about weak instruments. Its first-stage F -statistic is 8.0 for steps and 5.7 for sleep, both below the conventional rule-of-thumb threshold of 10 (Staiger and Stock, 1997). By comparison, the continuous-PM and $\text{PM}_{2.5} > 12$ steps models have first-stage F -statistics of 317.1 and 247.7, respectively. These low first-stage F -statistics for the $25 \mu\text{g}/\text{m}^3$ indicator warrant caution in interpreting the large estimates for the $\text{PM}_{2.5} > 25$ specification.

To further examine how activity responses vary across pollution levels, we separately compare days in five $\text{PM}_{2.5}$ concentration ranges with days below $5 \mu\text{g}/\text{m}^3$, as shown in Figure 4.¹⁸ Participants take an estimated 141 fewer steps at concentrations of 5–10 $\mu\text{g}/\text{m}^3$ and 95 fewer at 10–15 $\mu\text{g}/\text{m}^3$; both estimates are statistically significant at the 5% level. Estimates for the three higher ranges are also negative, at approximately 140, 290, and 644 fewer steps, but are not statistically distinguishable from zero. These results provide evidence of reduced activity even at relatively low pollution concentrations and suggestive, though not dispositive, evidence of larger reductions at higher pollution concentrations.

An alternative wind-direction instrument provides a further check on the activity findings. Table 5 reports estimates using our selected-direction instrument alongside estimates using wind-direction indicators interacted with geographic regions, following the approach of Deryugina et al. (2019). On the same estimation samples, the regional-interaction instruments yield reductions of 134 steps per 10 $\mu\text{g}/\text{m}^3$ and 266 steps on days with $\text{PM}_{2.5}$ above 12 $\mu\text{g}/\text{m}^3$ relative to days at or below that threshold, compared with 162 and 327 steps under our primary specification. Thus, the finding that pollution reduces daily steps holds under both instrument designs, with somewhat smaller magnitudes under the regional-interaction approach. At the $25 \mu\text{g}/\text{m}^3$ threshold, however, the regional-interaction estimate becomes positive and its first-stage F -statistic falls to 4.0, providing further reason for caution in interpreting estimates based on this higher cutoff.

Finally, we assess whether accounting for other pollutants changes the estimated effect of $\text{PM}_{2.5}$ on daily steps. Appendix Table H.23 reports specifications that control separately for ozone, carbon monoxide, sulfur dioxide, and nitrogen dioxide, as well as all four together. Adding ozone increases the estimated reduction from 162 to 245 steps per 10 $\mu\text{g}/\text{m}^3$, while the other individual controls produce smaller changes relative to estimates without those controls on the same samples. Including all four pollutants yields reductions of 241 steps per 10 $\mu\text{g}/\text{m}^3$ and 473 steps on days with $\text{PM}_{2.5}$ above 12 $\mu\text{g}/\text{m}^3$ relative to days at or below that

¹⁸We estimate these comparisons separately because our selected-direction instrument provides one excluded instrument. A model containing all five pollution-range indicators would require additional instruments or restrictions to identify their separate effects. Each comparison uses a different restricted sample, so the estimates should not be interpreted as a single dose-response function.

threshold. The corresponding estimates without these controls are 144 and 285 steps on the same sample. All estimates controlling for other pollutants remain negative and significant at the 1% level, indicating that the activity finding is robust to their inclusion.

Table 5: Daily steps: comparison of wind instruments

	Selected direction	Regional interactions
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$)</i>		
Daily steps	-161.755*** (48.547)	-134.007*** (32.729)
First-stage F	317.1	24.0
Adjusted R^2	0.460	0.460
<i>Panel B: $PM_{2.5} > 12$</i>		
Daily steps	-326.638*** (97.661)	-266.177*** (72.691)
First-stage F	247.7	12.1
Adjusted R^2	0.460	0.460
<i>Panel C: $PM_{2.5} > 25$</i>		
Daily steps	-6,004.456** (2,603.682)	93.404 (596.375)
First-stage F	8.0	4.0
Adjusted R^2	0.448	0.460
Observations	51,033,286	51,033,286
Mean outcome	7,838.291	7,838.291

Notes: The dependent variable is daily steps, and the two instrument designs use the same estimation sample within each panel. Panels distinguish exposure specifications. Column 1 reports our primary specification, which uses the cross-fitted selected-direction instrument following La Nauze and Severnini (2025). Column 2 uses 60 regional wind-direction interactions following Deryugina et al. (2019). Continuous-PM estimates are scaled to a 10 $\mu g/m^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu g/m^3$. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.6 Wildfire Smoke and Air Quality Categories

Wildfire smoke and public information about air quality may prompt activity adjustments that differ from responses to pollution more generally. We examine these responses using wildfire $PM_{2.5}$ concentrations and AQI categories. Both sets of models are estimated by OLS, so we interpret them as complementary evidence to the main IV results. Table 6 shows that participants take fewer steps during more intense smoke episodes. Relative to wildfire $PM_{2.5}$ concentrations of 0–5 $\mu g/m^3$, participants take 111 fewer steps at concentrations above 25 and up to 50, 451 fewer above 50 and up to 100, and 640 fewer above 100 and up to 400.

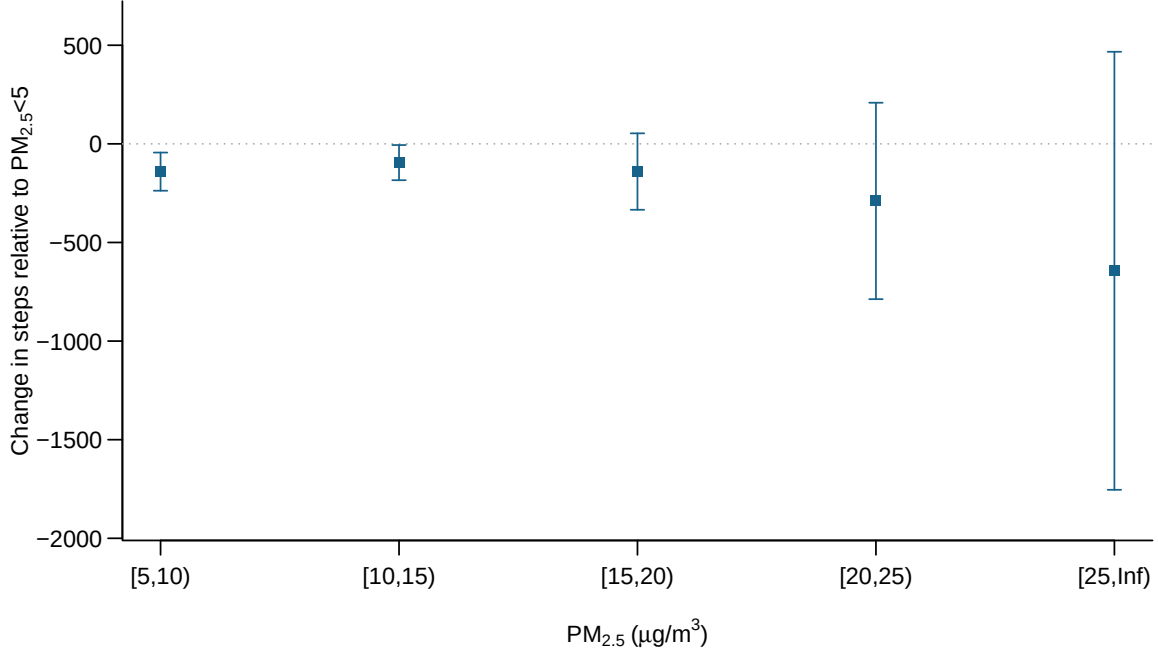


Figure 4: Daily steps across separate pollution intervals. Each estimate compares the indicated PM_{2.5} interval with $[0, 5)$ $\mu\text{g}/\text{m}^3$ in a separate restricted sample, instrumenting the comparison indicator with the cross-fitted selected-direction instrument. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

Participants also spend less time walking and running in these three higher concentration ranges.

Prior experience with wildfire smoke could influence these responses if participants learn how to adjust their activities or become accustomed to smoke exposure. Appendix Figures I.14 and I.15 compare estimates across quartiles of prior smoke experience. Participants in all four groups take fewer steps in the highest smoke category, but the estimated reductions do not vary monotonically with experience. These comparisons therefore provide no clear pattern of progressively stronger or weaker responses among participants living in areas with more prior smoke exposure.

Activity reductions associated with smoke also extend beyond the exposure day. Figure 5 shows the daily responses in the left panel and cumulative responses in the right panel following a one-day increase of $10 \mu\text{g}/\text{m}^3$ in wildfire PM_{2.5}. Both the unrestricted and seven-day spline models imply approximately 74 fewer steps cumulatively through day seven. Extending the spline horizon to fourteen days yields a similar cumulative reduction of 74 steps; all three estimates are significant at the 1% level (Appendix Table I.24). When we estimate separate responses by smoke-concentration category, the highest category shows a larger cumulative reduction in activity. For concentrations above 100 and up to $400 \mu\text{g}/\text{m}^3$,

relative to 0–5, the seven-day spline model implies 482 fewer steps on the exposure day and 1,078 fewer cumulatively through day seven. The fourteen-day cumulative estimate is 1,039 fewer steps (Appendix Table I.25). Thus, the smoke estimates suggest that subsequent activity does not fully compensate for the initial losses over these horizons. This contrasts with the rebound in the main PM IV results and may suggest that activity adjustments differ during wildfire smoke episodes.

Turning to public air-quality information, Figure 6 shows activity responses across overall and PM-specific AQI categories. Participants take 104 fewer steps on days classified as unhealthy, 380 fewer on very unhealthy days, and 453 fewer on hazardous days according to overall AQI, relative to days with good air quality. All three estimates are significant at the 5% level. However, the moderate and unhealthy-for-sensitive-groups categories have positive estimates, so activity does not decline consistently across the full index. Using PM-specific AQI yields larger reductions in the three highest categories—230, 488, and 891 steps, respectively.¹⁹ Appendix J reports the corresponding estimates for other daily outcomes. Because these category comparisons may capture responses to both pollution and information about air quality, they do not separately identify the effect of an air-quality warning.

Table 6: Daily outcomes by smoke category

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
(5, 25]	12.030 (8.506)	0.284*** (0.103)	-0.022*** (0.005)	0.730*** (0.175)	-0.268 (0.207)	5.191*** (0.828)	-0.391*** (0.106)
(25, 50]	-111.414*** (29.851)	-0.986*** (0.340)	-0.107*** (0.017)	-1.086* (0.588)	0.763 (0.808)	-4.973* (2.862)	0.206 (0.339)
(50, 100]	-451.474*** (50.793)	-4.820*** (0.577)	-0.231*** (0.030)	-6.676*** (0.809)	3.990*** (1.419)	-34.430*** (6.193)	-0.720 (0.602)
(100, 400]	-640.196*** (103.056)	-6.915*** (1.074)	-0.230*** (0.064)	-10.572*** (1.554)	9.291*** (2.292)	-40.800*** (11.172)	0.370 (0.610)
Observations	44,236,029	44,236,029	44,236,029	44,236,029	44,236,029	44,235,766	31,176,930
Mean outcome	7,966.809	87.045	1.699	262.205	816.701	2,330.242	390.686
Adjusted R^2	0.455	0.444	0.347	0.416	0.445	0.587	0.283

Notes: The dependent variable is given in each column heading. Entries are OLS coefficients for wildfire PM_{2.5} categories relative to [0, 5] $\mu\text{g}/\text{m}^3$. Category coefficients within each column are estimated jointly. Observations and mean outcomes refer to each model's final estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

¹⁹These estimates use fewer participant-days because some reporting areas provide AQI values for other pollutants without reporting PM_{2.5} AQI, so differences in the estimation samples may also contribute to the larger estimates.

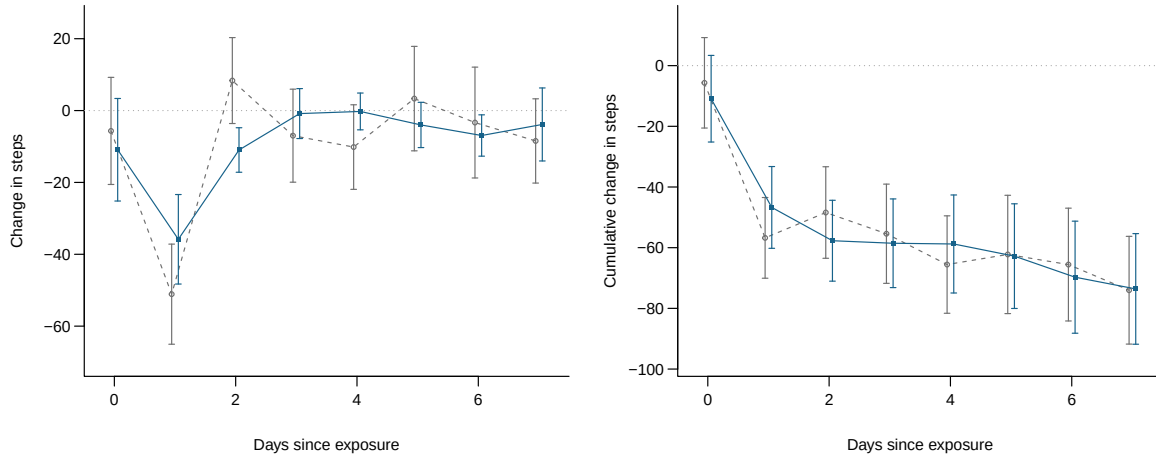


Figure 5: Dynamic step responses to wildfire smoke. The exposure is a one-day increase of $10 \mu\text{g}/\text{m}^3$ in wildfire $\text{PM}_{2.5}$. Gray dashed lines with open circles show unrestricted seven-day estimates; blue solid lines with squares show seven-day spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Both models are estimated by OLS with temperature and precipitation categories, daylight duration, participant, ZIP3-by-month-by-year, and calendar-date fixed effects. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

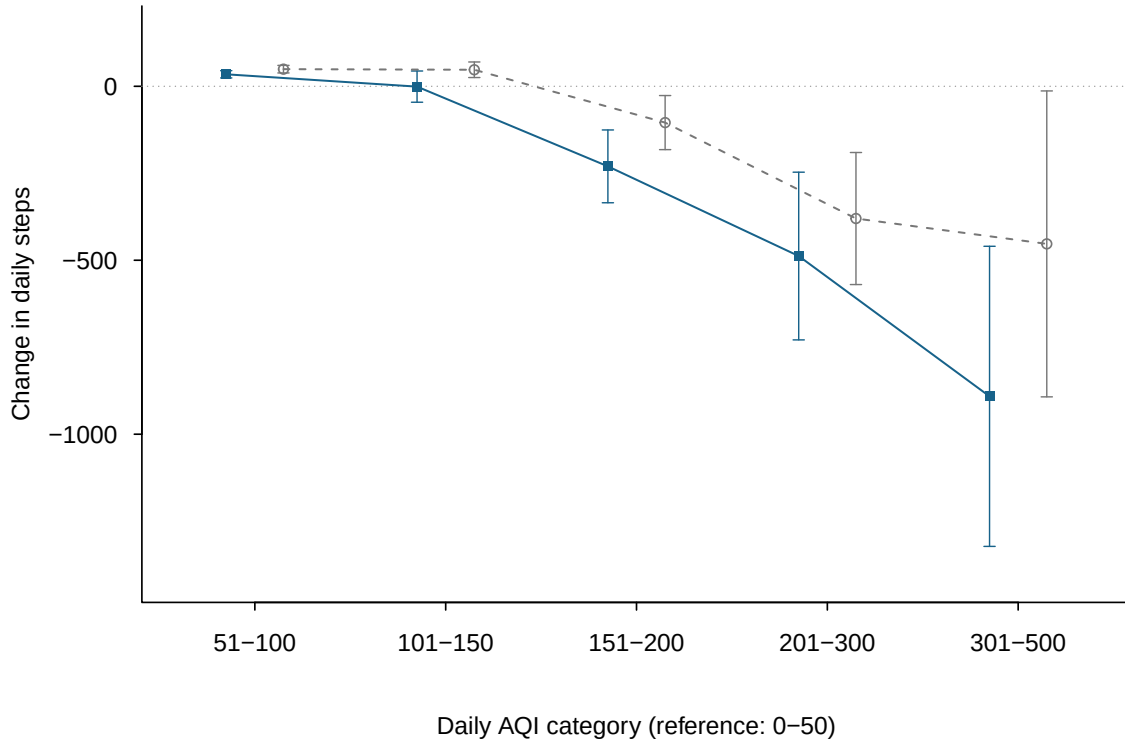


Figure 6: Daily steps across AQI categories. Blue squares show PM-specific AQI estimates; gray open circles show overall AQI estimates. Each series is estimated by OLS relative to AQI 0–50 with its respective sample, controls, and fixed effects. Bars show 95% confidence intervals based on standard errors clustered by reporting area and calendar date.

4.7 Heterogeneity and External Validity

Pollution responses may vary with participants’ characteristics, raising the question of how the composition of the Fitbit sample shapes our findings. Appendix Figures K.22 and K.23 report estimated differences in responses across demographic groups and between participants with and without a recorded asthma or chronic obstructive pulmonary disease diagnosis or a qualifying respiratory medication record. The latter comparison is restricted to participants with electronic health records. Older participants reduce their activity more in response to pollution. For a $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$, the reductions among adults aged 60 and older exceed those among adults aged 18–39 by 119 daily steps and 2.65 active minutes. In contrast, step reductions are smaller among Black, Asian, and Hispanic participants than among White participants, by 159, 243, and 205 steps, respectively. Asian and Hispanic participants also show smaller reductions in walking, while Black participants show smaller reductions in running. These differences are significant at the 5% level before adjustment for multiple comparisons. We find no statistically significant differences by gender, income, or evidence of respiratory conditions for any of the five outcomes. The estimates identify differences in responses but do not distinguish whether they arise from differences in exposure, opportunities to adjust activity, or other factors.

To assess how sample composition affects the main steps estimate, we reweight participants to match gender-by-age, race and ethnicity, and household-income distributions among U.S. adults living in households in the 2015–2019 American Community Survey, following Harris (2026a). Appendix Table L.32 reports weighted and unweighted estimates for participants with the information needed to construct these weights. Reweighting modestly reduces the estimated magnitudes: the reduction in steps per $10 \mu\text{g}/\text{m}^3$ changes from 166 to 148, while the estimated reduction for days with $\text{PM}_{2.5}$ above $12 \mu\text{g}/\text{m}^3$, relative to days at or below that threshold, changes from 336 to 295. Both weighted estimates remain significant at the 1% level and are similar to the full-sample estimates of 162 and 327 fewer steps.

Device ownership provides another potential source of selection. By supplying devices, the WEAR study broadens participation beyond individuals who contribute data from their own Fitbits. Appendix Tables A.1 and A.2 describe the WEAR and BYOD groups, and Appendix Table L.33 reports their pollution responses over overlapping calendar dates.²⁰ A $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$ reduces estimated daily steps by 149 among WEAR participants and 192 among BYOD participants. The corresponding estimates for the $\text{PM}_{2.5} > 12$ specification are 308 and 414 fewer steps. Although the WEAR point estimates are smaller, the differences between the WEAR and BYOD responses are not statistically significant

²⁰WEAR began in 2021. We restrict both groups to dates with WEAR observations through 2024 so that their estimated responses cover the same calendar dates.

under either specification. The reweighting exercise suggests that the sample’s observed demographic composition does not explain the main activity finding, while the WEAR comparison shows that activity reductions also occur among participants provided with a device. Neither exercise rules out selection on unobserved characteristics that could affect how pollution responses differ in the broader U.S. population.

5 Discussion

The economic consequences of pollution depend partly on people’s ability to adjust. Moving exercise indoors or postponing it may limit exposure while preserving activity. Participants reduce total movement on polluted days but subsequently increase steps and sleep, suggesting some recovery. Valuing every daily reduction as permanently forgone would overlook this recovery; treating recovery as costless would overlook the inconvenience or forgone enjoyment of changing schedules.

We estimate these responses in the United States, a comparatively low-pollution setting where air quality has improved substantially over several decades.²¹ Wildfire smoke, however, increasingly threatens this progress. Burke et al. (2023) find that wildfire smoke has slowed or reversed improvements in air quality in 35 states, undoing approximately one-quarter of earlier reductions in $\text{PM}_{2.5}$ concentrations on average across those states. Smoke can also affect communities far from the fires themselves. In September 2020, smoke from western U.S. wildfires traveled across the country and reached Europe; in July 2021, smoke from fires in western North America contributed to severe air pollution in New York City. Canadian wildfires again blanketed much of the northeastern United States in June 2023.²² These episodes demonstrate that communities can face substantial pollution originating far beyond their borders, despite long-run improvements in local air quality.

To illustrate the economic magnitude of these disruptions, we perform back-of-the-envelope calculations using the June 2023 Canadian wildfire smoke episode, when smoke severely degraded air quality in major population centers including New York City and Philadelphia.²³ Figure 7 shows the evolution of wildfire-attributable $\text{PM}_{2.5}$ across New York, New Jersey,

²¹See <https://www.epa.gov/air-trends/air-quality-national-summary>.

²²For descriptions and satellite observations of these episodes, see NASA’s accounts of September 2020 (<https://science.nasa.gov/mission/aura/air-quality-measurements-from-omi-during-the-2020-u-s-wildfire-season/>), July 2021 (<https://gmao.gsfc.nasa.gov/science-snapshots/an-early-start-to-the-north-american-biomass-burning-season-in-2021/>), and June 2023 (<https://science.nasa.gov/earth/earth-observatory/smoke-smothers-the-northeast-151433/>).

²³On June 7, 2023, smoke turned New York City’s skies orange, producing striking images that circulated widely in news coverage and on social media. The *New York Times* covered the event under the headline “Orange Skies and Burning Eyes as Smoke Shrouds New York City” (<https://www.nytimes.com/2023/06/07/nyregion/nyc-wildfire-smoke-scenes.html>).

and Pennsylvania during the episode. Population-weighted daily concentrations rose from $54 \mu\text{g}/\text{m}^3$ on June 6 to $125 \mu\text{g}/\text{m}^3$ on June 7, before falling to $74 \mu\text{g}/\text{m}^3$ on June 8. At the June 7 peak, ZIP3 codes containing approximately 79% of the region’s population experienced wildfire-attributable concentrations above $100 \mu\text{g}/\text{m}^3$. We combine these ZIP3-level concentrations during June 6–8 with our fourteen-day distributed lag estimates for wildfire smoke to predict how walking and running changed during and after these three days. The counterfactual replaces smoke concentrations on June 6–8 with the $0\text{--}5 \mu\text{g}/\text{m}^3$ reference category, leaving exposure on other dates unchanged. Averaging the projections across the three states using population weights yields cumulative losses of approximately 23 walking minutes and 0.69 running minutes per person through June 22, accounting for subsequent recovery. Following the recreational valuation approach in Harris (2026a), we apply values of \$17.30 per walking hour and \$80.48 per running hour from the Recreation Use Values Database (Rosenberger, 2016), expressed in 2023 dollars. The resulting value of forgone activity is approximately \$7.56 per person.

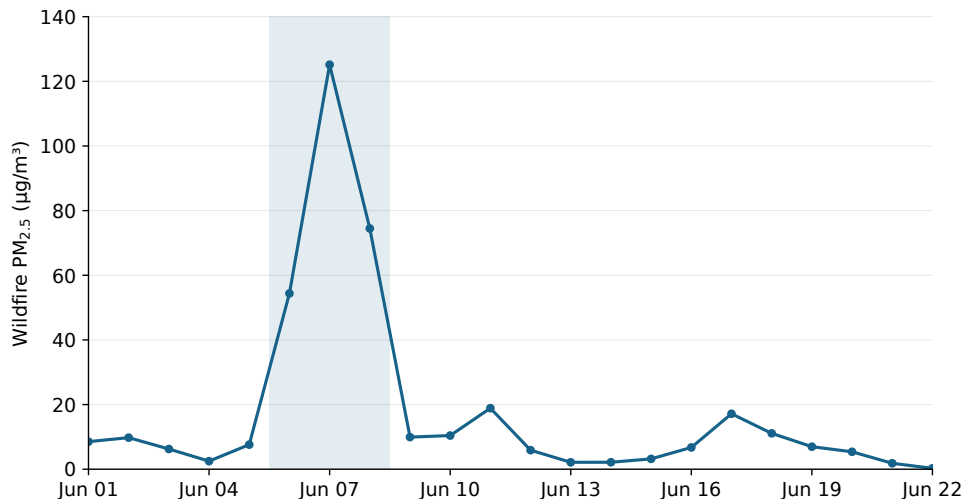


Figure 7: Wildfire smoke exposure in New York, New Jersey, and Pennsylvania, June 2023. The figure shows population-weighted daily wildfire-attributable $\text{PM}_{2.5}$ concentrations across ZIP3 areas in the three states. Weights are ZIP3 population totals constructed from 2010 Census ZCTA populations. Shading marks June 6–8, the exposure dates used in the back-of-the-envelope calculation.

Multiplying these projections by the population in the three-state region yields approximately \$309 million in forgone recreational value.²⁴ This calculation extrapolates the adult sample’s estimated responses to residents of all ages and assumes that recreational walking and running values apply to the movement captured by the devices, including activity undertaken indoors or for purposes other than recreation. The projections also rely on our wildfire

²⁴We use 2010 Census population data, totaling 40.87 million residents across the ZIP3 areas in New York, New Jersey, and Pennsylvania.

estimates from section 4.6 rather than a separately identified effect of this episode. They therefore provide an illustrative valuation of net forgone activity, rather than a comprehensive measure of welfare loss. Furthermore, they do not separately account for the value of alternative uses of time, the costs of rescheduling activity, or the health benefits of avoiding exposure. Nonetheless, these estimates suggest that forgone recreational activity can represent a meaningful cost alongside the health damages associated with wildfire smoke: for comparison, the Canadian Climate Institute estimates about \$1 billion (USD) in mortality and morbidity damages in Ontario during June 4–8 of the same episode.²⁵

An important question for future research is how much people protect their health by adjusting their behavior on polluted days. Reducing outdoor exertion may lower inhaled pollution, but it can also displace activity that contributes to health. Although our data link wearable measurements to electronic health records, identifying the health benefits of avoidance requires more than observing behavior and health together. Pollution can affect health directly while also inducing changes in activity, and people may adjust their behavior in response to symptoms or anticipated health risks. Using wind direction to instrument for pollution in a health-outcome regression would not resolve this problem: wind-induced pollution can affect health both directly and through changes in behavior. Estimating the protection provided by avoidance would require an additional source of variation that changes protective behavior without independently affecting health, or a model with explicit assumptions that allow these effects to be distinguished. Information on indoor and outdoor activity and personal pollution exposure would further help establish how behavioral adjustments change exposure. Measuring the resulting health benefits alongside the costs of forgone activity would help assess the tradeoffs people face when responding to polluted air.

6 Conclusion

Air pollution causes people to change their everyday behaviors, including physical activity and sleep. Using a large panel of Fitbit records linked to daily pollution exposure and an instrument based on wind direction, we find that participants respond to higher PM_{2.5} by taking fewer steps, spending less time walking and running, and spending more time sedentary. Most of the reduction in active time occurs in light activity, indicating that these responses extend beyond vigorous exercise to routine movement. When participants run, they do so for less time and at a lower cadence. Participants also sleep less, particularly in light and REM sleep. Because the devices record movement across indoor and outdoor settings, these findings capture changes in total activity after any substitution between them.

²⁵See <https://climateinstitute.ca/with-the-forest-ablaze-the-health-costs-hit-home/>.

Following responses over time changes the interpretation of these daily losses. Participants increase their steps and sleep after the initial reductions, suggesting that some activity and rest are postponed rather than permanently forgone. Although the estimates do not establish whether participants fully make up these reductions, they caution against treating a single day's response as a lasting loss. Complementary analyses of wildfire smoke show more persistent activity reductions. Recovery may also entail costs if participants exercise or sleep at less preferred times. Accounting for both forgone activity and the costs of rescheduling it would provide a fuller measure of the benefits of cleaner air.

References

- Adams, Swann Arp, Charles E. Matthews, Cara B. Ebbeling, Charity G. Moore, Joan E. Cunningham, Jeanette Fulton, and James R. Hebert. 2005. “The Effect of Social Desirability and Social Approval on Self-Reports of Physical Activity.” *American Journal of Epidemiology*, 161(4): 389–398.
- Aguilar-Gomez, Sandra, Holt Dwyer, Joshua Graff Zivin, and Matthew Neidell. 2022. “This Is Air: The “Nonhealth” Effects of Air Pollution.” *Annual Review of Resource Economics*, 14(1): 403–425.
- Archsmith, James, Anthony Heyes, and Soodeh Saberian. 2018. “Air Quality and Error Quantity: Pollution and Performance in a High-Skilled, Quality-Focused Occupation.” *Journal of the Association of Environmental and Resource Economists*, 5(4): 827–863.
- Avery, Mallory, Osea Giuntella, and Peiran Jiao. 2025. “Why Don’t We Sleep Enough? A Field Experiment among College Students.” *The Review of Economics and Statistics*, 107(1): 65–77.
- Bailey, Caitlin P., Kevin W. Dodd, James J. McClain, Isabell Seo, William Wheeler, and Dana L. Wolff-Hughes. 2025. “Fitbit Physical Activity and Sleep Data in the All of Us Research Program: Data Exploration and Processing Considerations for Research.” *Medicine and science in sports and exercise*, 57(12): 2946–2953.
- Banzhaf, H. Spencer, and Randall P. Walsh. 2008. “Do People Vote with Their Feet? An Empirical Test of Tiebout.” *American Economic Review*, 98(3): 843–863.
- Bartik, Timothy J. 1988. “Evaluating the benefits of non-marginal reductions in pollution using information on defensive expenditures.” *Journal of Environmental Economics and Management*, 15(1): 111–127.
- Barwick, Panle Jia, Shanjun Li, Deyu Rao, and Nahim Bin Zahur. 2026. “The Health Care Cost of Air Pollution: Evidence from the World’s Largest Payment Network.” *The Review of Economics and Statistics*, 108(1): 110–128.
- Barwick, Panle Jia, Shanjun Li, Liguang Lin, and Eric Yongchen Zou. 2024. “From Fog to Smog: The Value of Pollution Information.” *American Economic Review*, 114(5): 1338–1381.
- Bayer, Patrick, Nathaniel Keohane, and Christopher Timmins. 2009. “Migration and hedonic valuation: The case of air quality.” *Journal of Environmental Economics and Management*, 58(1): 1–14.
- Berestycki, Clara, and M. Keith Chen. 2026. “Behavioral responses to wildfire smoke: Insights from smartphone location data.” *Proceedings of the National Academy of Sciences*, 123(18): e2527320123.
- Biddle, Jeff E., and Daniel S. Hamermesh. 1990. “Sleep and the Allocation of Time.” *Journal of Political Economy*, 98(5, Part 1): 922–943.
- Billings, Martha E., Diane Gold, Adam Szpiro, Carrie P. Aaron, Neal Jorgensen, Amanda Gassett, Peter J. Leary, Joel D. Kaufman, and Susan R. Redline. 2019. “The Association of Ambient Air Pollution with Sleep Apnea: The Multi-Ethnic Study of Atherosclerosis.” *Annals of the American Thoracic Society*, 16(3): 363–370.
- Breshears, Elise, Dylan Brewer, and Andrew Earle. 2027. “Practical Considerations When Using Wildfire Smoke Data in Economic Analyses.” *Review of Environmental Eco-*

- Bresnahan, Brian W., Mark Dickie, and Shelby Gerking.** 1997. “Averting behavior and urban air pollution.” *Land Economics*, 73(3): 340.
- Brewer, Dylan, Daniel Dench, and Laura O. Taylor.** 2023. “Advances in Causal Inference at the Intersection of Air Pollution and Health Outcomes.” *Annual Review of Resource Economics*, 15: 455–469.
- Burke, Marshall, Marissa L. Childs, Brandon de la Cuesta, Minghao Qiu, Jessica Li, Carlos F. Gould, Sam Heft-Neal, and Michael Wara.** 2023. “The contribution of wildfire to PM2.5 trends in the USA.” *Nature*, 622(7984): 761–766.
- Burke, Marshall, Sam Heft-Neal, Jessica Li, Anne Driscoll, Patrick Baylis, Matthieu Stigler, Joakim A. Weill, Jennifer A. Burney, Jeff Wen, Marissa L. Childs, and Carlos F. Gould.** 2022. “Exposures and behavioural responses to wildfire smoke.” *Nature Human Behaviour*, 6(10): 1351–1361.
- Chang, Tom, Joshua Graff Zivin, Tal Gross, and Matthew Neidell.** 2016. “Particulate Pollution and the Productivity of Pear Packers.” *American Economic Journal: Economic Policy*, 8(3): 141–169.
- Chang, Tom Y., Joshua Graff Zivin, Tal Gross, and Matthew Neidell.** 2019. “The Effect of Pollution on Worker Productivity: Evidence from Call Center Workers in China.” *American Economic Journal: Applied Economics*, 11(1): 151–172.
- Chay, Kenneth Y, and Michael Greenstone.** 2003. “Air Quality, Infant Mortality, and the Clean Air Act of 1970.” National Bureau of Economic Research Working Paper 10053.
- Childs, Marissa L., Jessica Li, Jeffrey Wen, Sam Heft-Neal, Anne Driscoll, Sherrie Wang, Carlos F. Gould, Minghao Qiu, Jennifer Burney, and Marshall Burke.** 2022. “Daily Local-Level Estimates of Ambient Wildfire Smoke PM2.5 for the Contiguous US.” *Environmental Science & Technology*, 56(19): 13607–13621.
- Childs, Marissa, Mariana Martins, Andrew Wilson, Minghao Qiu, Sam Heft-Neal, and Marshall Burke.** 2025. “Growing wildfire-derived PM2.5 across the contiguous U.S. and implications for air quality regulation.”
- Currie, Janet, and Matthew Neidell.** 2005. “Air Pollution and Infant Health: What Can We Learn from California’s Recent Experience?” *The Quarterly Journal of Economics*, 120(3): 1003–1030.
- Denny, Joshua C, Joni L Rutter, David B Goldstein, Anthony Philppakis, Jordan W Smoller, Gwynne Jenkins, Eric Dishman, and All of Us Program Investigators.** 2019. “The “All of Us” Research Program.” *New England Journal of Medicine*, 381(7): 668–676.
- Deryugina, Tatyana, Garth Heutel, Nolan H. Miller, David Molitor, and Julian Reif.** 2019. “The Mortality and Medical Costs of Air Pollution: Evidence from Changes in Wind Direction.” *American Economic Review*, 109(12): 4178–4219.
- Deschênes, Olivier, and Enrico Moretti.** 2009. “Extreme Weather Events, Mortality, and Migration.” *The Review of Economics and Statistics*, 91(4): 659–681.
- Deschênes, Olivier, Huixia Wang, Si Wang, and Peng Zhang.** 2020. “The effect of air pollution on body weight and obesity: Evidence from China.” *Journal of Development Economics*, 145: 102461.
- Deschênes, Olivier, Michael Greenstone, and Joseph S. Shapiro.** 2017. “Defensive Investments and the Demand for Air Quality: Evidence from the NOx Budget Program.”

- American Economic Review*, 107(10): 2958–2989.
- Fan, Yichun.** 2024. “Social cost of lifestyle adaptation: Air pollution and outdoor physical exercise.” *Journal of Environmental Economics and Management*, 127: 103042.
- Gibson, Matthew, and Jeffrey Shrader.** 2018. “Time Use and Labor Productivity: The Returns to Sleep.” *The Review of Economics and Statistics*, 100(5): 783–798.
- Giuntella, Osea, and Fabrizio Mazzonna.** 2019. “Sunset time and the economic effects of social jetlag: evidence from US time zone borders.” *Journal of Health Economics*, 65: 210–226.
- Giuntella, Osea, Silvia Saccardo, and Sally Sadoff.** 2026. “Sleep: Educational Impact and Habit Formation.” *Journal of Political Economy*.
- Gneezy, Uri, John A. List, Jeffrey A. Livingston, Xiangdong Qin, Sally Sadoff, and Yang Xu.** 2019. “Measuring Success in Education: The Role of Effort on the Test Itself.” *American Economic Review: Insights*, 1(3): 291–308.
- Graff Zivin, Joshua, and Matthew Neidell.** 2012. “The Impact of Pollution on Worker Productivity.” *American Economic Review*, 102(7): 3652–3673.
- Harris, Robert I.** 2026a. “Extreme temperatures, physical activity, and adaptation.” *Journal of Public Economics*, 257: 105638.
- Harris, Robert I.** 2026b. “Sleep in America: New Facts from 39 Million Nights of Wearable Data.” *Working Paper*.
- Harris, Robert I.** 2026c. “Wide Awake: The Effects of Salient Events on Sleep.” *Working Paper*.
- Heft-Neal, Sam, Carlos F. Gould, Marissa L. Childs, Mathew V. Kiang, Kari C. Nadeau, Mark Duggan, Eran Bendavid, and Marshall Burke.** 2023. “Emergency department visits respond nonlinearly to wildfire smoke.” *Proceedings of the National Academy of Sciences*, 120(39): e2302409120.
- Heyes, Anthony, and Mingying Zhu.** 2019. “Air pollution as a cause of sleeplessness: Social media evidence from a panel of Chinese cities.” *Journal of Environmental Economics and Management*, 98: 102247.
- Ito, Koichiro, and Shuang Zhang.** 2020. “Willingness to Pay for Clean Air: Evidence from Air Purifier Markets in China.” *Journal of Political Economy*, 128(5): 1627–1672.
- Janzen, Benedikt, and Patrick Bigler.** 2026. “Hazy Dreams: The Impact of Air Pollution on Sleep.”
- Keiser, David, Gabriel Lade, and Ivan Rudik.** 2018. “Air pollution and visitation at U.S. national parks.” *Science Advances*, 4(7): eaat1613.
- Krebs, Benjamin, and Simon Luechinger.** 2024. “Air Pollution, Cognitive Performance, and the Role of Task Proficiency.” *Journal of the Association of Environmental and Resource Economists*, 11(4): 921–958.
- Künn, Steffen, Juan Palacios, and Nico Pestel.** 2023. “Indoor Air Quality and Strategic Decision Making.” *Management Science*, 69(9): 5354–5377.
- La Nauze, Andrea, and Edson Severnini.** 2025. “Air Pollution and Adult Cognition: Evidence from Brain Training.” *Journal of the Association of Environmental and Resource Economists*, 12(1): 221–255.
- Lauderdale, Diane S., Kristen L. Knutson, Lijing L. Yan, Kiang Liu, and Paul J. Rathouz.** 2008. “Self-Reported and Measured Sleep Duration: How Similar Are They?” *Epidemiology*, 19(6): 838.

- Lichter, Andreas, Nico Pestel, and Eric Sommer.** 2017. "Productivity effects of air pollution: Evidence from professional soccer." *Labour Economics*, 48: 54–66.
- Miller, Nolan H., David Molitor, and Eric Zou.** 2024. "The Nonlinear Effects of Air Pollution on Health: Evidence from Wildfire Smoke."
- Molitor, David, Jamie T. Mullins, and Corey White.** 2023. "Air pollution and suicide in rural and urban America: Evidence from wildfire smoke." *Proceedings of the National Academy of Sciences*, 120(38): e2221621120.
- Mullins, Jamie T.** 2018. "Ambient air pollution and human performance: Contemporaneous and acclimatization effects of ozone exposure on athletic performance." *Health Economics*, 27(8): 1189–1200.
- Neidell, Matthew.** 2009. "Information, Avoidance Behavior, and Health: The Effect of Ozone on Asthma Hospitalizations." *Journal of Human Resources*, 44(2): 450–478.
- Neidell, Matthew J.** 2004. "Air pollution, health, and socio-economic status: the effect of outdoor air quality on childhood asthma." *Journal of Health Economics*, 23(6): 1209–1236.
- Nunn, Emily Baumgaertner.** 2026. "N.I.H. Announces World's Largest Integrated Health Database." *The New York Times*.
- Rasch, Björn, and Jan Born.** 2013. "About Sleep's Role in Memory." *Physiological Reviews*, 93(2): 681–766.
- Rosenberger, Randall S.** 2016. "Recreation Use Values Database." Oregon State University, College of Forestry, Corvallis, OR.
- Schlenker, Wolfram, and W. Reed Walker.** 2016. "Airports, Air Pollution, and Contemporaneous Health." *The Review of Economic Studies*, 83(2): 768–809.
- Schwartz, Joel.** 2000. "Harvesting and Long Term Exposure Effects in the Relation between Air Pollution and Mortality." *American Journal of Epidemiology*, 151(5): 440–448.
- Staiger, Douglas, and James H. Stock.** 1997. "Instrumental Variables Regression with Weak Instruments." *Econometrica*, 65(3): 557–586.
- Van Dongen, Hans P. A., Greg Maislin, Janet M. Mullington, and David F. Dinges.** 2003. "The cumulative cost of additional wakefulness: dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restriction and total sleep deprivation." *Sleep*, 26(2): 117–126.
- Wang, Zhenxuan, and Junjie Zhang.** 2023. "The value of information disclosure: Evidence from mask consumption in China." *Journal of Environmental Economics and Management*, 122: 102865.
- Zanobetti, Antonella, Susan Redline, Joel Schwartz, Dennis Rosen, Sanjay Patel, George T. O'Connor, Michael Lebowitz, Brent A. Coull, and Diane R. Gold.** 2010. "Associations of PM10 with Sleep and Sleep-disordered Breathing in Adults from Seven U.S. Urban Areas." *American Journal of Respiratory and Critical Care Medicine*, 182(6): 819–825.
- Zhang, Junjie, and Quan Mu.** 2018. "Air pollution and defensive expenditures: Evidence from particulate-filtering facemasks." *Journal of Environmental Economics and Management*, 92: 517–536.
- Zhang, Xin, Xi Chen, and Xiaobo Zhang.** 2018. "The impact of exposure to air pollution on cognitive performance." *Proceedings of the National Academy of Sciences*, 115(37): 9193–9197.
- Zhou, Peining, Jing Ma, Xueying Li, Yixue Zhao, Kunyao Yu, Rui Su, Rui**

Zhou, Hui Wang, and Guangfa Wang. 2023. “The long-term and short-term effects of ambient air pollutants on sleep characteristics in the Chinese population: big data analysis from real world by sleep records of consumer wearable devices.” *BMC Medicine*, 21(1): 83.

Online Appendix

Air Pollution and Behavior: Evidence from Wearable Devices

Robert I. Harris and Maghfira Ramadhani

A Participant Geography and WEAR Comparisons

Figure A.1 shows the geographic distribution of participants in the activity sample. Tables A.1 and A.2 compare participant characteristics and Fitbit outcomes by WEAR participation.

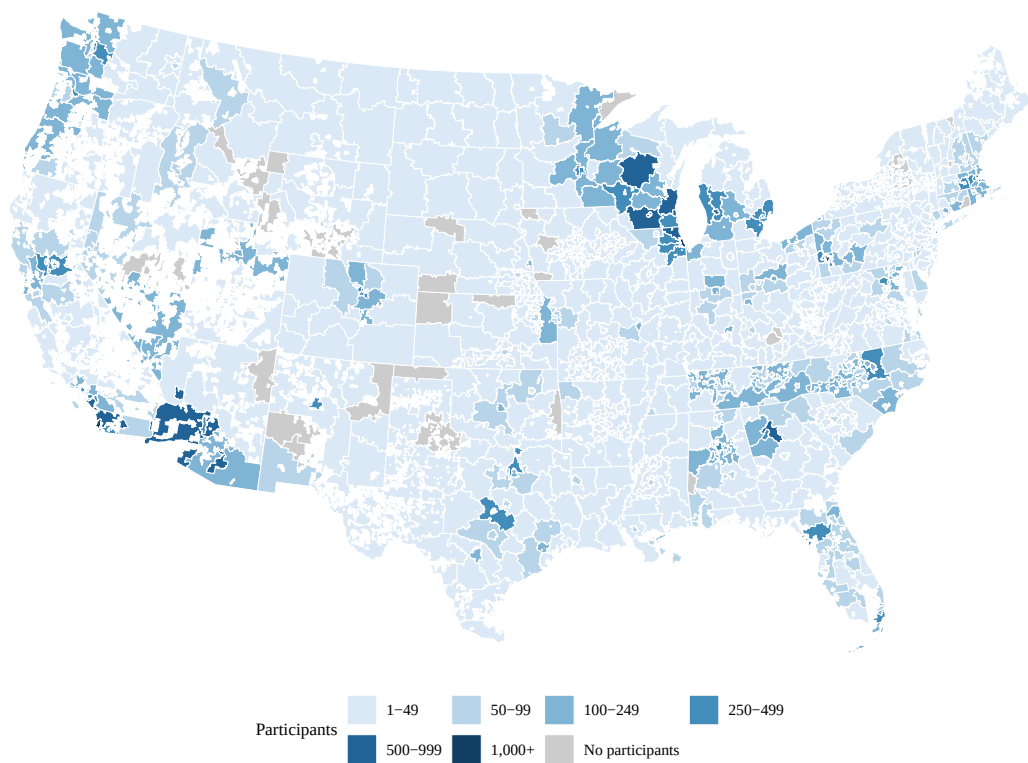


Figure A.1: Geographic distribution of participants. Each participant is counted once in the ZIP3 area recorded on their first eligible activity day. Shading indicates participant-count ranges.

Table A.1: Participant characteristics by WEAR participation

Characteristic	All participants	WEAR	BYOD	Other/unknown
Participants	61,338	28,931	31,580	827
Share of participants (%)	100.0	47.2	51.5	1.3
	50.8 (16.4)	52.7 (17.2)	49.1 (15.5)	50.6 (16.4)
Age (years)	61,338	28,931	31,580	827
	834.1 (889.9)	422.8 (488.1)	1,223.1 (1,000.5)	367.3 (506.6)
Observed days	61,338	28,931	31,580	827
	168.4 (9.5)	168.2 (9.6)	168.6 (9.4)	167.4 (9.2)
Height (cm)	56,007	27,940	27,250	817
	83.9 (23.0)	84.6 (24.0)	83.1 (21.8)	86.9 (23.2)
Weight (kg)	55,849	27,873	27,162	814
	29.6 (7.6)	29.9 (8.0)	29.2 (7.1)	30.9 (7.8)
Body mass index (kg/m ²)	55,729	27,815	27,100	814
Female	40,562 (66.1)	18,446 (63.8)	21,526 (68.2)	590 (71.3)
<i>Race/ethnicity</i>				
Black	4,807 (7.8)	3,090 (10.7)	1,587 (5.0)	130 (15.7)
White	43,162 (70.4)	16,860 (58.3)	25,860 (81.9)	442 (53.4)
Asian	2,976 (4.9)	1,984 (6.9)	934 (3.0)	58 (7.0)
Hispanic	6,791 (11.1)	4,715 (16.3)	1,939 (6.1)	137 (16.6)
Other	1,870 (3.0)	1,208 (4.2)	633 (2.0)	29 (3.5)
Missing/unknown	1,732 (2.8)	1,074 (3.7)	627 (2.0)	31 (3.7)
<i>Household income</i>				
<25k	9,222 (15.0)	6,573 (22.7)	2,466 (7.8)	183 (22.1)
25-75k	19,552 (31.9)	9,905 (34.2)	9,353 (29.6)	294 (35.6)
75-150k	18,301 (29.8)	7,142 (24.7)	10,975 (34.8)	184 (22.2)
>150k	10,318 (16.8)	3,252 (11.2)	6,964 (22.1)	102 (12.3)
Missing/unknown	3,945 (6.4)	2,059 (7.1)	1,822 (5.8)	64 (7.7)
<i>Educational attainment</i>				
HS or less	–	–	–	–
Some college	16,272 (26.5)	8,765 (30.3)	7,230 (22.9)	277 (33.5)
BA+	38,718 (63.1)	16,318 (56.4)	21,960 (69.5)	440 (53.2)
Missing/unknown	–	–	–	–
<i>Employment status</i>				
Employed	29,569 (48.2)	11,552 (39.9)	17,686 (56.0)	331 (40.0)
Retired	14,313 (23.3)	7,327 (25.3)	6,771 (21.4)	215 (26.0)
Other employment	9,513 (15.5)	5,910 (20.4)	3,436 (10.9)	167 (20.2)
Missing/unknown	7,943 (12.9)	4,142 (14.3)	3,687 (11.7)	114 (13.8)

Notes: Continuous characteristics report means (standard deviations), with valid participant counts beneath. Categorical entries report counts (percentages); percentages include missing responses in the denominator. Age is measured at the first eligible observation. WEAR participants received a Fitbit through the All of Us WEAR study, which provides devices at no cost. BYOD (“bring your own device”) participants contributed records from their own Fitbits. Hispanic ethnicity takes precedence over race; other race includes multiracial responses. Height, weight, and BMI use earliest valid adult measurements within 100–250 cm, 20–400 kg, and 10–100 kg/m², respectively. Dashes denote unavailable or suppressed statistics, including companion cells suppressed to prevent recovery of small counts.

Table A.2: Physical activity and sleep by WEAR participation

Outcome	All participants	WEAR	BYOD	Other/unknown
Daily steps	7,838.2 (5,032.9); 51,161,203	6,940.3 (4,886.8); 12,231,413	8,124.2 (5,045.4); 38,625,993	7,625.3 (4,977.3); 303,797
Walking minutes	85.6 (58.4); 51,161,203	76.0 (58.4); 12,231,413	88.7 (58.0); 38,625,993	84.7 (59.8); 303,797
Running minutes	1.6 (8.3); 51,161,203	1.1 (6.3); 12,231,413	1.8 (8.8); 38,625,993	1.3 (6.9); 303,797
Active minutes	259.0 (119.0); 51,161,202	241.0 (120.8); 12,231,413	264.7 (117.7); 38,625,992	255.6 (124.1); 303,797
Sedentary minutes	817.7 (268.7); 51,161,202	839.6 (270.5); 12,231,413	810.4 (267.3); 38,625,992	860.7 (318.6); 303,797
Lightly active minutes	218.7 (100.4); 51,161,202	207.4 (101.3); 12,231,413	222.3 (99.8); 38,625,992	215.5 (102.6); 303,797
Fairly active minutes	19.6 (31.3); 51,161,202	16.8 (29.2); 12,231,413	20.5 (31.8); 38,625,992	21.7 (37.0); 303,797
Very active minutes	20.7 (34.1); 51,161,203	16.9 (31.3); 12,231,413	22.0 (34.9); 38,625,993	18.4 (32.9); 303,797
Calories expended (kcal/day)	2,320.1 (736.9); 51,161,202	2,262.7 (865.2); 12,231,413	2,338.3 (690.8); 38,625,992	2,315.7 (677.0); 303,797
Sleep duration (minutes)	389.5 (110.5); 36,904,607	380.7 (117.8); 9,879,536	392.9 (107.4); 26,824,702	376.3 (123.8); 200,369
Time in bed (minutes)	442.8 (124.3); 36,904,485	434.5 (133.4); 9,879,513	445.9 (120.5); 26,824,606	428.0 (137.4); 200,366
Light sleep (minutes)	259.9 (69.1); 28,801,295	259.7 (73.4); 8,212,819	260.1 (67.2); 20,476,222	248.8 (71.9); 112,254
Deep sleep (minutes)	59.7 (26.9); 28,801,295	58.0 (27.6); 8,212,819	60.4 (26.6); 20,476,222	59.9 (27.9); 112,254
REM sleep (minutes)	80.7 (36.1); 28,801,295	77.7 (37.0); 8,212,819	81.8 (35.6); 20,476,222	78.4 (37.7); 112,254

Notes: Entries report means, with standard deviations in parentheses, followed by the number of observations. Each participant-day receives equal weight. Activity measures use the daily activity sample; sleep measures use eligible nights, with sleep-stage measures restricted to nights with valid stage records. WEAR participants received a Fitbit through the All of Us WEAR study, which provides devices at no cost. BYOD ("bring your own device") participants contributed records from their own Fitbits.

B Regions for the Wind-Direction Instruments

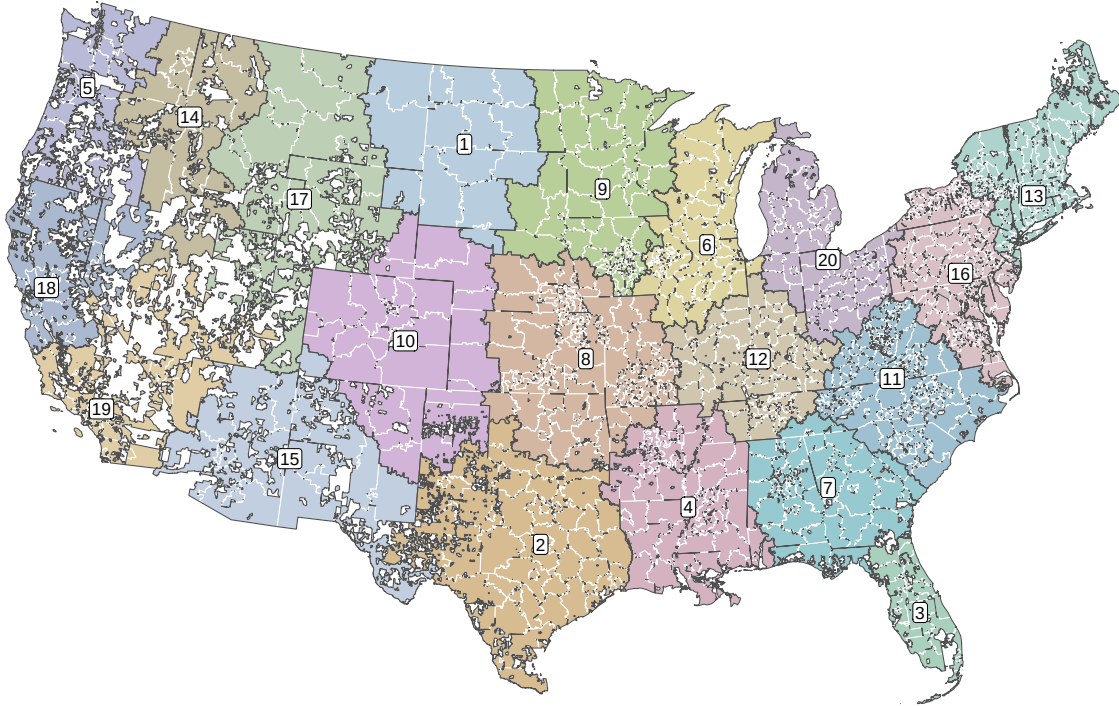


Figure B.2: Regions for the wind-direction instruments. ZIP3 areas are grouped into 20 regions using k -means clustering of their population-weighted centroids. Numbers identify the region assignments used in the analysis. Thin white lines mark ZIP3 boundaries; dark lines mark region boundaries.

C Activity Intensity

Table C.3 separates lightly, fairly, and very active minutes. Each exposure specification uses the same cross-fitted instrument as the main daily-outcome table.

Table C.3: Pollution and activity intensity

	Lightly active minutes	Fairly active minutes	Very active minutes
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$)</i>			
Estimate	-2.21*** (0.49)	0.16 (0.24)	-0.23 (0.29)
First-stage F	317.1	317.1	317.1
Adjusted R^2	0.416	0.339	0.352
<i>Panel B: $PM_{2.5} > 9$</i>			
Estimate	-2.59*** (0.57)	0.18 (0.28)	-0.26 (0.34)
First-stage F	499.3	499.3	499.3
Adjusted R^2	0.416	0.339	0.352
<i>Panel C: $PM_{2.5} > 12$</i>			
Estimate	-4.45*** (0.99)	0.32 (0.48)	-0.46 (0.58)
First-stage F	247.7	247.7	247.7
Adjusted R^2	0.416	0.339	0.352
<i>Panel D: $PM_{2.5} > 25$</i>			
Estimate	-81.87** (32.23)	5.84 (9.26)	-8.37 (10.73)
First-stage F	8.0	8.0	8.0
Adjusted R^2	0.410	0.338	0.351
Observations	51,033,285	51,033,285	51,033,286
Mean outcome	218.67	19.62	20.74

Notes: The dependent variable is given in each column heading. Activity outcomes are in minutes. Panels A–D report continuous $PM_{2.5}$, $PM_{2.5} > 9$, $PM_{2.5} > 12$, and $PM_{2.5} > 25$, respectively. Continuous-PM estimates are scaled to a 10 $\mu g/m^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu g/m^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. The $PM_{2.5} > 25$ models have substantially weaker first stages. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Additional Within-Day Estimates

Table D.4: Within-day activity: 12:00 a.m.–5:59 a.m.

	Steps	Walking minutes	Running minutes
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$)</i>			
Estimate	-4.553*	-0.044*	-0.003
	(2.665)	(0.026)	(0.005)
First-stage F	317.6	317.6	317.6
Adjusted R^2	0.334	0.299	0.358
<i>Panel B: $PM_{2.5} > 12$</i>			
Estimate	-9.191*	-0.089*	-0.007
	(5.347)	(0.053)	(0.011)
First-stage F	248.0	248.0	248.0
Adjusted R^2	0.334	0.299	0.358
Observations	51,033,286	51,033,286	51,033,286
Mean outcome	260.432	2.368	0.113

Notes: The dependent variable is given in each column heading. Steps and walking and running minutes are totals for 12:00 a.m.–5:59 a.m. Panels distinguish continuous PM and $PM_{2.5} > 12$. All windows use the same daily pollution measure. Continuous-PM estimates are scaled to a 10 $\mu g/m^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu g/m^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Regressions additionally control for daylight minutes within the window. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.5: Within-day activity: 6:00 a.m.–9:59 a.m.

	Steps	Walking minutes	Running minutes
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>			
Estimate	-0.168 (11.852)	0.037 (0.124)	-0.013 (0.015)
First-stage F	317.8	317.8	317.8
Adjusted R^2	0.373	0.372	0.261
<i>Panel B: PM_{2.5} > 12</i>			
Estimate	-0.339 (23.923)	0.074 (0.250)	-0.026 (0.030)
First-stage F	248.2	248.2	248.2
Adjusted R^2	0.373	0.372	0.261
Observations	51,033,247	51,033,247	51,033,247
Mean outcome	1,643.076	17.417	0.609

Notes: The dependent variable is given in each column heading. Steps and walking and running minutes are totals for 6:00 a.m.–9:59 a.m. Panels distinguish continuous PM and PM_{2.5} > 12. All windows use the same daily pollution measure. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Regressions additionally control for daylight minutes within the window. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.6: Within-day activity: 10:00 a.m.–1:59 p.m.

	Steps	Walking minutes	Running minutes
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>			
Estimate	-13.153 (17.083)	-0.061 (0.207)	-0.036*** (0.012)
First-stage F	317.1	317.1	317.1
Adjusted R^2	0.254	0.263	0.163
<i>Panel B: PM_{2.5} > 12</i>			
Estimate	-26.559 (34.397)	-0.123 (0.417)	-0.072*** (0.025)
First-stage F	247.7	247.7	247.7
Adjusted R^2	0.254	0.263	0.163
Observations	51,033,017	51,033,017	51,033,017
Mean outcome	2,192.612	24.781	0.366

Notes: The dependent variable is given in each column heading. Steps and walking and running minutes are totals for 10:00 a.m.–1:59 p.m. Panels distinguish continuous PM and PM_{2.5} > 12. All windows use the same daily pollution measure. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Regressions additionally control for daylight minutes within the window. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.7: Within-day activity: 2:00 p.m.–5:59 p.m.

	Steps	Walking minutes	Running minutes
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>			
Estimate	-80.570*** (18.874)	-0.796*** (0.221)	-0.038*** (0.012)
First-stage F	317.1	317.1	317.1
Adjusted R^2	0.240	0.250	0.133
<i>Panel B: PM_{2.5} > 12</i>			
Estimate	-162.692*** (38.793)	-1.607*** (0.450)	-0.077*** (0.024)
First-stage F	247.7	247.7	247.7
Adjusted R^2	0.239	0.249	0.133
Observations	51,032,692	51,032,692	51,032,692
Mean outcome	2,088.591	23.558	0.304

Notes: The dependent variable is given in each column heading. Steps and walking and running minutes are totals for 2:00 p.m.–5:59 p.m. Panels distinguish continuous PM and PM_{2.5} > 12. All windows use the same daily pollution measure. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Regressions additionally control for daylight minutes within the window. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D.8: Within-day activity: 6:00 p.m.–11:59 p.m.

	Steps	Walking minutes	Running minutes
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>			
Estimate	-67.092*** (12.934)	-0.718*** (0.153)	-0.018** (0.008)
First-stage F	317.1	317.1	317.1
Adjusted R^2	0.299	0.306	0.133
<i>Panel B: PM_{2.5} > 12</i>			
Estimate	-135.485*** (25.874)	-1.449*** (0.305)	-0.036** (0.016)
First-stage F	247.7	247.7	247.7
Adjusted R^2	0.299	0.306	0.133
Observations	51,031,648	51,031,648	51,031,648
Mean outcome	1,647.937	17.522	0.248

Notes: The dependent variable is given in each column heading. Steps and walking and running minutes are totals for 6:00 p.m.–11:59 p.m. Panels distinguish continuous PM and PM_{2.5} > 12. All windows use the same daily pollution measure. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Regressions additionally control for daylight minutes within the window. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E Additional Dynamic PM Responses

Tables E.9 and E.10 report cumulative responses to continuous PM_{2.5} and the PM_{2.5} > 12 indicator, respectively, using seven-day unrestricted and spline models and fourteen-day spline models.

Figure E.3 shows fourteen-day responses for steps. Figures E.4–E.12 report seven-day responses under both exposure specifications and fourteen-day continuous-PM responses for walking, active time, and sleep. The left panels show responses at each lag, while the right panels show cumulative responses.

Table E.9: Cumulative responses: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
<i>Panel A: Unrestricted, Lags 0–7</i>							
Cumulative effect	41.208 (97.603)	0.167 (1.159)	0.027 (0.059)	0.056 (1.903)	1.710 (2.220)	8.490 (8.304)	-0.708 (1.121)
First-stage F	34.0–47.5	34.0–47.5	34.0–47.5	34.0–47.5	34.0–47.5	34.0–47.5	28.1–40.5
Adjusted R^2	0.460	0.449	0.349	0.418	0.436	0.586	0.284
<i>Panel B: Spline, Lags 0–7</i>							
Cumulative effect	12.668 (94.422)	-0.163 (1.123)	0.019 (0.059)	-0.378 (1.860)	1.924 (2.186)	6.705 (8.106)	-0.809 (1.116)
First-stage F	39.2–69.8	39.2–69.8	39.2–69.8	39.2–69.8	39.2–69.8	39.2–69.8	34.4–59.0
Adjusted R^2	0.460	0.449	0.349	0.418	0.436	0.586	0.284
<i>Panel C: Spline, Lags 0–14</i>							
Cumulative effect	5.765 (231.827)	-1.568 (2.801)	0.239 (0.152)	-3.614 (4.732)	3.714 (5.897)	4.867 (20.395)	4.818 (2.970)
First-stage F	27.7–65.6	27.7–65.6	27.7–65.6	27.7–65.6	27.7–65.6	27.7–65.6	26.9–56.3
Adjusted R^2	0.460	0.449	0.349	0.418	0.436	0.586	0.284
Observations	49,947,893	49,947,893	49,947,893	49,947,892	49,947,892	49,947,576	36,156,808
Mean outcome	7,840.419	85.668	1.640	259.073	817.669	2,320.398	389.523

Notes: The dependent variable is given in each column heading. Samples require complete exposure histories for the indicated horizon. Panels distinguish unrestricted seven-day, spline seven-day, and spline fourteen-day specifications. Entries are sums of the contemporaneous and lag coefficients, measuring cumulative responses to a one-day exposure change. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Current and lagged exposures are instrumented with the cross-fitted dynamic wind histories. Observations and mean outcomes refer to each model's final estimation sample. The two seven-day specifications share a sample; fourteen-day specifications need not. First-stage F -statistics are cluster-robust Wald statistics. Where more than one exposure term is endogenous, the entry reports the range across first-stage equations; it is not a conditional-strength test. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Dynamic models also control for the weather histories described in the methods. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. Standard errors for cumulative effects use the full covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E.10: Cumulative responses: PM_{2.5} > 12

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
<i>Panel A: Unrestricted, Lags 0–7</i>							
Cumulative effect	117.711 (215.784)	0.728 (2.539)	0.044 (0.127)	0.690 (4.054)	3.299 (4.885)	20.789 (17.481)	-1.680 (2.557)
First-stage F	25.4–34.1	25.4–34.1	25.4–34.1	25.4–34.1	25.4–34.1	25.4–34.1	21.9–30.3
Adjusted R^2	0.457	0.447	0.349	0.417	0.436	0.586	0.284
<i>Panel B: Spline, Lags 0–7</i>							
Cumulative effect	18.402 (199.272)	-0.367 (2.373)	0.018 (0.126)	-0.775 (3.865)	4.363 (4.694)	15.521 (16.815)	-2.196 (2.489)
First-stage F	36.6–53.9	36.6–53.9	36.6–53.9	36.6–53.9	36.6–53.9	36.6–53.9	31.8–46.7
Adjusted R^2	0.459	0.448	0.349	0.418	0.436	0.586	0.284
<i>Panel C: Spline, Lags 0–14</i>							
Cumulative effect	-88.310 (500.408)	-4.578 (6.102)	0.470 (0.338)	-9.732 (10.387)	10.635 (13.160)	10.225 (43.755)	8.954 (6.670)
First-stage F	23.6–51.5	23.6–51.5	23.6–51.5	23.6–51.5	23.6–51.5	23.6–51.5	21.4–44.9
Adjusted R^2	0.460	0.449	0.349	0.418	0.436	0.586	0.284
Observations	49,947,893	49,947,893	49,947,893	49,947,892	49,947,892	49,947,576	36,156,808
Mean outcome	7,840.419	85.668	1.640	259.073	817.669	2,320.398	389.523

Notes: The dependent variable is given in each column heading. Samples require complete exposure histories for the indicated horizon. Panels distinguish unrestricted seven-day, spline seven-day, and spline fourteen-day specifications. Entries are sums of the contemporaneous and lag coefficients, measuring cumulative responses to a one-day exposure change. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Current and lagged exposures are instrumented with the cross-fitted dynamic wind histories. Observations and mean outcomes refer to each model's final estimation sample. The two seven-day specifications share a sample; fourteen-day specifications need not. First-stage F -statistics are cluster-robust Wald statistics. Where more than one exposure term is endogenous, the entry reports the range across first-stage equations; it is not a conditional-strength test. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Dynamic models also control for the weather histories described in the methods. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. Standard errors for cumulative effects use the full covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

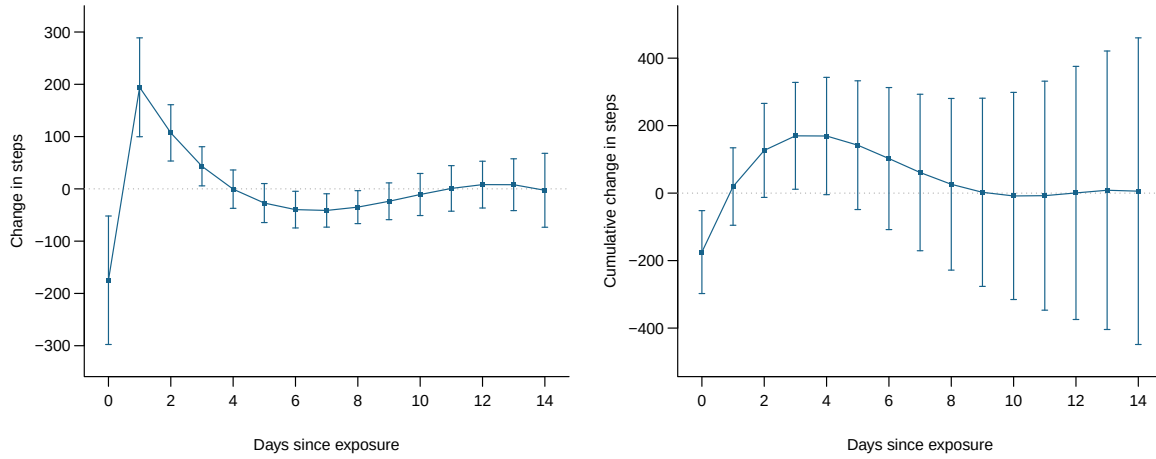


Figure E.3: Fourteen-day spline responses: total steps. The exposure is a one-day increase of $10 \mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

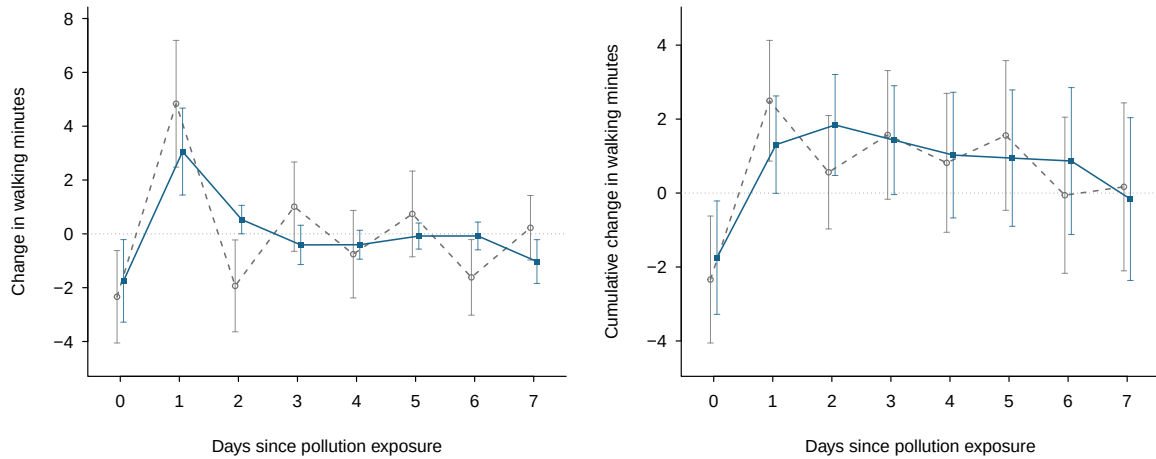


Figure E.4: Dynamic responses: walk minutes. Continuous PM is scaled to a one-day increase of $10 \mu\text{g}/\text{m}^3$. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

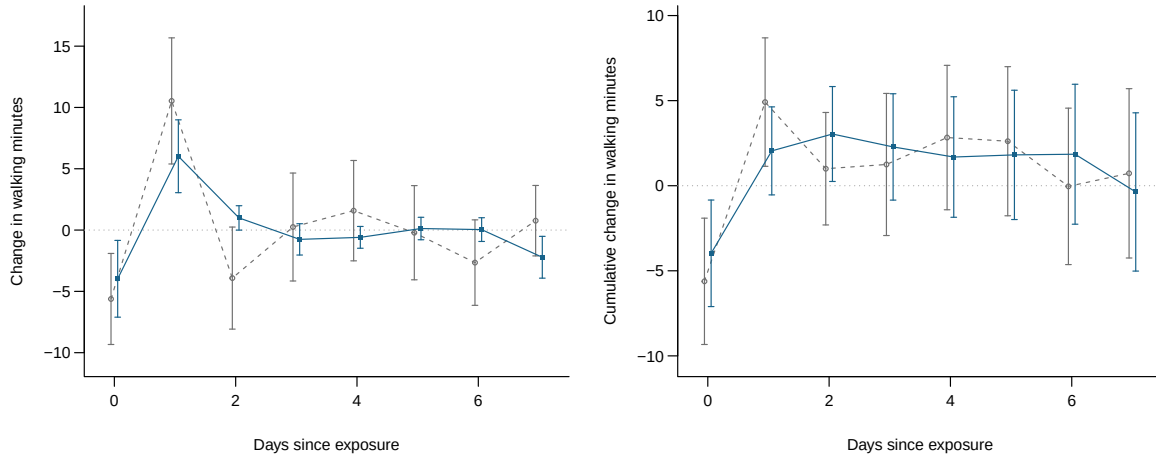


Figure E.5: Dynamic responses to $\text{PM}_{2.5} > 12$: walk minutes. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

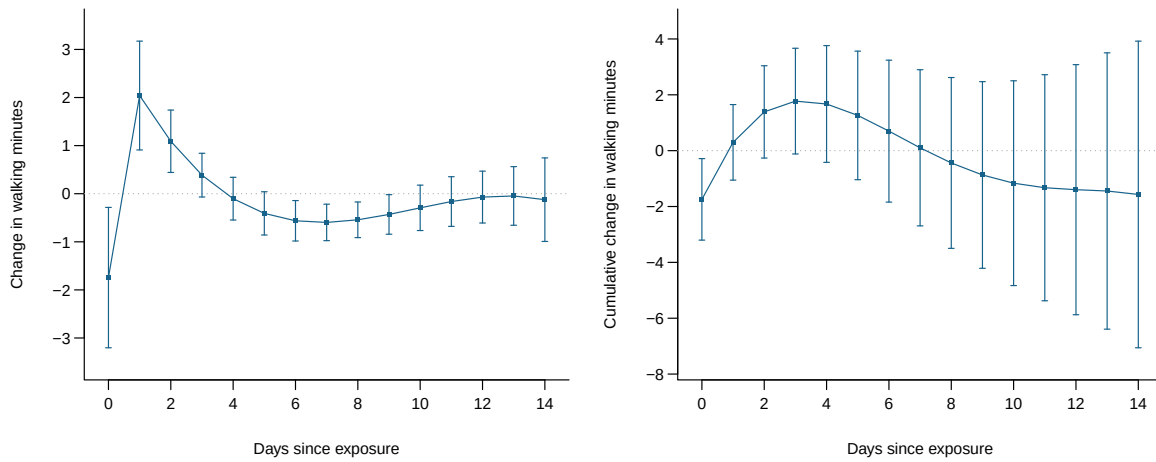


Figure E.6: Fourteen-day spline responses: walk minutes. The exposure is a one-day increase of $10 \mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

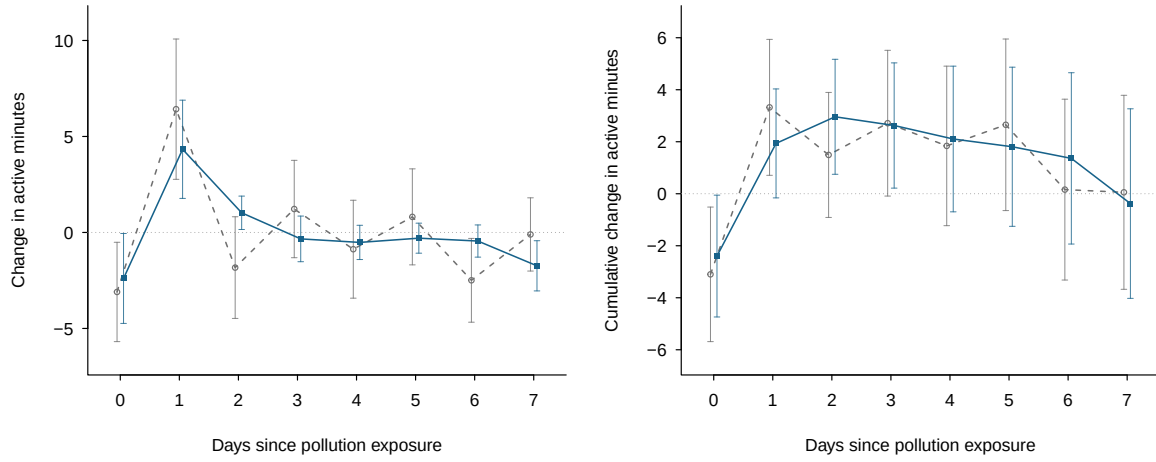


Figure E.7: Dynamic responses: active minutes. Continuous PM is scaled to a one-day increase of $10 \mu\text{g}/\text{m}^3$. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

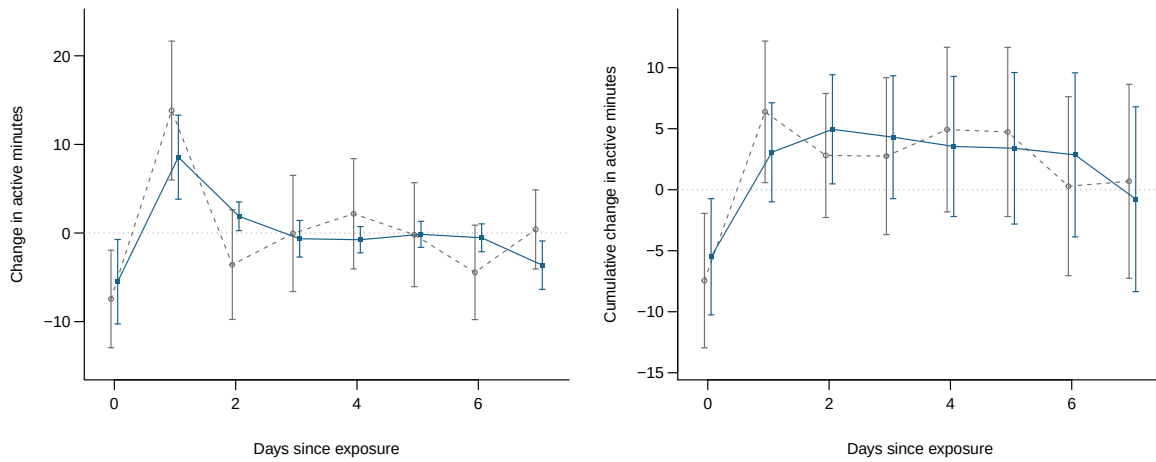


Figure E.8: Dynamic responses to $\text{PM}_{2.5} > 12$: active minutes. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

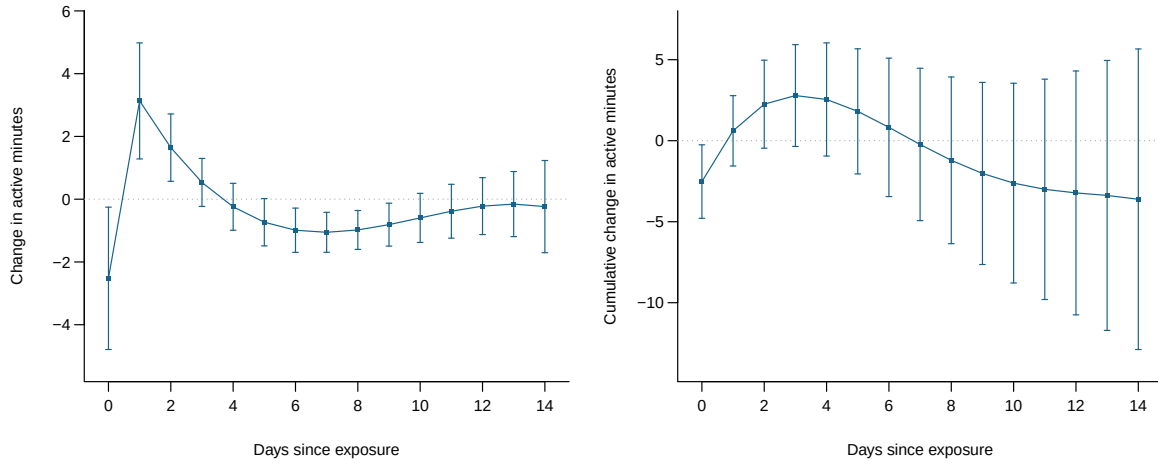


Figure E.9: Fourteen-day spline responses: active minutes. The exposure is a one-day increase of $10 \mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

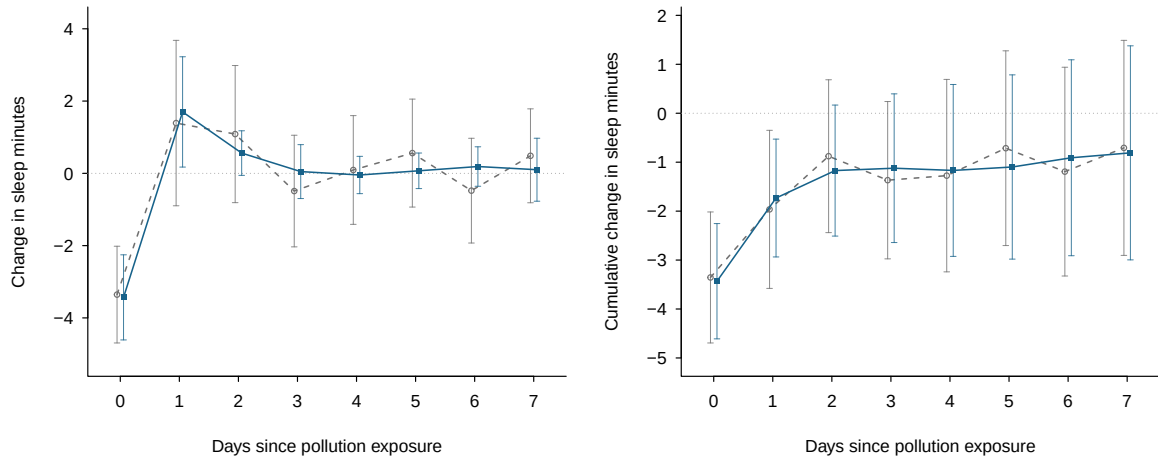


Figure E.10: Dynamic responses: sleep minutes. Continuous PM is scaled to a one-day increase of $10 \mu\text{g}/\text{m}^3$. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

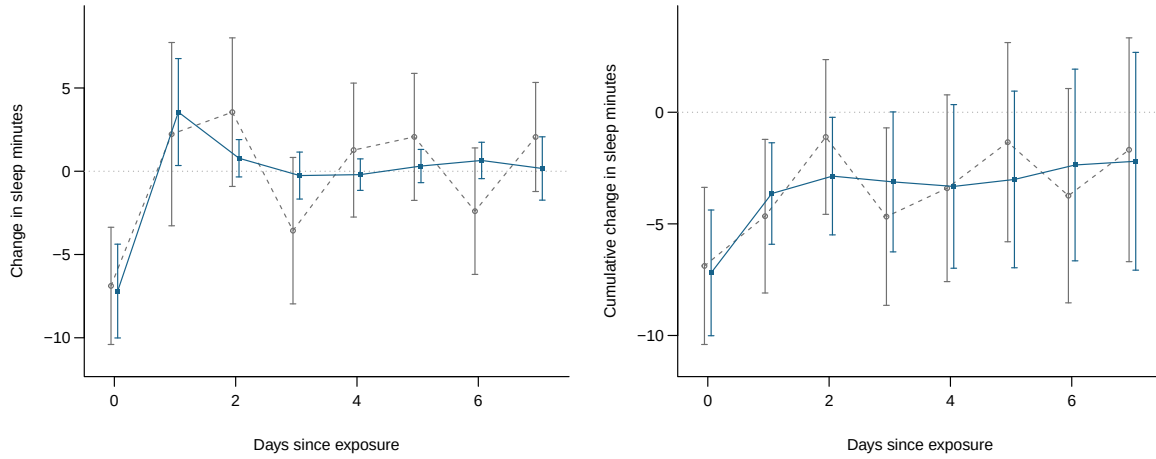


Figure E.11: Dynamic responses to $\text{PM}_{2.5} > 12$: sleep minutes. Gray dashed lines with open circles show unrestricted lag estimates; blue solid lines with squares show spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

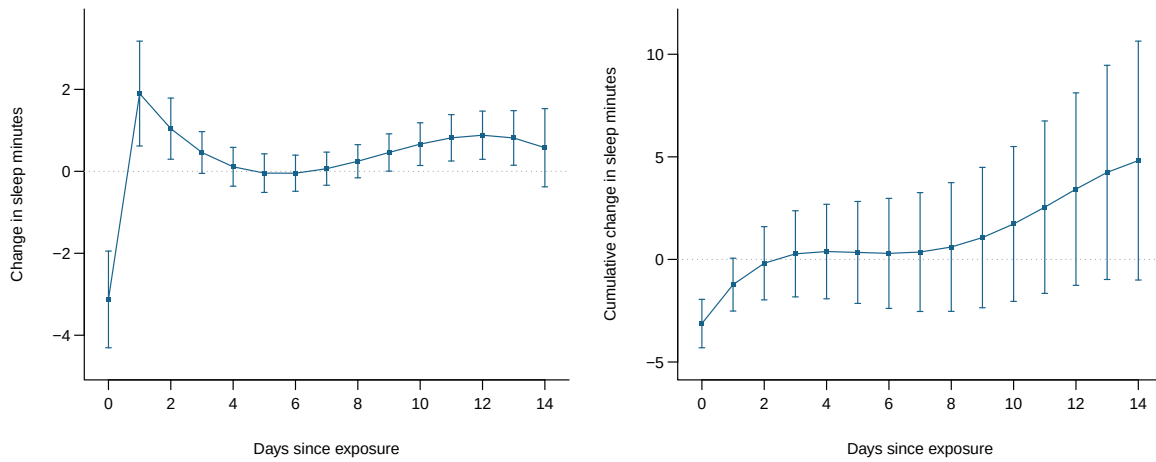


Figure E.12: Fourteen-day spline responses: sleep minutes. The exposure is a one-day increase of $10 \mu\text{g}/\text{m}^3$ in $\text{PM}_{2.5}$. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

E.1 Activity-Intensity Responses

Table E.11: Cumulative responses: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)

	Lightly active minutes	Fairly active minutes	Very active minutes
<i>Panel A: Unrestricted, Lags 0–7</i>			
Cumulative effect	-0.783 (1.231)	0.126 (0.468)	0.713 (0.526)
First-stage F	34.0–47.5	34.0–47.5	34.0–47.5
Adjusted R^2	0.416	0.339	0.352
<i>Panel B: Spline, Lags 0–7</i>			
Cumulative effect	-0.968 (1.214)	0.013 (0.456)	0.577 (0.513)
First-stage F	39.2–69.8	39.2–69.8	39.2–69.8
Adjusted R^2	0.416	0.339	0.352
<i>Panel C: Spline, Lags 0–14</i>			
Cumulative effect	-4.605 (3.157)	-0.527 (1.125)	1.518 (1.244)
First-stage F	27.7–65.6	27.7–65.6	27.7–65.6
Adjusted R^2	0.416	0.339	0.352
Observations	49,947,892	49,947,892	49,947,893
Mean outcome	218.711	19.618	20.744

Notes: The dependent variable is given in each column heading. Samples require complete exposure histories for the indicated horizon. Panels distinguish unrestricted seven-day, spline seven-day, and spline fourteen-day specifications. Entries are sums of the contemporaneous and lag coefficients, measuring cumulative responses to a one-day exposure change. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Current and lagged exposures are instrumented with the cross-fitted dynamic wind histories. Observations and mean outcomes refer to each model's final estimation sample. The two seven-day specifications share a sample; fourteen-day specifications need not. First-stage F -statistics are cluster-robust Wald statistics. Where more than one exposure term is endogenous, the entry reports the range across first-stage equations; it is not a conditional-strength test. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Dynamic models also control for the weather histories described in the methods. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. Standard errors for cumulative effects use the full covariance matrix. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table E.12: Cumulative responses: PM_{2.5} > 12

	Lightly active minutes	Fairly active minutes	Very active minutes
<i>Panel A: Unrestricted, Lags 0–7</i>			
Cumulative effect	-1.482 (2.608)	0.428 (1.003)	1.744 (1.142)
First-stage <i>F</i>	25.4–34.1	25.4–34.1	25.4–34.1
Adjusted <i>R</i> ²	0.415	0.338	0.351
<i>Panel B: Spline, Lags 0–7</i>			
Cumulative effect	-2.128 (2.523)	0.106 (0.969)	1.247 (1.084)
First-stage <i>F</i>	36.6–53.9	36.6–53.9	36.6–53.9
Adjusted <i>R</i> ²	0.415	0.339	0.352
<i>Panel C: Spline, Lags 0–14</i>			
Cumulative effect	-11.610 (7.104)	-1.303 (2.511)	3.181 (2.645)
First-stage <i>F</i>	23.6–51.5	23.6–51.5	23.6–51.5
Adjusted <i>R</i> ²	0.415	0.339	0.352
Observations	49,947,892	49,947,892	49,947,893
Mean outcome	218.711	19.618	20.744

Notes: The dependent variable is given in each column heading. Samples require complete exposure histories for the indicated horizon. Panels distinguish unrestricted seven-day, spline seven-day, and spline fourteen-day specifications. Entries are sums of the contemporaneous and lag coefficients, measuring cumulative responses to a one-day exposure change. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Current and lagged exposures are instrumented with the cross-fitted dynamic wind histories. Observations and mean outcomes refer to each model's final estimation sample. The two seven-day specifications share a sample; fourteen-day specifications need not. First-stage *F*-statistics are cluster-robust Wald statistics. Where more than one exposure term is endogenous, the entry reports the range across first-stage equations; it is not a conditional-strength test. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Dynamic models also control for the weather histories described in the methods. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. Standard errors for cumulative effects use the full covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Running Definitions and Additional Outcomes

Figure F.13 compares the estimated effect of $\text{PM}_{2.5} > 12$ on running cadence across definitions that vary the minimum running duration, cadence threshold, and permitted interruptions.

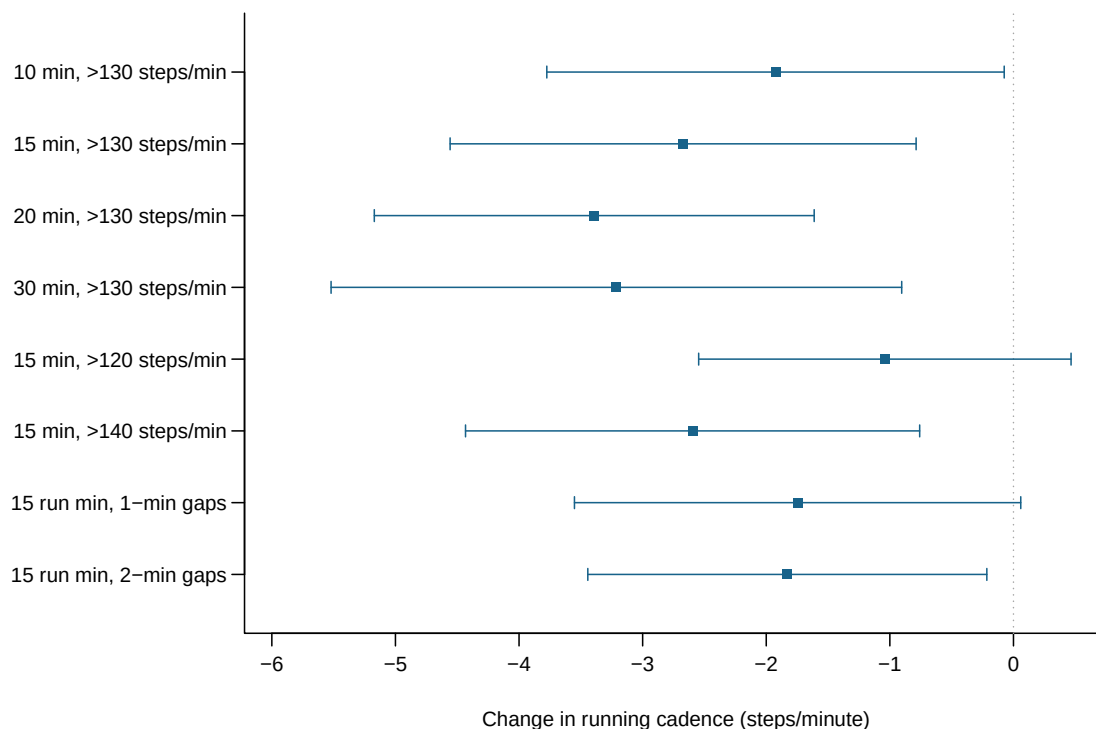


Figure F.13: Running cadence under alternative definitions. Coefficients instrument $\text{PM}_{2.5} > 12$ with the cross-fitted selected-direction indicator. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date. Cadence is measured during running minutes on days with qualifying runs. Gap definitions permit interruptions while retaining at least 15 running minutes; the uninterrupted definitions require the indicated consecutive duration. Samples differ across definitions.

Tables F.13–F.20 report participation, duration, and cadence estimates under each definition. Table F.21 additionally reports cadence averaged over the entire running period, including interruptions, and the share of that period spent in breaks.

Table F.13: Running responses: at least 10 consecutive minutes above 130 steps per minute

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$)</i>				
Estimate	-0.100** (0.049)	-0.051*** (0.018)	-0.947*** (0.356)	-0.976** (0.477)
First-stage F	317.1	317.1	403.3	403.3
Adjusted R^2	0.258	0.313	0.499	0.339
<i>Panel B: $PM_{2.5} > 12$</i>				
Estimate	-0.202** (0.100)	-0.104*** (0.037)	-1.866*** (0.717)	-1.925** (0.944)
First-stage F	247.7	247.7	273.2	273.2
Adjusted R^2	0.258	0.313	0.499	0.339
Observations	51,033,286	51,033,286	1,386,631	1,386,631
Mean outcome	2.737	0.889	32.566	158.241

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous $PM_{2.5}$ and $PM_{2.5} > 12$, respectively. Continuous-PM estimates are scaled to a 10 $\mu g/m^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu g/m^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.14: Running responses: at least 15 consecutive minutes above 130 steps per minute

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>				
Estimate	-0.086** (0.042)	-0.052*** (0.017)	-1.167*** (0.407)	-1.339*** (0.477)
First-stage <i>F</i>	317.1	317.1	409.9	409.9
Adjusted <i>R</i> ²	0.249	0.300	0.495	0.332
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.173** (0.085)	-0.104*** (0.035)	-2.331*** (0.822)	-2.673*** (0.962)
First-stage <i>F</i>	247.7	247.7	273.8	273.8
Adjusted <i>R</i> ²	0.249	0.300	0.494	0.332
Observations	51,033,286	51,033,286	1,033,181	1,033,181
Mean outcome	2.043	0.744	36.499	159.428

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage *F*-statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table F.15: Running responses: at least 20 consecutive minutes above 130 steps per minute

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 µg/m³)</i>				
Estimate	-0.088** (0.036)	-0.053*** (0.016)	-1.512*** (0.497)	-1.705*** (0.445)
First-stage <i>F</i>	317.1	317.1	400.1	400.1
Adjusted <i>R</i> ²	0.239	0.289	0.498	0.519
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.179** (0.072)	-0.106*** (0.033)	-3.008*** (1.003)	-3.392*** (0.908)
First-stage <i>F</i>	247.7	247.7	269.0	269.0
Adjusted <i>R</i> ²	0.239	0.289	0.498	0.519
Observations	51,033,286	51,033,286	803,221	803,221
Mean outcome	1.592	0.636	40.046	160.463

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 µg/m³ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in µg/m³. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage *F*-statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table F.16: Running responses: at least 30 consecutive minutes above 130 steps per minute

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 µg/m³)</i>				
Estimate	-0.071** (0.028)	-0.047*** (0.015)	-2.049*** (0.696)	-1.604*** (0.581)
First-stage <i>F</i>	317.1	317.1	355.6	355.6
Adjusted <i>R</i> ²	0.214	0.266	0.513	0.549
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.144** (0.057)	-0.094*** (0.032)	-4.103*** (1.394)	-3.212*** (1.178)
First-stage <i>F</i>	247.7	247.7	252.7	252.7
Adjusted <i>R</i> ²	0.214	0.266	0.512	0.548
Observations	51,033,286	51,033,286	469,845	469,845
Mean outcome	0.938	0.449	47.984	162.167

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 µg/m³ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in µg/m³. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage *F*-statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table F.17: Running responses: at least 15 consecutive minutes above 120 steps per minute

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>				
Estimate	-0.252*** (0.062)	-0.119*** (0.024)	-1.313*** (0.328)	-0.518 (0.381)
First-stage F	317.1	317.1	445.4	445.4
Adjusted R^2	0.255	0.315	0.472	0.351
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.509*** (0.126)	-0.240*** (0.049)	-2.640*** (0.678)	-1.040 (0.769)
First-stage F	247.7	247.7	287.2	287.2
Adjusted R^2	0.255	0.315	0.471	0.351
Observations	51,033,286	51,033,286	1,745,030	1,745,030
Mean outcome	3.440	1.259	36.659	147.309

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.18: Running responses: at least 15 consecutive minutes above 140 steps per minute

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>				
Estimate	-0.070*	-0.051***	-1.576***	-1.290***
	(0.037)	(0.014)	(0.473)	(0.456)
First-stage F	317.1	317.1	401.2	401.2
Adjusted R^2	0.241	0.300	0.529	0.362
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.142*	-0.102***	-3.171***	-2.596***
	(0.075)	(0.030)	(0.967)	(0.937)
First-stage F	247.7	247.7	262.5	262.5
Adjusted R^2	0.241	0.300	0.529	0.362
Observations	51,033,286	51,033,286	772,424	772,424
Mean outcome	1.531	0.551	36.059	165.417

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table F.19: Running responses: at least 15 running minutes above 130 steps per minute, allowing one-minute interruptions

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>				
Estimate	-0.097** (0.046)	-0.057*** (0.019)	-1.144*** (0.362)	-0.876* (0.460)
First-stage <i>F</i>	317.1	317.1	404.5	404.5
Adjusted <i>R</i> ²	0.256	0.308	0.490	0.338
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.195** (0.094)	-0.114*** (0.038)	-2.279*** (0.730)	-1.746* (0.921)
First-stage <i>F</i>	247.7	247.7	271.3	271.3
Adjusted <i>R</i> ²	0.256	0.308	0.490	0.338
Observations	51,033,286	51,033,286	1,233,764	1,233,764
Mean outcome	2.437	0.905	37.209	157.702

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage *F*-statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table F.20: Running responses: at least 15 running minutes above 130 steps per minute, allowing two-minute interruptions

	Any run (pp)	Running minutes	Running minutes conditional on running	Cadence (steps/min)
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>				
Estimate	-0.089* (0.047)	-0.052*** (0.020)	-0.990*** (0.345)	-0.918** (0.409)
First-stage <i>F</i>	317.1	317.1	405.8	405.8
Adjusted <i>R</i> ²	0.258	0.313	0.494	0.335
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-0.179* (0.096)	-0.104*** (0.040)	-1.974*** (0.703)	-1.830** (0.824)
First-stage <i>F</i>	247.7	247.7	273.9	273.9
Adjusted <i>R</i> ²	0.258	0.313	0.494	0.335
Observations	51,033,286	51,033,286	1,350,811	1,350,811
Mean outcome	2.666	0.995	37.403	156.879

Notes: The dependent variable is given in each column heading. Running participation is measured in percentage points, duration in minutes, and cadence in steps per minute. Participation and unconditional minutes include eligible days without a qualifying run. Conditional minutes and cadence use days with a qualifying run; conditional minutes sum running time across all qualifying bouts that day. Conditional estimates may also reflect changes in which participants run. Qualifying runs follow the cadence, duration, and interruption definition indicated in the table title. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage *F*-statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. **p* < 0.10, ***p* < 0.05, ****p* < 0.01.

Table F.21: Cadence and breaks within running bouts

	1-min gaps Bout cadence	1-min gaps Break share (pp)	2-min gaps Bout cadence	2-min gaps Break share (pp)
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>				
Estimate	-0.906* (0.463)	0.048 (0.084)	-0.997** (0.423)	0.300** (0.117)
First-stage F	404.5	404.5	405.8	405.8
Adjusted R^2	0.338	0.348	0.334	0.401
<i>Panel B: PM_{2.5} > 12</i>				
Estimate	-1.805* (0.929)	0.095 (0.168)	-1.987** (0.852)	0.598** (0.237)
First-stage F	271.3	271.3	273.9	273.9
Adjusted R^2	0.338	0.348	0.334	0.401
Observations	1,233,764	1,233,764	1,350,811	1,350,811
Mean outcome	156.577	3.148	154.557	6.075

Notes: The dependent variable is given in each column heading. The sample is conditional on a qualifying bout with at least 15 running minutes above 130 steps per minute, allowing the stated interruptions. Bout cadence includes break minutes; break shares are in percentage points. Panels A and B report continuous PM_{2.5} and PM_{2.5} > 12, respectively. Continuous-PM estimates are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase; binary estimates measure a change in the indicated daily exposure indicator. Concentrations and thresholds are in $\mu\text{g}/\text{m}^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Alternative Pollution Thresholds

Table G.22 reports the $PM_{2.5} > 9$ and $PM_{2.5} > 25$ specifications. The $PM_{2.5} > 9$ estimates imply 190 fewer daily steps and 3.68 fewer minutes of sleep. The wind-direction instrument is much less predictive of whether $PM_{2.5}$ exceeds $25 \mu g/m^3$, as indicated by the low first-stage F -statistics, making the large estimated effects in this specification less reliable.

Table G.22: Daily outcomes: alternative pollution thresholds

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
<i>Panel A: $PM_{2.5} > 9$</i>							
Estimate	-189.923*** (56.762)	-1.861*** (0.682)	-0.128*** (0.029)	-2.670** (1.072)	3.069*** (1.071)	0.551 (5.119)	-3.679*** (0.595)
First-stage F	499.3	499.3	499.3	499.3	499.3	499.3	451.7
Adjusted R^2	0.460	0.449	0.349	0.418	0.436	0.587	0.284
<i>Panel B: $PM_{2.5} > 25$</i>							
Estimate	-6,004.456** (2,603.682)	-58.838** (28.168)	-4.042** (1.619)	-84.398** (42.404)	97.037** (47.416)	17.411 (162.621)	-124.081** (56.008)
First-stage F	8.0	8.0	8.0	8.0	8.0	8.0	5.7
Adjusted R^2	0.448	0.441	0.347	0.414	0.435	0.587	0.273
Observations	51,033,286	51,033,286	51,033,286	51,033,285	51,033,285	51,032,966	36,952,488
Mean outcome	7,838.291	85.645	1.639	259.022	817.699	2,320.103	389.539

Notes: The dependent variable is given in each column heading. Activity-duration and sleep outcomes are in minutes; samples can differ across outcomes. Panels A and B report $PM_{2.5} > 9$ and $PM_{2.5} > 25$, respectively. Binary estimates measure a change in the indicated daily exposure indicator; thresholds are in $\mu g/m^3$. Models are estimated by two-stage least squares using the cross-fitted selected-direction instrument. Observations and mean outcomes refer to each model's final estimation sample. First-stage F -statistics are cluster-robust Wald statistics. The $PM_{2.5} > 25$ models have substantially weaker first stages. Regressions include temperature and precipitation categories, daylight duration, wind speed, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

H Controls for Other Pollutants

Following La Nauze and Severnini (2025), we add ozone, carbon monoxide, sulfur dioxide, and nitrogen dioxide as controls, separately and jointly. Table H.23 reports continuous-PM estimates in Panel A and estimates for the $PM_{2.5} > 12$ indicator in Panel B. Each specification is compared with a baseline regression using the same estimation sample. The estimated step reductions remain negative and statistically significant across all specifications.

Table H.23: Daily steps with controls for other pollutants

	(1)	(2)	(3)	(4)	(5)
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$)</i>					
Daily steps	-244.724*** (53.341)	-168.128*** (49.579)	-148.836*** (51.249)	-146.743*** (51.911)	-240.695*** (55.469)
First-stage F	320.0	259.0	537.5	459.5	374.7
Adjusted R^2	0.460	0.460	0.445	0.445	0.445
<i>Panel B: $PM_{2.5} > 12$</i>					
Daily steps	-487.662*** (107.065)	-337.350*** (99.454)	-294.630*** (101.535)	-290.940*** (103.261)	-473.343*** (111.690)
First-stage F	233.3	219.1	284.1	242.8	234.3
Adjusted R^2	0.460	0.460	0.445	0.445	0.444
Observations	51,033,286	51,033,286	34,905,302	34,905,302	34,905,302
Mean outcome	7,838.291	7,838.291	8,156.938	8,156.938	8,156.938
<i>Controls for other pollutants</i>					
Ozone	Yes				Yes
Carbon monoxide		Yes			Yes
Sulfur dioxide			Yes		Yes
Nitrogen dioxide				Yes	Yes

Notes: The dependent variable is daily steps. Standard errors clustered by ZIP3 and calendar date are in parentheses. All models include participant, ZIP3-by-month-by-year, and calendar-date fixed effects, temperature and precipitation categories, daylight duration, and wind speed. Continuous-PM coefficients and standard errors are scaled to a 10 $\mu g/m^3$ increase. Panels distinguish continuous PM (per 10 $\mu g/m^3$) and $PM_{2.5} > 12$. First-stage F -statistics are cluster-robust Wald statistics. Observations and mean outcomes describe final estimation samples. Columns include ozone, carbon monoxide, sulfur dioxide, nitrogen dioxide, and all four pollutants, respectively, as linear controls, following La Nauze and Severnini (2025). The control indicators apply to both panels. PM exposure alone is instrumented. Each column uses observations with available measurements for its included pollutants; samples may differ across columns. These specifications do not identify separate causal effects of all pollutants. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

I Wildfire Smoke Results

I.1 Prior Wildfire Smoke Experience

Figures I.14 and I.15 report responses to wildfire smoke by quartiles of prior smoke exposure. Activity reductions do not vary systematically with prior exposure, providing no clear evidence that participants with more smoke experience respond differently.

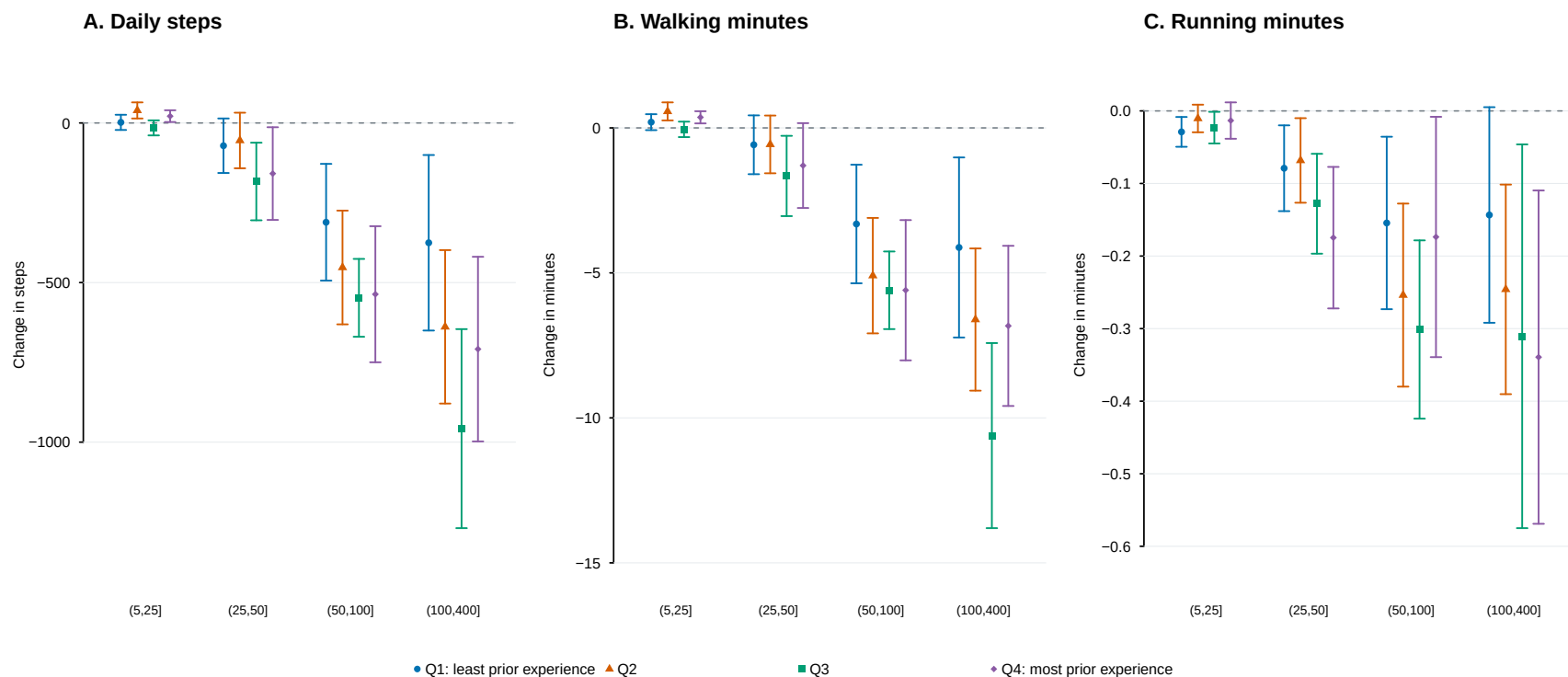


Figure I.14: Wildfire smoke and movement by prior smoke experience

Notes: Dots show OLS estimates for wildfire PM_{2.5} categories relative to 0–5 $\mu\text{g}/\text{m}^3$, estimated separately within each prior-experience quartile. The horizontal axis gives wildfire PM_{2.5} in $\mu\text{g}/\text{m}^3$; parentheses exclude the lower endpoint and brackets include the upper endpoint. Quartiles are based on the ZIP3's smoke-episode count through the preceding calendar year; Q1 has the least prior experience and Q4 the most. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date. Models include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. These are separate group estimates, not interaction coefficients or formal tests of differences across quartiles. Vertical scales differ across outcomes.

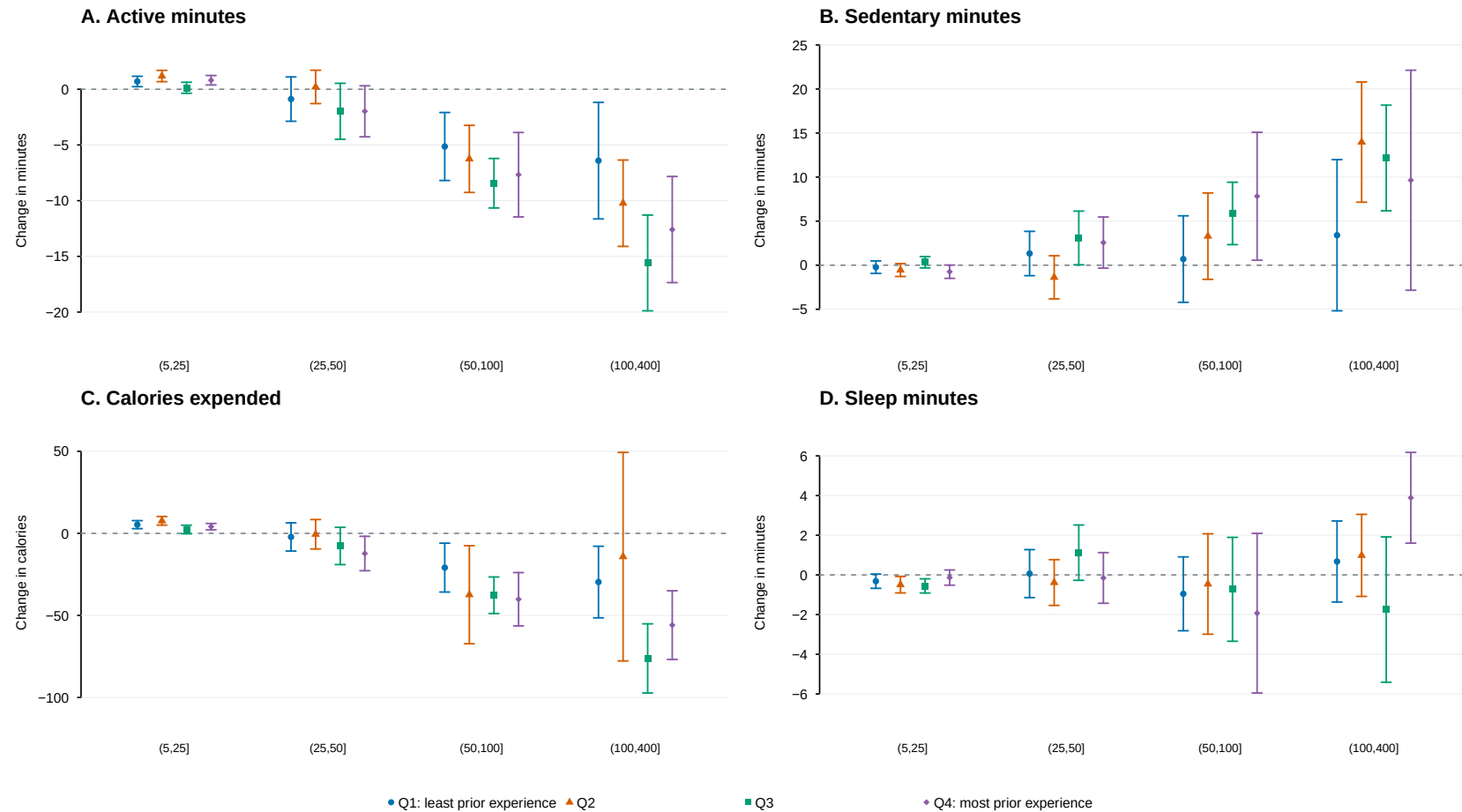


Figure I.15: Wildfire smoke, activity intensity, energy expenditure, and sleep by prior smoke experience

Notes: Dots show OLS estimates for wildfire PM_{2.5} categories relative to 0–5 μg/m³, estimated separately within each prior-experience quartile. The horizontal axis gives wildfire PM_{2.5} in μg/m³; parentheses exclude the lower endpoint and brackets include the upper endpoint. Quartiles are based on the ZIP3’s smoke-episode count through the preceding calendar year; Q1 has the least prior experience and Q4 the most. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date. Models include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. These are separate group estimates, not interaction coefficients or formal tests of differences across quartiles. Vertical scales differ across outcomes.

I.2 Dynamic Responses

Tables I.24 and I.25 report cumulative responses to wildfire smoke using continuous concentrations and concentration categories, respectively. The estimates indicate that subsequent activity does not fully offset the initial reductions over seven or fourteen days.

Figures I.16–I.19 show the daily and cumulative responses for active time, sedentary time, steps, and walking. The left panels show responses at each lag, while the right panels show cumulative responses.

Table I.24: Cumulative responses: wildfire smoke

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
<i>Panel A: Unrestricted, days 0–7</i>							
Cumulative effect	-74.015*** (9.058)	-0.798*** (0.108)	-0.025*** (0.005)	-1.073*** (0.186)	1.240*** (0.208)	-3.971*** (1.302)	-0.155 (0.102)
Adjusted R^2	0.455	0.444	0.347	0.416	0.445	0.587	0.283
<i>Panel B: Spline, days 0–7</i>							
Cumulative effect	-73.590*** (9.299)	-0.793*** (0.111)	-0.025*** (0.005)	-1.059*** (0.193)	1.233*** (0.209)	-3.903*** (1.329)	-0.158 (0.103)
Adjusted R^2	0.455	0.444	0.347	0.416	0.445	0.587	0.283
<i>Panel C: Spline, days 0–14</i>							
Cumulative effect	-73.660*** (12.465)	-0.804*** (0.153)	-0.019*** (0.007)	-1.213*** (0.275)	1.254*** (0.354)	-4.944*** (1.405)	-0.036 (0.144)
Adjusted R^2	0.455	0.444	0.347	0.416	0.445	0.587	0.283
Observations	44,236,029	44,236,029	44,236,029	44,236,029	44,236,029	44,235,766	31,176,930
Mean outcome	7,966.809	87.045	1.699	262.205	816.701	2,330.242	390.686

Notes: Dependent variables are given in the column headings. Samples require complete exposure histories. Panels distinguish unrestricted seven-day, spline seven-day, and spline fourteen-day models. Entries sum responses from day zero through the stated horizon. Estimates describe a one-day increase of $10 \mu\text{g}/\text{m}^3$ in wildfire $\text{PM}_{2.5}$. Models are estimated by OLS. Observations and mean outcomes refer to each model's estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are clustered by ZIP3 and calendar date. Cumulative standard errors use the full clustered covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table I.25: Cumulative responses: wildfire smoke categories

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
<i>Panel A: Unrestricted, days 0–7</i>							
(5, 25]	-46.014** (21.040)	-0.463* (0.260)	-0.008 (0.012)	-0.708 (0.432)	1.760*** (0.523)	-4.624** (2.015)	-0.192 (0.262)
(25, 50]	-310.959*** (70.821)	-3.201*** (0.851)	-0.115*** (0.045)	-4.036*** (1.551)	3.042 (2.255)	-26.159*** (8.313)	0.930 (1.105)
(50, 100]	-765.752*** (114.927)	-8.390*** (1.327)	-0.271*** (0.089)	-10.792*** (2.271)	12.880*** (3.552)	-24.062 (18.361)	-4.713*** (1.289)
(100, 400]	-1,079.468*** (152.906)	-12.441*** (1.726)	-0.307*** (0.098)	-17.721*** (2.587)	15.472*** (4.474)	-69.053*** (18.667)	-0.845 (1.322)
Adjusted R^2	0.455	0.444	0.347	0.416	0.445	0.587	0.283
<i>Panel B: Spline, days 0–7</i>							
(5, 25]	-45.353** (21.075)	-0.454* (0.260)	-0.008 (0.012)	-0.697 (0.431)	1.748*** (0.523)	-4.643** (1.999)	-0.188 (0.261)
(25, 50]	-312.977*** (70.632)	-3.226*** (0.847)	-0.115*** (0.044)	-4.087*** (1.539)	3.092 (2.252)	-26.387*** (8.264)	0.905 (1.113)
(50, 100]	-766.515*** (115.285)	-8.396*** (1.340)	-0.272*** (0.087)	-10.807*** (2.291)	12.906*** (3.581)	-24.243 (18.288)	-4.769*** (1.308)
(100, 400]	-1,077.844*** (158.216)	-12.407*** (1.817)	-0.312*** (0.097)	-17.546*** (2.898)	15.268*** (4.535)	-68.025*** (20.221)	-0.876 (1.351)
Adjusted R^2	0.455	0.444	0.347	0.416	0.445	0.587	0.283
<i>Panel C: Spline, days 0–14</i>							
(5, 25]	-74.036** (30.087)	-0.898** (0.368)	0.013 (0.019)	-1.637*** (0.606)	1.482* (0.805)	-9.611*** (2.739)	0.651* (0.385)
(25, 50]	-221.786* (116.949)	-1.846 (1.378)	-0.129* (0.078)	-2.211 (2.434)	4.195 (3.711)	-14.883 (11.597)	0.018 (1.554)
(50, 100]	-704.521*** (170.369)	-7.484*** (2.018)	-0.314** (0.157)	-8.738** (3.471)	11.980* (6.180)	-20.250 (19.921)	-6.176*** (2.084)
(100, 400]	-1,038.544*** (223.199)	-12.042*** (2.697)	-0.215 (0.140)	-20.871*** (4.940)	18.046** (8.655)	-96.597*** (23.464)	0.630 (2.716)
Adjusted R^2	0.455	0.444	0.347	0.416	0.445	0.587	0.283
Observations	44,236,029	44,236,029	44,236,029	44,236,029	44,236,029	44,235,766	31,176,930
Mean outcome	7,966.809	87.045	1.699	262.205	816.701	2,330.242	390.686

Notes: Dependent variables are given in the column headings. Samples require complete exposure histories. Panels distinguish unrestricted seven-day, spline seven-day, and spline fourteen-day models. Entries sum responses from day zero through the stated horizon. Rows indicate wildfire PM_{2.5} intervals in $\mu\text{g}/\text{m}^3$; the reference is [0, 5]. Each category response describes a one-day change relative to the reference, holding other days fixed. Models are estimated by OLS. Observations and mean outcomes refer to each model's estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are clustered by ZIP3 and calendar date. Cumulative standard errors use the full clustered covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

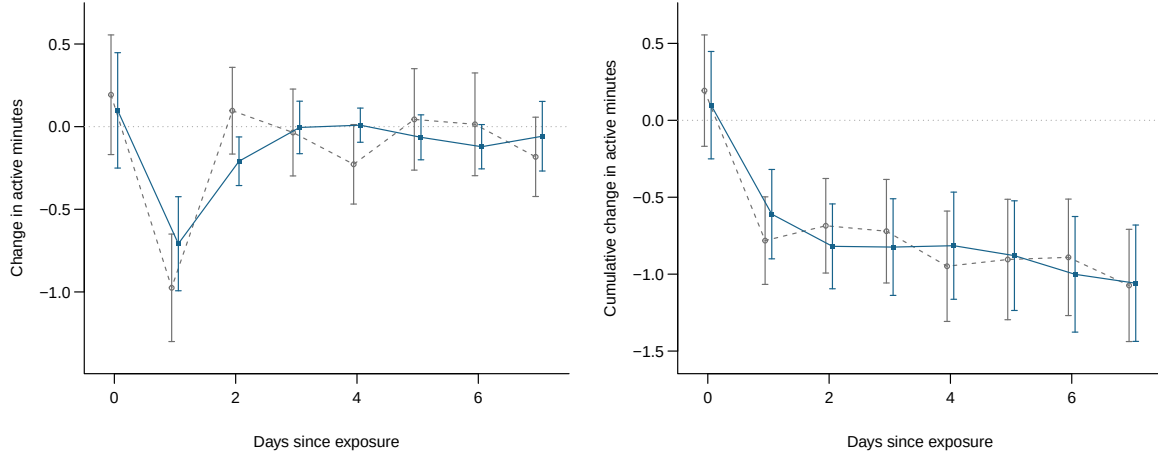


Figure I.16: Dynamic wildfire smoke responses: active minutes. OLS responses to a one-day increase of $10 \mu\text{g}/\text{m}^3$ in wildfire $\text{PM}_{2.5}$. Gray dashed lines with open circles show unrestricted seven-day estimates; blue solid lines with squares show seven-day spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

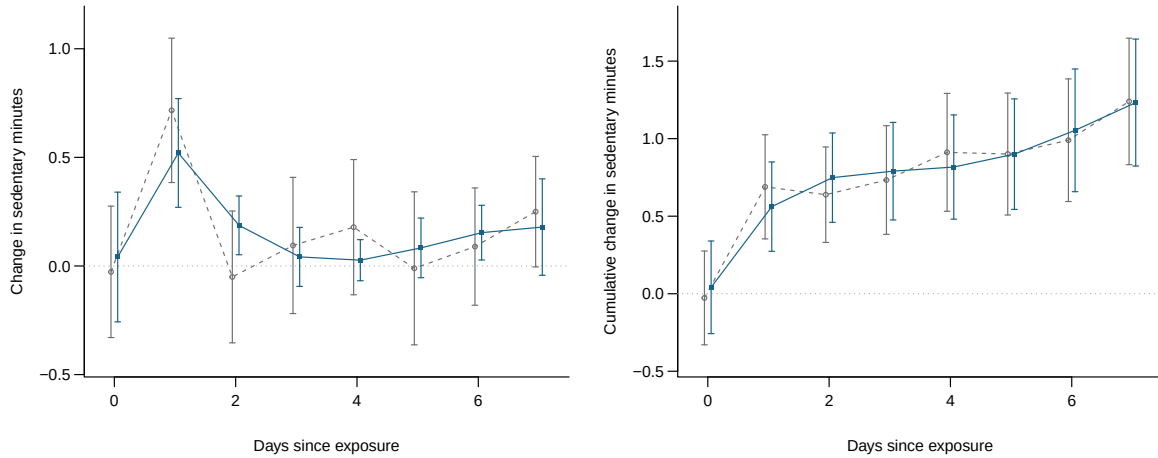


Figure I.17: Dynamic wildfire smoke responses: sedentary minutes. OLS responses to a one-day increase of $10 \mu\text{g}/\text{m}^3$ in wildfire $\text{PM}_{2.5}$. Gray dashed lines with open circles show unrestricted seven-day estimates; blue solid lines with squares show seven-day spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

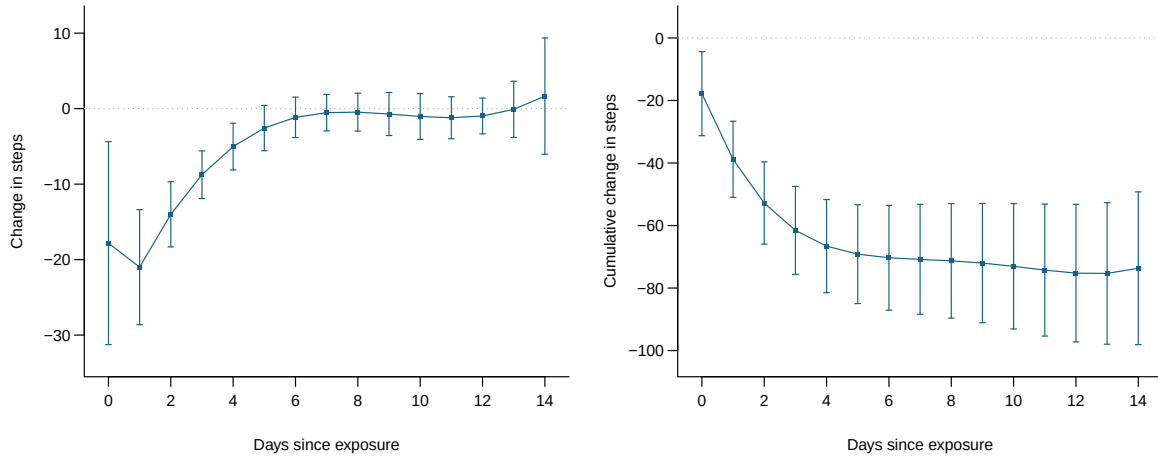


Figure I.18: Fourteen-day wildfire smoke responses: steps. OLS responses to a one-day increase of $10 \mu\text{g}/\text{m}^3$ in wildfire $\text{PM}_{2.5}$. Blue solid lines with squares show fourteen-day spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Models include temperature and precipitation categories, daylight duration, participant, ZIP3-by-month-by-year, and calendar-date fixed effects. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

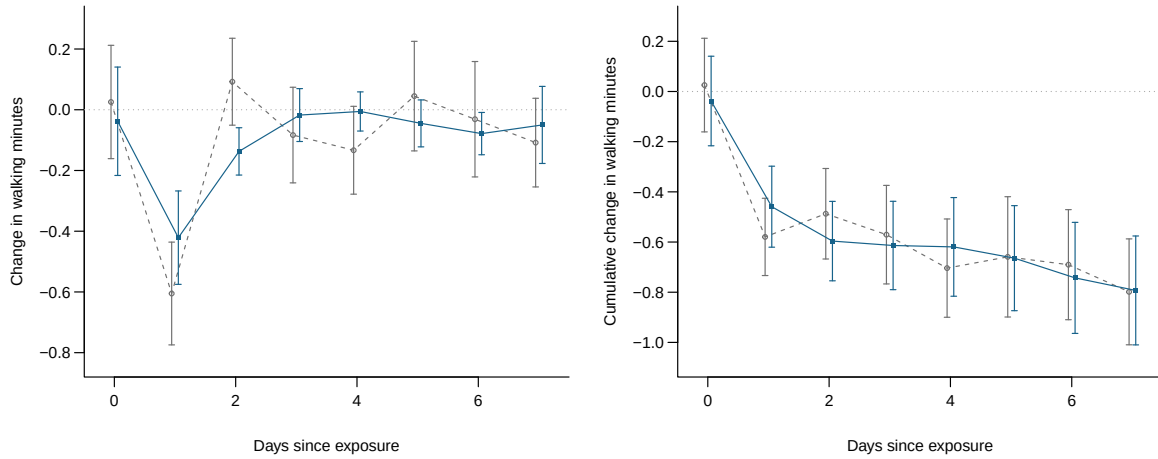


Figure I.19: Wildfire smoke responses: walking minutes. OLS responses to a one-day increase of $10 \mu\text{g}/\text{m}^3$ in wildfire $\text{PM}_{2.5}$. Gray dashed lines with open circles show unrestricted seven-day estimates; blue solid lines with squares show seven-day spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Models include temperature and precipitation categories, daylight duration, participant, ZIP3-by-month-by-year, and calendar-date fixed effects. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date.

I.3 Activity-Intensity Responses

Table I.26: Daily outcomes by smoke category

	Lightly active minutes	Fairly active minutes	Very active minutes
(5, 25]	0.144 (0.105)	0.386*** (0.047)	0.200*** (0.050)
(25, 50]	-0.264 (0.331)	-0.079 (0.169)	-0.743*** (0.201)
(50, 100]	-2.326*** (0.543)	-1.644*** (0.235)	-2.707*** (0.348)
(100, 400]	-4.181*** (0.711)	-2.449*** (0.385)	-3.942*** (0.680)
Observations	44,236,029	44,236,029	44,236,029
Mean outcome	220.725	20.270	21.210
Adjusted R^2	0.415	0.350	0.353

Notes: The dependent variable is given in each column heading. Entries are OLS coefficients for wildfire PM_{2.5} categories relative to [0, 5] $\mu\text{g}/\text{m}^3$. Category coefficients within each column are estimated jointly. Observations and mean outcomes refer to each model's final estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Additional AQI Results

J.1 Category Estimates

Table J.27: Daily outcomes by PM-specific AQI category

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
51–100	34.416*** (5.304)	0.482*** (0.063)	-0.013*** (0.003)	0.702*** (0.106)	-0.449*** (0.117)	4.615*** (0.456)	-0.353*** (0.057)
101–150	-0.967 (22.895)	0.219 (0.266)	-0.057*** (0.017)	0.349 (0.447)	-0.397 (0.556)	2.270 (2.145)	0.038 (0.321)
151–200	-230.130*** (53.381)	-2.462*** (0.577)	-0.130*** (0.038)	-3.668*** (0.918)	1.898* (1.063)	-19.220*** (4.918)	-0.145 (0.490)
201–300	-487.882*** (122.960)	-5.127*** (1.320)	-0.243*** (0.071)	-8.033*** (2.052)	4.706 (3.034)	-25.300* (15.166)	0.770 (0.723)
301–500	-891.184*** (220.044)	-10.637*** (2.404)	-0.072 (0.144)	-15.925*** (4.370)	17.970*** (6.091)	-66.698*** (17.124)	4.569 (3.369)
Observations	38,307,693	38,307,693	38,307,693	38,307,692	38,307,692	38,307,382	28,207,828
Mean outcome	7,819.787	85.256	1.626	257.531	816.660	2,317.681	389.131
Adjusted R^2	0.462	0.451	0.346	0.420	0.437	0.574	0.288

Notes: The dependent variable is given in each column heading. Entries are OLS coefficients for PM-specific AQI categories relative to AQI 0–50. Category coefficients within each column are estimated jointly. AQI is unitless; these comparisons are not discontinuity estimates. Observations and mean outcomes refer to each model's final estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by reporting area and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table J.28: Daily outcomes by overall AQI category

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
51–100	49.213*** (5.553)	0.661*** (0.066)	-0.010*** (0.003)	1.048*** (0.104)	-0.652*** (0.107)	6.238*** (0.463)	-0.438*** (0.055)
101–150	47.533*** (11.405)	0.706*** (0.142)	-0.030*** (0.010)	1.461*** (0.225)	-0.740*** (0.277)	8.667*** (0.987)	-0.884*** (0.163)
151–200	-104.346*** (39.658)	-0.993** (0.449)	-0.100*** (0.027)	-1.087 (0.721)	0.983 (0.727)	-4.218 (3.676)	-0.798** (0.399)
201–300	-380.087*** (96.822)	-4.011*** (1.055)	-0.191*** (0.056)	-6.126*** (1.705)	3.015 (2.523)	-18.175 (11.888)	-0.312 (0.735)
301–500	-452.975** (224.306)	-5.662** (2.531)	-0.027 (0.112)	-9.074** (3.874)	7.879 (5.504)	-35.562** (16.225)	2.664 (2.496)
Observations	43,645,202	43,645,202	43,645,202	43,645,201	43,645,201	43,644,886	31,506,171
Mean outcome	7,873.272	85.884	1.671	258.843	818.794	2,319.719	389.426
Adjusted R^2	0.457	0.445	0.350	0.416	0.436	0.579	0.285

Notes: The dependent variable is given in each column heading. Entries are OLS coefficients for overall AQI (the maximum across reported pollutants) categories relative to AQI 0–50. Category coefficients within each column are estimated jointly. AQI is unitless; these comparisons are not discontinuity estimates. Observations and mean outcomes refer to each model's final estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by reporting area and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table J.29: Daily outcomes by PM-specific AQI category

	Lightly active minutes	Fairly active minutes	Very active minutes
51–100	0.151** (0.065)	0.292*** (0.024)	0.258*** (0.029)
101–150	-0.030 (0.310)	0.389*** (0.108)	-0.010 (0.131)
151–200	-1.395*** (0.513)	-0.842*** (0.218)	-1.431*** (0.328)
201–300	-2.864** (1.111)	-1.956*** (0.440)	-3.213*** (0.773)
301–500	-6.732** (2.805)	-4.384*** (0.704)	-4.809*** (1.141)
Observations	38,307,692	38,307,692	38,307,693
Mean outcome	217.295	19.317	20.919
Adjusted R^2	0.418	0.330	0.355

Notes: The dependent variable is given in each column heading. Entries are OLS coefficients for PM-specific AQI categories relative to AQI 0–50. Category coefficients within each column are estimated jointly. AQI is unitless; these comparisons are not discontinuity estimates. Observations and mean outcomes refer to each model's final estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by reporting area and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table J.30: Daily outcomes by overall AQI category

	Lightly active minutes	Fairly active minutes	Very active minutes
51–100	0.335*** (0.061)	0.375*** (0.025)	0.339*** (0.031)
101–150	0.478*** (0.138)	0.574*** (0.061)	0.408*** (0.062)
151–200	-0.453 (0.415)	-0.102 (0.164)	-0.532** (0.232)
201–300	-2.342** (0.944)	-1.404*** (0.387)	-2.380*** (0.601)
301–500	-4.144* (2.127)	-2.252** (0.923)	-2.678** (1.052)
Observations	43,645,201	43,645,201	43,645,202
Mean outcome	218.142	19.680	21.021
Adjusted R^2	0.413	0.336	0.351

Notes: The dependent variable is given in each column heading. Entries are OLS coefficients for overall AQI (the maximum across reported pollutants) categories relative to AQI 0–50. Category coefficients within each column are estimated jointly. AQI is unitless; these comparisons are not discontinuity estimates. Observations and mean outcomes refer to each model's final estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are two-way clustered by reporting area and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J.2 Dynamic Responses

Table J.31 reports cumulative responses to a ten-point increase in PM-specific AQI. Figures J.20 and J.21 show the daily and cumulative responses for steps and running minutes, respectively.

Table J.31: Cumulative responses: PM-specific AQI

	Daily steps	Walking minutes	Running minutes	Active minutes	Sedentary minutes	Calories	Sleep minutes
<i>Panel A: Unrestricted, days 0–7</i>							
Cumulative effect	-6.327*	-0.042	-0.010***	-0.112	0.204**	-0.278	-0.115***
	(3.700)	(0.044)	(0.002)	(0.073)	(0.088)	(0.415)	(0.043)
Adjusted R^2	0.462	0.452	0.345	0.421	0.437	0.577	0.288
<i>Panel B: Spline, days 0–7</i>							
Cumulative effect	-6.093*	-0.039	-0.010***	-0.107	0.202**	-0.268	-0.114***
	(3.694)	(0.044)	(0.002)	(0.073)	(0.088)	(0.415)	(0.043)
Adjusted R^2	0.462	0.452	0.345	0.421	0.437	0.577	0.288
<i>Panel C: Spline, days 0–14</i>							
Cumulative effect	-8.470*	-0.075	-0.009***	-0.195**	0.206*	-0.581	-0.058
	(4.703)	(0.056)	(0.002)	(0.093)	(0.124)	(0.448)	(0.055)
Adjusted R^2	0.462	0.452	0.345	0.421	0.437	0.577	0.288
Observations	36,158,751	36,158,751	36,158,751	36,158,751	36,158,751	36,158,443	26,734,259
Mean outcome	7,811.045	85.139	1.619	257.232	816.061	2,317.191	389.000

Notes: Dependent variables are given in the column headings. Samples require complete exposure histories. Panels distinguish unrestricted seven-day, spline seven-day, and spline fourteen-day models. Entries sum responses from day zero through the stated horizon. Estimates describe a one-day increase of 10 points in PM-specific AQI. Models are estimated by OLS. Observations and mean outcomes refer to each model's estimation sample. Regressions include temperature and precipitation categories, daylight duration, participant fixed effects, ZIP3-by-month-by-year fixed effects, and calendar-date fixed effects. Standard errors (in parentheses) are clustered by reporting area and calendar date. Cumulative standard errors use the full clustered covariance matrix. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

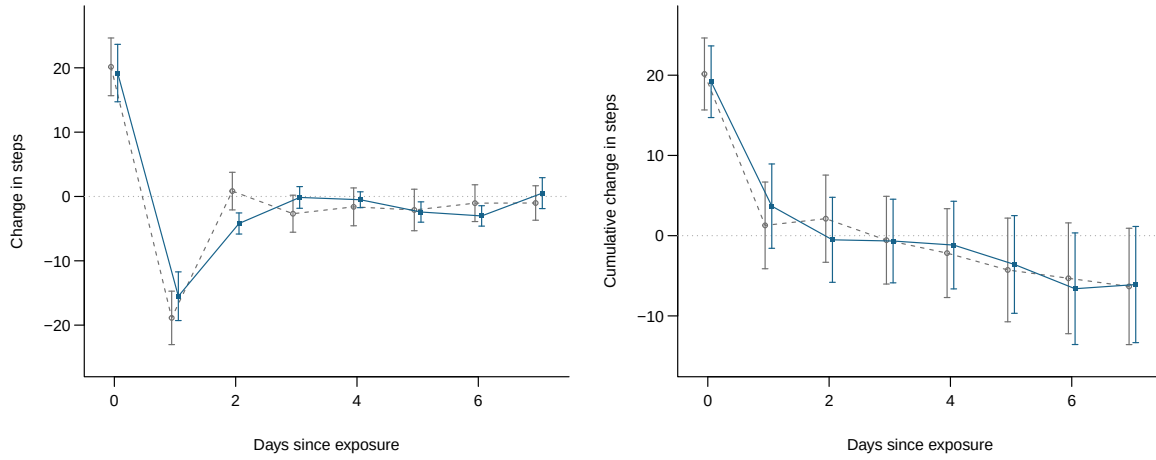


Figure J.20: PM-specific AQI dynamics: steps. OLS responses to a one-day increase of ten points in PM-specific AQI. Gray dashed lines with open circles show unrestricted seven-day estimates; blue solid lines with squares show seven-day spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Models include temperature and precipitation categories, daylight duration, participant, ZIP3-by-month-by-year, and calendar-date fixed effects. Bars show 95% confidence intervals based on standard errors clustered by reporting area and calendar date.

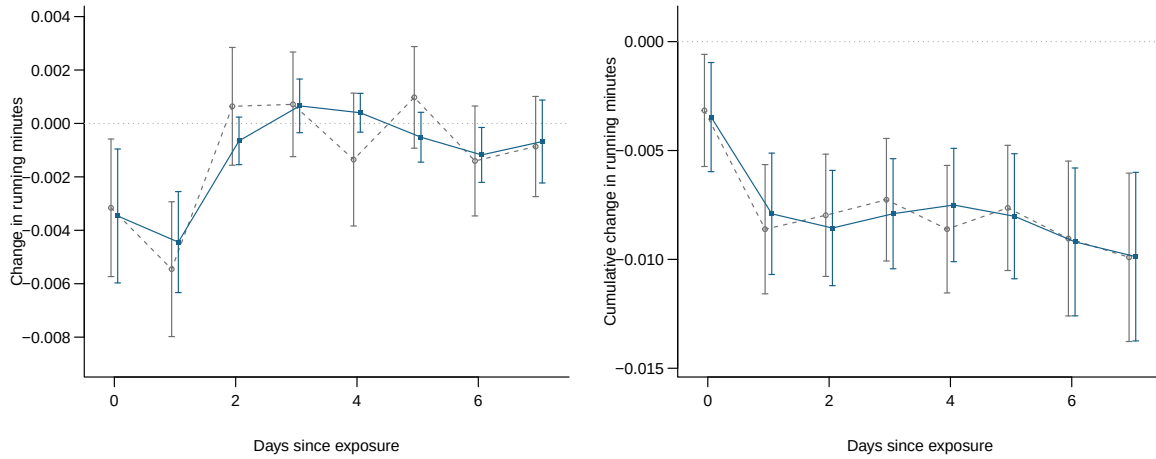


Figure J.21: PM-specific AQI dynamics: running minutes. OLS responses to a one-day increase of ten points in PM-specific AQI. Gray dashed lines with open circles show unrestricted seven-day estimates; blue solid lines with squares show seven-day spline estimates. The left panel shows the response at each lag. The right panel shows the cumulative response through each lag. Models include temperature and precipitation categories, daylight duration, participant, ZIP3-by-month-by-year, and calendar-date fixed effects. Bars show 95% confidence intervals based on standard errors clustered by reporting area and calendar date.

K Heterogeneity in Pollution Responses

Figures K.22 and K.23 report differences in pollution responses by age, gender, household income, race and ethnicity, and evidence of respiratory conditions. The plotted interaction coefficients measure each group's response relative to the indicated reference group. For respiratory conditions, we compare participants with a recorded asthma or chronic obstructive pulmonary disease diagnosis or qualifying respiratory medication record with participants who have electronic health records but no such evidence.

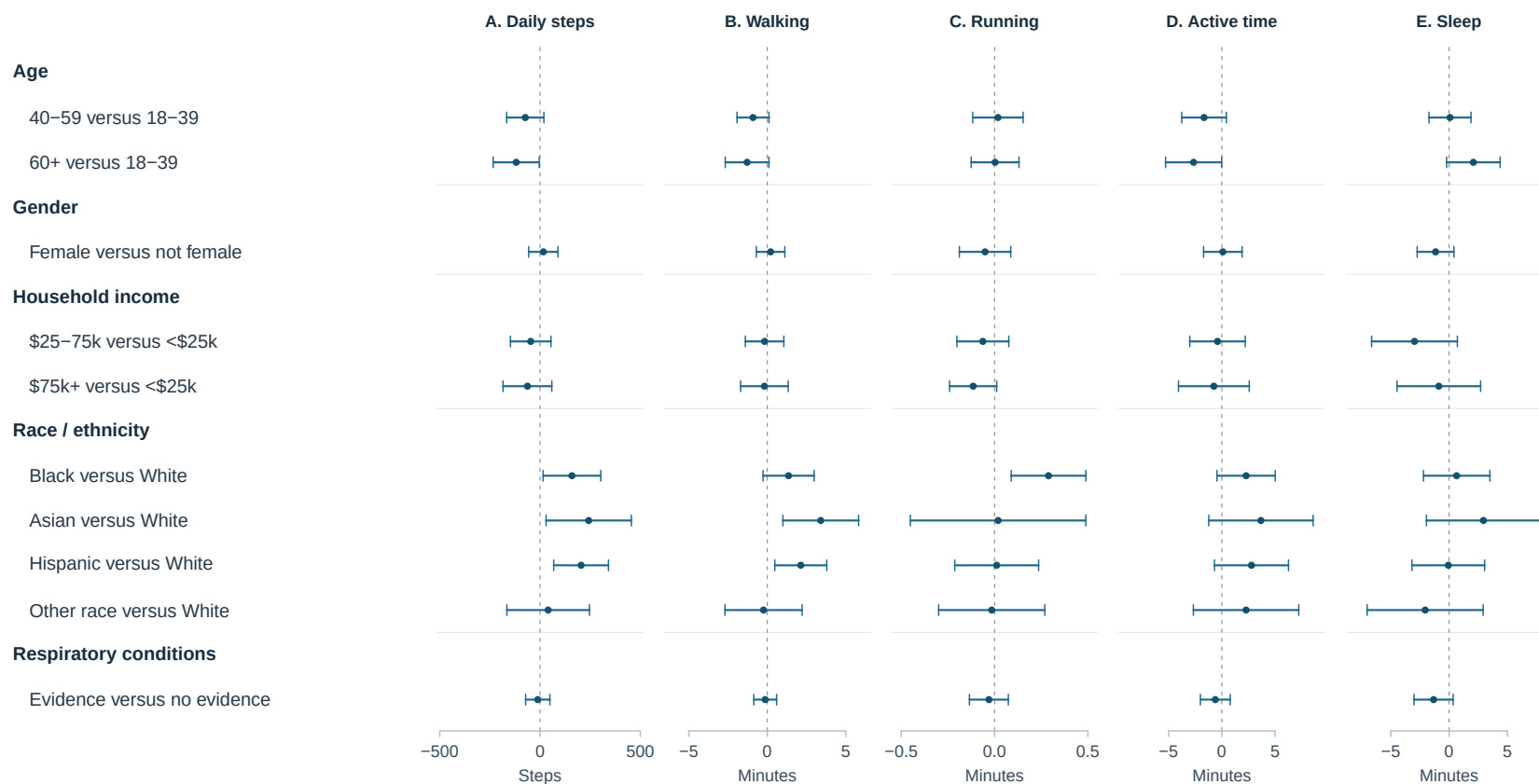


Figure K.22: Differences in responses to a $10 \mu\text{g}/\text{m}^3$ increase in $\text{PM}_{2.5}$.

Notes: Dots show the estimated interaction terms between pollution exposure and each group indicator in the IV models. Each coefficient measures the difference in the pollution response between the indicated group and its reference group. Positive values indicate a more positive response than in the reference group; zero indicates no difference. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date. Intervals are not adjusted for multiple comparisons. Panel scales differ. For respiratory conditions, evidence denotes a recorded asthma or chronic obstructive pulmonary disease diagnosis, or a qualifying respiratory medication record (rescue or maintenance treatment), at any time through the study end. The comparison group has EHR records but no such evidence; participants without EHR records are excluded from this comparison.

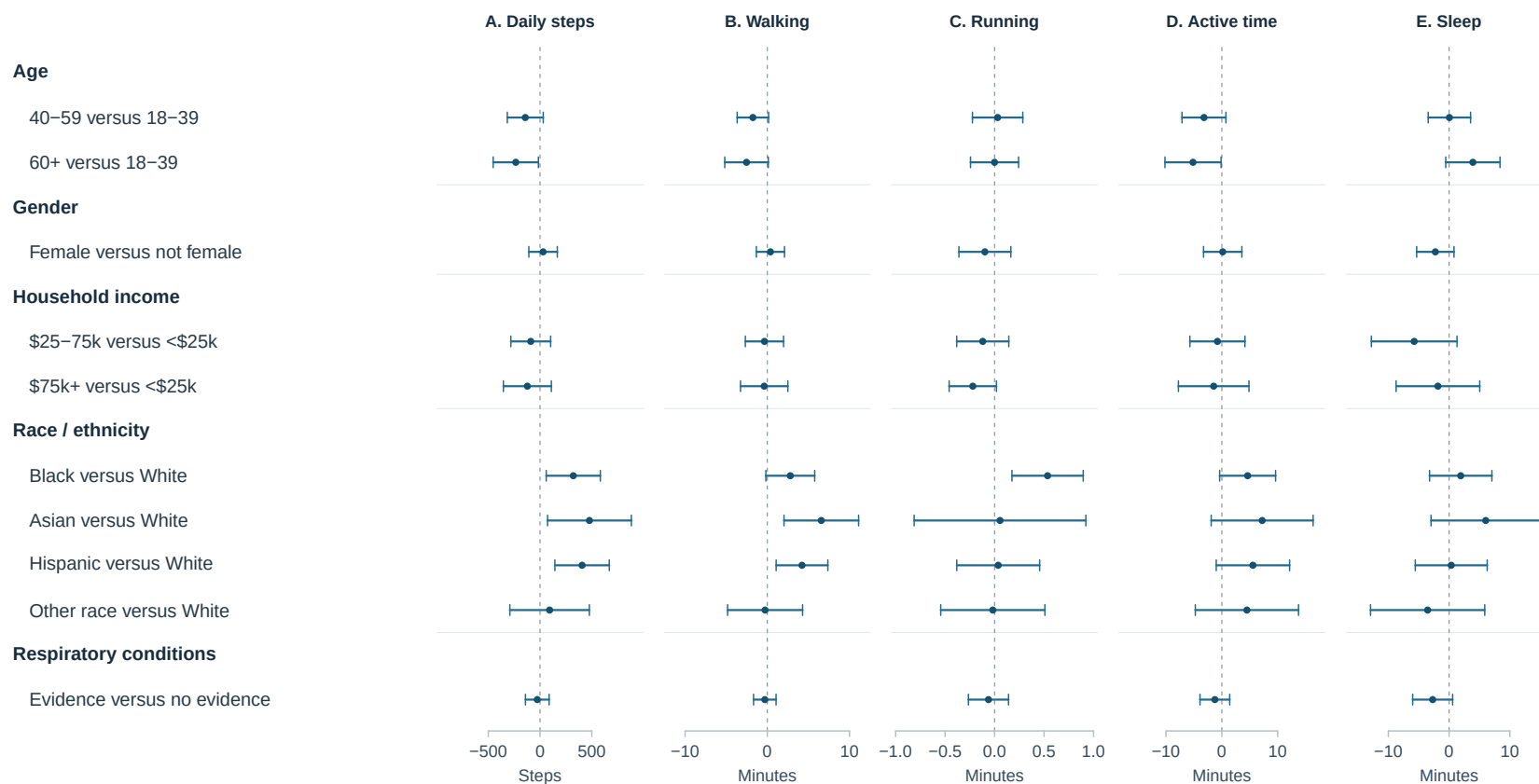


Figure K.23: Differences in responses to $PM_{2.5} > 12$.

Notes: Dots show the estimated interaction terms between pollution exposure and each group indicator in the IV models. Each coefficient measures the difference in the pollution response between the indicated group and its reference group. Positive values indicate a more positive response than in the reference group; zero indicates no difference. Bars show 95% confidence intervals based on standard errors clustered by ZIP3 and calendar date. Intervals are not adjusted for multiple comparisons. Panel scales differ. For respiratory conditions, evidence denotes a recorded asthma or chronic obstructive pulmonary disease diagnosis, or a qualifying respiratory medication record (rescue or maintenance treatment), at any time through the study end. The comparison group has EHR records but no such evidence; participants without EHR records are excluded from this comparison.

L Population Reweighting and WEAR Estimates

We calibrate participant weights to pooled 2015–2019 ACS distributions of gender-by-age, race and ethnicity, and household income among adults living in households. Male and female gender responses are matched to the corresponding ACS sex categories, and age is measured at the first eligible activity observation. We trim weights at the 0.5th and 99.5th percentiles and apply each participant’s weight to all their daily observations. Table L.32 compares weighted and unweighted estimates using the same sample of participants with complete demographic information.

Table L.33 compares pollution responses among WEAR and BYOD participants over overlapping calendar dates. We also estimate a pooled model interacting pollution with WEAR participation to test whether responses differ between the two groups.

Table L.32: Daily steps: population reweighting

	Original sample	Complete demographics	ACS weighted
<i>Panel A: Continuous PM_{2.5} (per 10 $\mu\text{g}/\text{m}^3$)</i>			
Daily steps	-161.755*** (48.547)	-166.402*** (48.358)	-147.931*** (51.702)
First-stage F	317.1	325.8	304.7
Adjusted R^2	0.460	0.461	0.447
<i>Panel B: PM_{2.5} > 12</i>			
Daily steps	-326.638*** (97.661)	-335.778*** (97.227)	-294.865*** (102.193)
First-stage F	247.7	250.2	248.9
Adjusted R^2	0.460	0.461	0.447
Observations	51,033,286	45,882,420	45,882,420
Mean outcome	7,838.291	7,868.735	8,076.355

Notes: The dependent variable is daily steps. Standard errors clustered by ZIP3 and calendar date are in parentheses. All models include participant, ZIP3-by-month-by-year, and calendar-date fixed effects, temperature and precipitation categories, daylight duration, and wind speed. Continuous-PM coefficients and standard errors are scaled to a 10 $\mu\text{g}/\text{m}^3$ increase. Panels distinguish continuous PM (per 10 $\mu\text{g}/\text{m}^3$) and PM_{2.5} > 12. First-stage F -statistics are cluster-robust Wald statistics. Observations and mean outcomes describe final estimation samples. The *Complete demographics* column reports unweighted estimates restricted to participants with an age of at least 18 at their first eligible activity observation, a male or female gender response, and race/ethnicity and household-income information that can be assigned to the ACS categories used for weighting. Participants with missing or unclassifiable values for these variables are excluded from this sample. The *ACS weighted* column applies population weights to exactly the same participant-days. Comparing these two columns isolates the effect of reweighting from the effect of restricting the sample. Raking targets use 2015–2019 ACS adults in households, with gender-by-age, race/ethnicity, and household-income margins. Male/female gender responses are matched to ACS sex categories. Weights are trimmed at the 0.5th and 99.5th percentiles and applied uniformly to each participant’s days. Trimming and unequal observation spans can leave residual demographic imbalance. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table L.33: Daily steps: WEAR and BYOD comparison

	BYOD	WEAR	WEAR minus BYOD
<i>Panel A: Continuous $PM_{2.5}$ (per 10 $\mu g/m^3$)</i>			
Daily steps	-191.711*** (70.889)	-148.715* (79.405)	88.145 (64.024)
First-stage F	145.1	51.9	28.0–85.2
Adjusted R^2	0.502	0.517	0.510
<i>Panel B: $PM_{2.5} > 12$</i>			
Daily steps	-413.681*** (154.002)	-308.368* (163.721)	185.608 (127.059)
First-stage F	111.8	52.1	28.6–55.7
Adjusted R^2	0.501	0.517	0.510
Observations	17,777,875	8,179,587	25,957,640
Mean outcome	7,799.728	6,622.075	7,428.616

Notes: The dependent variable is daily steps. WEAR identifies participants who consented to join the All of Us study providing Fitbits at no cost; BYOD identifies participants with no recorded WEAR enrollment. WEAR observations begin on the participant's consent date, and both groups are restricted to calendar dates represented in the WEAR sample. The first two columns report separate regressions for BYOD and WEAR participants. The third reports the WEAR-minus-BYOD difference from a pooled model that interacts pollution with WEAR participation, allows group-specific weather and wind-speed controls, and imposes common fixed effects. Pollution and its interaction with WEAR participation are instrumented using the selected-direction indicator and its interaction with WEAR participation. Panel A reports continuous-PM estimates, scaled to a 10 $\mu g/m^3$ increase; Panel B reports estimates for $PM_{2.5} > 12$. All models include participant, ZIP3-by-month-by-year, and calendar-date fixed effects, temperature and precipitation categories, daylight duration, and wind speed. Observations and mean outcomes refer to the final estimation samples; those in the third column describe the pooled sample. First-stage F -statistics are cluster-robust Wald statistics; the third column reports the range across the two first-stage equations. Standard errors (in parentheses) are clustered by ZIP3 and calendar date. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.